

8

7

6

5

4

3

2

1

1. ALL RESISTANCE VALUES ARE IN OHMS, 0.1 WATT +/- 5%.

2. ALL CAPACITANCE VALUES ARE IN MICROFARADS.

3. ALL CRYSTALS & OSCILLATOR VALUES ARE IN HERTZ.

REV

ECN

DESCRIPTION OF REVISION

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J11 MLB PIB SCHEMATIC

2.6.0

02/23/12

FX

CHINA

FDX

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(02/16/2010)

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(K09_MLB)

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01/11/2012

Schematic / PCB #'s

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
051-9276	1	SCHEM, MLB, J11	SCH	CRITICAL	
020-3208	1	PCBF, MLB, J11	PCB	CRITICAL	

DRAWING

TITLE=MLB

ABBREV=DRAWING

LAST MODIFIED=Thu Jan 23 12:05:40 2012

PRODUCT SAFETY REQUIREMENTS:

PCB, UL RECOGNIZED, MIN. 130-C TEMP. RATING AND V-0 FLAME RATING PER UL 796 & UL 94.

PCB TO BE SILK-SCREENED WITH UL/CUL RECOGNITION MARK, MANUFACTURER'S UL FILE NUMBER, UL PCB MATERIAL DESIGNATION, 130-C TEMP. RATING AND V-0 FLAME RATING.

DRAWING TITLE

SCHEM, MLB, J11

Apple Inc.

051-9276

2.7.0

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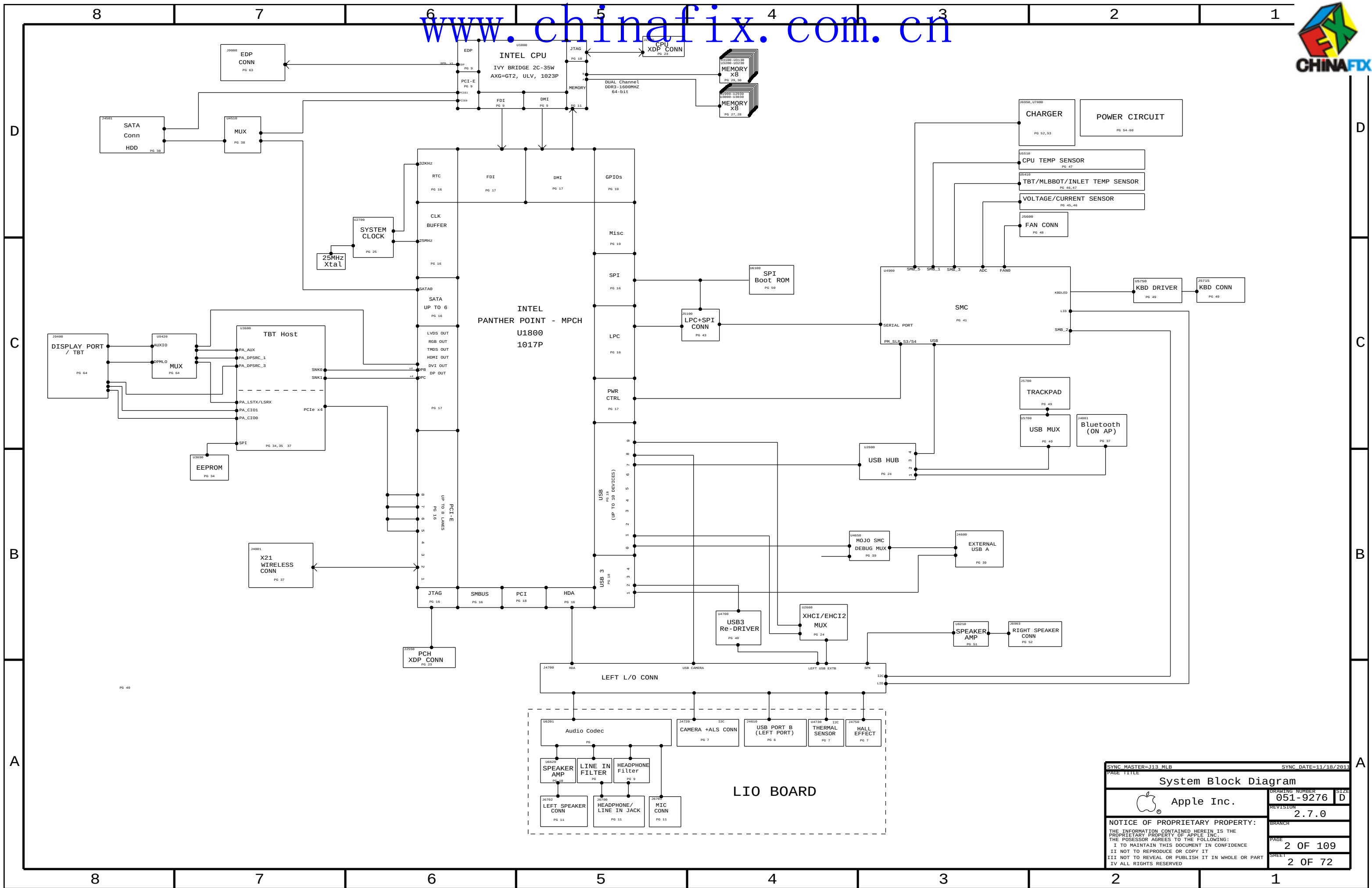
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[illegible][illegible]

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_DYKL]	CRITICAL	EEEE: DYKL
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_DYKH]	CRITICAL	EEEE: DYKH
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_DYKK]	CRITICAL	EEEE: DYKK
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_DYKF]	CRITICAL	EEEE: DYKF
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_DYKJ]	CRITICAL	EEEE: DYKJ
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_DYKG]	CRITICAL	EEEE: DYKG
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F0V3]	CRITICAL	EEEE: F0V3
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F0V4]	CRITICAL	EEEE: F0V4
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F279]	CRITICAL	EEEE: F279
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27C]	CRITICAL	EEEE: F27C
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27D]	CRITICAL	EEEE: F27D
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27F]	CRITICAL	EEEE: F27F
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27G]	CRITICAL	EEEE: F27G
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27H]	CRITICAL	EEEE: F27H
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27J]	CRITICAL	EEEE: F27J
825-7670	1	LABEL, TEXT, ML8, K21/K78	[EEEE_F27K]	CRITICAL	EEEE: F27K

SYNC MASTER-K23 MLB		SYNC DATE=11/16/2016	
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 Apple Inc.		DRAWING NUMBER 051-9276	
		SIZE D	
		REVISION 2.7.0	
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		BRANCH PAGE 4 OF 109	
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Module Parts

PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
33754197	1	IVB, QBPB, ES2, K0, 1.5, 17W, 2+2, 0.95, 4M, ULVB	U1000	CRITICAL	CPU:1.5GHZ
33754299	1	IVB, QC9E, QS, L1, 1.7, 17W, 2+2, 1.05, 3M, ULVBG	U1000	CRITICAL	CPU:1.7GHZ
33754296	1	IVB, QCSB, QS, L1, 2.0, 17W, 2+2, 1.15, 4M, ULVBG	U1000	CRITICAL	CPU:2.0GHZ
33754198	1	IVB, QBTB, ES2, K0, 1.5, 17W, 2+2, 0.95, 4M, ULVBGA	U1000	CRITICAL	CPU:1.5GHZTDP
33754299	1	IVB, QC9E, QS, L1, 1.7, 17W, 2+2, 1.05, 3M, ULVBG	U1000	CRITICAL	CPU:1.7GHZTDP
33754296	1	IVB, QC9B, QS, L1, 2.0, 17W, 2+2, 1.15, 4M, ULVBG	U1000	CRITICAL	CPU:2.0GHZTDP
33754297	1	IVB, QC9C, QS, L1, 1.9, 17W, 2+2, 1.15, 4M, ULVBG	U1000	CRITICAL	CPU:1.9GHZ
33754165	1	IC, PCH, PPT-MB, SFF, ES1	U1000	CRITICAL	PCH_ES1
33754180	1	IC, PCH, PPT-MB, SFF, ES2, B0	U1000	CRITICAL	PCH_ES2
33754235	1	IC, PCH, PPT-MB, SFF, P-QS, C0	U1000	CRITICAL	PCH_C0
33754275	1	IC, PCH, PPT-MB, SFF, C1, QS	U1000	CRITICAL	PCH_C1
33754275	1	IC, PCH, PPT-MB, SFF, C1, QS	U1000	CRITICAL	PCH_C1TDP
33851108	1	IC, TBT, CR-4C, LP, ES3, 288 FCBG6, 12X12MM	U3600	CRITICAL	TBT

333S0622	4	IC, SDRAM, 2GBIT, DDR3L-1600, GEMMA, 78P FPGA	U2900, U2910, U2920, U2930	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0622	4	IC, SDRAM, 2GBIT, DDR3L-1600, GEMMA, 78P FPGA	U3000, U3010, U3020, U3030	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0622	4	IC, SDRAM, 2GBIT, DDR3L-1600, GEMMA, 78P FPGA	U3100, U3110, U3120, U3130	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0622	4	IC, SDRAM, 2GBIT, DDR3L-1600, GEMMA, 78P FPGA	U3200, U3210, U3220, U3230	CRITICAL	DRAM_TYPE:HYNIX_4GB
333S0625	4	IC, SDRAM, 4GBIT, 512MX8, DDR3-1600, 82 FBGA	U2900, U2910, U2920, U2930	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S0625	4	IC, SDRAM, 4GBIT, 512MX8, DDR3-1600, 82 FBGA	U3000, U3010, U3020, U3030	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S0625	4	IC, SDRAM, 4GBIT, 512MX8, DDR3-1600, 82 FBGA	U3100, U3110, U3120, U3130	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S0625	4	IC, SDRAM, 4GBIT, 512MX8, DDR3-1600, 82 FBGA	U3200, U3210, U3220, U3230	CRITICAL	DRAM_TYPE:HYNIX_8GB
333S0623	4	IC, SDRAM, 2GBIT, DDR3-1600, D35, 78P FBGA	U2900, U2910, U2920, U2930	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0623	4	IC, SDRAM, 2GBIT, DDR3-1600, D35, 78P FBGA	U3000, U3010, U3020, U3030	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0623	4	IC, SDRAM, 2GBIT, DDR3-1600, D35, 78P FBGA	U3100, U3110, U3120, U3130	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0623	4	IC, SDRAM, 2GBIT, DDR3-1600, D35, 78P FBGA	U3200, U3210, U3220, U3230	CRITICAL	DRAM_TYPE:SAMSUNG_4GB
333S0642	4	IC, SDRAM, 4GBIT, DDR3-1600, C-DIE, 78P FBGA	U2900, U2910, U2920, U2930	CRITICAL	DRAM_TYPE:SAMSUNG_8GB
333S0642	4	IC, SDRAM, 4GBIT, DDR3-1600, C-DIE, 78P FBGA	U3000, U3010, U3020, U3030	CRITICAL	DRAM_TYPE:SAMSUNG_8GB
333S0642	4	IC, SDRAM, 4GBIT, DDR3-1600, C-DIE, 78P FBGA	U3100, U3110, U3120, U3130	CRITICAL	DRAM_TYPE:SAMSUNG_8GB
333S0642	4	IC, SDRAM, 4GBIT, DDR3-1600, C-DIE, 78P FBGA	U3200, U3210, U3220, U3230	CRITICAL	DRAM_TYPE:SAMSUNG_8GB
333S0628	4	IC, SDRAM, 2GBIT, DDR3L-1600, REV D, 78P FBGA	U2900, U2910, U2920, U2930	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0628	4	IC, SDRAM, 2GBIT, DDR3L-1600, REV D, 78P FBGA	U3000, U3010, U3020, U3030	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0628	4	IC, SDRAM, 2GBIT, DDR3L-1600, REV D, 78P FBGA	U3100, U3110, U3120, U3130	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0628	4	IC, SDRAM, 2GBIT, DDR3L-1600, REV D, 78P FBGA	U3200, U3210, U3220, U3230	CRITICAL	DRAM_TYPE:ELPIDA_4GB
333S0629	4	IC, SDRAM, 4GBIT, DDR3L-1600, REV B, 78P FBGA	U2900, U2910, U2920, U2930	CRITICAL	DRAM_TYPE:ELPIDA_8GB
333S0629	4	IC, SDRAM, 4GBIT, DDR3L-1600, REV B, 78P FBGA	U3000, U3010, U3020, U3030	CRITICAL	DRAM_TYPE:ELPIDA_8GB
333S0629	4	IC, SDRAM, 4GBIT, DDR3L-1600, REV B, 78P FBGA	U3100, U3110, U3120, U3130	CRITICAL	DRAM_TYPE:ELPIDA_8GB
333S0629	4	IC, SDRAM, 4GBIT, DDR3L-1600, REV B, 78P FBGA	U3200, U3210, U3220, U3230	CRITICAL	DRAM_TYPE:ELPIDA_8GB

607-6811	1	ASSEMBLY,SUBASSY,PCBA,HALL EFFECT,K99	J6955	CRITICAL
353S2929	1	IC,ISL6259,BATCHARGER,3%,4C4MM,QFN28	U7000	CRITICAL
946-3116	1	MLB,DYMAX UV EB 0.22 GRAM,K78	GLUE	CRITICAL

DRAM CFG CHART

SIZE	CFG 2
4GB	0
8GB	1

DIE REV	CFG 3
A	0
B	1

806-3706	1	CAN, TOPSIDE, COVER, ALT, 311/313	TBTTOPSIDE_2P_COVER	CRITICAL	
806-3705	1	CAN, TOPSIDE, FENCE, ALT, 311/313	TBTTOPSIDE_2P_FENCE	CRITICAL	
806-3214	1	CAN, TOPSIDE, 311/313	TBTTOPSIDE_1P	CRITICAL	
806-3216	1	CAN, MDP, 311/313	MDPCAN	CRITICAL	
806-3083	1	SHLD, USB, MLB, 311/313	USBCAN	CRITICAL	
806-3142	1	CAN, TBT, 311/313	TBT FENCE	CRITICAL	
806-3215	1	CAN, COVER, TBT, 311/313	TBT COVER	CRITICAL	

A

Misc Voltages & Control Signals

NC EDP TYP<0..3>	TRUE	==	TP EDP TX P<0..3>	
MAKE_BASE=TRUE				
NC EDP TXN<0..3>	TRUE	==	TP EDP TX N<0..3>	
MAKE_BASE=TRUE				
NC EDP AUXP	TRUE	==	TP EDP AUX P	
MAKE_BASE=TRUE				
NC EDP AUXN	TRUE	==	TP EDP AUX N	
MAKE_BASE=TRUE				
NC CPU THERMDA	TRUE	==	TP CPU THERMDA	
MAKE_BASE=TRUE				
NC CPU THERMDC	TRUE	==	TP CPU THERMDC	
MAKE_BASE=TRUE				
NC CPU RSTD<30..45>	TRUE	==	TP CPU RSTD<30..45>	
MAKE_BASE=TRUE				
NC CPU RSTD<8..27>	TRUE	==	TP CPU RSTD<8..27>	
MAKE_BASE=TRUE				
NC PEG R2D CP<15..2>	TRUE	==	=PEG R2D C P<15..2>	9
MAKE_BASE=TRUE				
NC PEG R2D CN<15..2>	TRUE	==	=PEG R2D C N<15..2>	9
MAKE_BASE=TRUE				
NC PEG D2RP<15..2>	TRUE	==	=PEG D2R P<15..2>	9
MAKE_BASE=TRUE				
NC PEG D2RN<15..2>	TRUE	==	=PEG D2R N<15..2>	9
MAKE_BASE=TRUE				

Pin	Signal	Notes
1	TRUE	PP3V3_TBTLIC 7 (Need to add 27 GND TPs)
2	TRUE	PP1V05_TBTLIC 7 35
3	TRUE	PP1V05_S0_PCH_VCCADPLL 7
4	TRUE	PPVCORE_S0_CPU 7
5	TRUE	PPVCORE_S0_AXG 7 16
6	TRUE	PP1V5_S3_CPU_VCCDQ 7 16
7	TRUE	PP1V05_S0_CPU_VCCPOE 7
8	TRUE	PP1V8_S0_CPU_VCCPLL_R 7
9	TRUE	PP1V05_TBTCIO 7
10	TRUE	PPBUS_S5_HS_OTHER_ISNS 7
11	TRUE	PPDCIN_G3H_ISOL 7
12	TRUE	PP5V_S3 7
13	TRUE	PP5V_S0 7
14	TRUE	PP3V3_S4 7

NO_TEST Nets

POWER SIGNALS

HEB3	TRUE	VCCSAS0_SREF	53
HEB9	TRUE	VCCSAS0_SET1_R	53
HEB3	TRUE	VCCSAS0_SET0	53
HEB3	TRUE	VCCSAS0_SET1	53

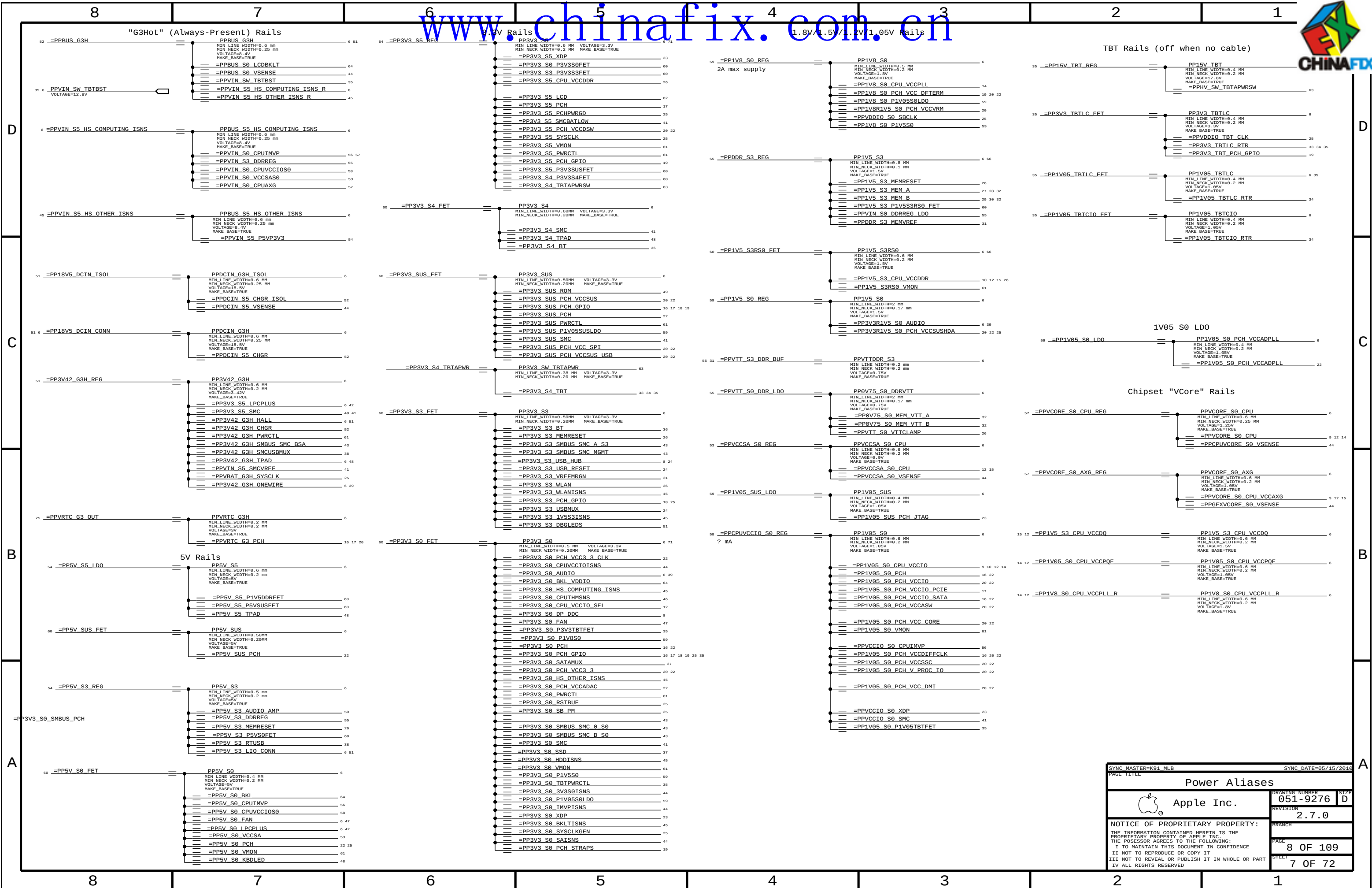
NO_TEST

17	TP_CRT_IG_BLUE	TRUE	NC_CRT_IG_BLUE
		MAKE_BASE=TRUE	
17	TP_CRT_IG_GREEN	TRUE	NC_CRT_IG_GREEN
		MAKE_BASE=TRUE	
17	TP_CRT_IG_RED	TRUE	NC_CRT_IG_RED
		MAKE_BASE=TRUE	
17	TP_CRT_IG_DDC_CLK	TRUE	NC_CRT_IG_DDC_CLK
		MAKE_BASE=TRUE	
17	TP_CRT_IG_DDC_DATA	TRUE	NC_CRT_IG_DDC_DATA
		MAKE_BASE=TRUE	
17	TP_CRT_IG_HSYNC	TRUE	NC_CRT_IG_HSYNC
		MAKE_BASE=TRUE	
17	TP_CRT_IG_VSYNC	TRUE	NC_CRT_IG_VSYNC
		MAKE_BASE=TRUE	
	TP_LVDS_IG_CTRL_CLK	TRUE	NC_LVDS_IG_CTRL_CLK
		MAKE_BASE=TRUE	
	TP_LVDS_IG_CTRL_DATA	TRUE	NC_LVDS_IG_CTRL_DATA
		MAKE_BASE=TRUE	
	TP_PCH_LVDS_VBG	TRUE	NC_PCH_LVDS_VBG
		MAKE_BASE=TRUE	

16	TP_CLKIN_CLK	=====	TRUE	NC_CLKIN_CLK
		=====	MAKE_BASE+TRUE	
16	TP_CLKIN_DATA	=====	TRUE	NC_CLKIN_DATA
		=====	MAKE_BASE+TRUE	
16	TP_CLKIN_RESET_L	=====	TRUE	NC_CLKIN_RESET_L
		=====	MAKE_BASE+TRUE	
16	TP_PCIE_CLK100M_PEBN	=====	TRUE	NC_PCIE_CLK100M_PEBN
		=====	MAKE_BASE+TRUE	
16	TP_PCIE_CLK100M_PEBP	=====	TRUE	NC_PCIE_CLK100M_PEBP
		=====	MAKE_BASE+TRUE	
17	TP_SDVO_TVCLKINN	=====	TRUE	NC_SDVO_TVCLKINN
		=====	MAKE_BASE+TRUE	
17	TP_SDVO_TVCLKINP	=====	TRUE	NC_SDVO_TVCLKINP
		=====	MAKE_BASE+TRUE	
17	TP_SDVO_STALLN	=====	TRUE	NC_SDVO_STALLN
		=====	MAKE_BASE+TRUE	
17	TP_SDVO_STALLP	=====	TRUE	NC_SDVO_STALLP
		=====	MAKE_BASE+TRUE	
17	TP_SDVO_INTN	=====	TRUE	NC_SDVO_INTN
		=====	MAKE_BASE+TRUE	
17	TP_SDVO_INTP	=====	TRUE	NC_SDVO_INTP
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_OBSFN A<0...1>	=====	TRUE	NC_TP_XDP_PCH_OBSFN A<0...1>
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_OBSFN B<0...1>	=====	TRUE	NC_TP_XDP_PCH_OBSFN B<0...1>
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_HOOK2	=====	TRUE	NC_TP_XDP_PCH_HOOK2
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_HOOK3	=====	TRUE	NC_TP_XDP_PCH_HOOK3
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_OBSFN D<0...1>	=====	TRUE	NC_TP_XDP_PCH_OBSFN D<0...1>
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_HOOK4	=====	TRUE	NC_TP_XDP_PCH_HOOK4
		=====	MAKE_BASE+TRUE	
23	TP_XDP_PCH_HOOK5	=====	TRUE	NC_TP_XDP_PCH_HOOK5
		=====	MAKE_BASE+TRUE	
16	TP_PCH_GPI064_CLKOUTFLEX0	=====	TRUE	NC_PCH_GPI064_CLKOUTFLEX0
		=====	MAKE_BASE+TRUE	
16	TP_PCH_GPI065_CLKOUTFLEX1	=====	TRUE	NC_PCH_GPI065_CLKOUTFLEX1
		=====	MAKE_BASE+TRUE	
16	TP_PCH_GPI066_CLKOUTFLEX2	=====	TRUE	NC_PCH_GPI066_CLKOUTFLEX2
		=====	MAKE_BASE+TRUE	
16	TP_PCH_GPI067_CLKOUTFLEX3	=====	TRUE	NC_PCH_GPI067_CLKOUTFLEX3
		=====	MAKE_BASE+TRUE	

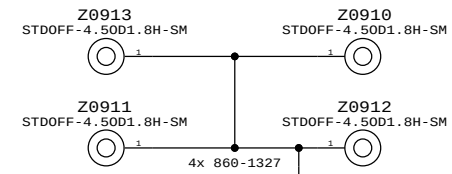
10	TP_PCIE_CLK100M_PE4N	==	TRUE	NC_PCIE_CLK100M_PE4N
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE4P	==	TRUE	NC_PCIE_CLK100M_PE4P
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE5N	==	TRUE	NC_PCIE_CLK100M_PE5N
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE5P	==	TRUE	NC_PCIE_CLK100M_PE5P
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE6N	==	TRUE	NC_PCIE_CLK100M_PE6N
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE6P	==	TRUE	NC_PCIE_CLK100M_PE6P
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE7N	==	TRUE	NC_PCIE_CLK100M_PE7N
		==	MAKE_BASE=TRUE	
10	TP_PCIE_CLK100M_PE7P	==	TRUE	NC_PCIE_CLK100M_PE7P
		==	MAKE_BASE=TRUE	
10	TP_PSDC_P1_3	==	TRUE	NC_PSDC_P1_3
		==	MAKE_BASE=TRUE	
10	TP_SATA_B_D2RN	==	TRUE	NC_SATA_B_D2RN
		==	MAKE_BASE=TRUE	
10	TP_SATA_B_D2RP	==	TRUE	NC_SATA_B_D2RP
		==	MAKE_BASE=TRUE	
10	TP_SATA_B_R2D_CN	==	TRUE	NC_SATA_B_R2D_CN
		==	MAKE_BASE=TRUE	
10	TP_SATA_B_R2D_CP	==	TRUE	NC_SATA_B_R2D_CP
		==	MAKE_BASE=TRUE	
10	TP_SATA_D_D2RN	==	TRUE	NC_SATA_D_D2RN
		==	MAKE_BASE=TRUE	
10	TP_SATA_D_D2RP	==	TRUE	NC_SATA_D_D2RP
		==	MAKE_BASE=TRUE	
10	TP_SATA_D_R2D_CN	==	TRUE	NC_SATA_D_R2D_CN
		==	MAKE_BASE=TRUE	
10	TP_SATA_D_R2D_CP	==	TRUE	NC_SATA_D_R2D_CP
		==	MAKE_BASE=TRUE	
10	TP_SATA_E_D2RN	==	TRUE	NC_SATA_E_D2RN
		==	MAKE_BASE=TRUE	
10	TP_SATA_E_D2RP	==	TRUE	NC_SATA_E_D2RP
		==	MAKE_BASE=TRUE	
10	TP_SATA_E_R2D_CN	==	TRUE	NC_SATA_E_R2D_CN
		==	MAKE_BASE=TRUE	
10	TP_SATA_E_R2D_CP	==	TRUE	NC_SATA_E_R2D_CP
		==	MAKE_BASE=TRUE	
10	TP_SATA_F_D2RN	==	TRUE	NC_SATA_F_D2RN
		==	MAKE_BASE=TRUE	
10	TP_SATA_F_D2RP	==	TRUE	NC_SATA_F_D2RP
		==	MAKE_BASE=TRUE	
10	TP_SATA_F_R2D_CN	==	TRUE	NC_SATA_F_R2D_CN
		==	MAKE_BASE=TRUE	
10	TP_SATA_F_R2D_CP	==	TRUE	NC_SATA_F_R2D_CP
		==	MAKE_BASE=TRUE	
	TP_PCH_TP18	==	TRUE	NC_PCH_TP18
		==	MAKE_BASE=TRUE	
	TP_PCH_TP17	==	TRUE	NC_PCH_TP17
		==	MAKE_BASE=TRUE	
	TP_PCH_TP16	==	TRUE	NC_PCH_TP16
		==	MAKE_BASE=TRUE	
	TP_PCH_TP15	==	TRUE	NC_PCH_TP15
		==	MAKE_BASE=TRUE	
	TP_PCH_TP14	==	TRUE	NC_PCH_TP14
		==	MAKE_BASE=TRUE	
	TP_PCH_TP13	==	TRUE	NC_PCH_TP13
		==	MAKE_BASE=TRUE	
	TP_PCH_TP12	==	TRUE	NC_PCH_TP12
		==	MAKE_BASE=TRUE	
	TP_PCH_TP10	==	TRUE	NC_PCH_TP10
		==	MAKE_BASE=TRUE	
	TP_PCH_TP9	==	TRUE	NC_PCH_TP9
		==	MAKE_BASE=TRUE	
	TP_PCH_TP8	==	TRUE	NC_PCH_TP8
		==	MAKE_BASE=TRUE	
	TP_PCH_TP7	==	TRUE	NC_PCH_TP7
		==	MAKE_BASE=TRUE	
	TP_PCH_TP6	==	TRUE	NC_PCH_TP6
		==	MAKE_BASE=TRUE	
	TP_PCH_TP5	==	TRUE	NC_PCH_TP5
		==	MAKE_BASE=TRUE	
	TP_PCH_TP4	==	TRUE	NC_PCH_TP4
		==	MAKE_BASE=TRUE	
	TP_PCH_TP3	==	TRUE	NC_PCH_TP3
		==	MAKE_BASE=TRUE	
	TP_PCH_TP2	==	TRUE	NC_PCH_TP2
		==	MAKE_BASE=TRUE	
	TP_PCH_TP1	==	TRUE	NC_PCH_TP1
		==	MAKE_BASE=TRUE	
8	TP_LVDS_IG_B_CLKN	==	TRUE	NC_LVDS_IG_B_CLKN
		==	MAKE_BASE=TRUE	
8	TP_LVDS_IG_B_CLKP	==	TRUE	NC_LVDS_IG_B_CLKP
		==	MAKE_BASE=TRUE	
	TP_LVDS_IG_BKL_PwM	==	TRUE	NC_LVDS_IG_BKL_PwM
		==	MAKE_BASE=TRUE	
	SMC_BS_ALRT_L	==	TRUE	NC_SMC_BS_ALRT_L

SYNC MASTER=(K99 MLB)		SYNC DATE=(02/16/2019)	
PAGE TITLE			
Functional Test / No Test			
 Apple Inc.		DRAWING NUMBER	051-9276
		SIZE	D
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		BRANCH	
		PAGE	7 OF 109
		SHEET	6 OF 72

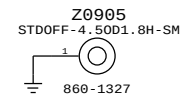




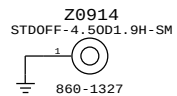
CPU Heat Sink Mounting Bosses



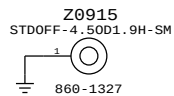
Fan Boss



X21 Boss

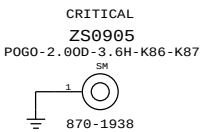


SSD Boss

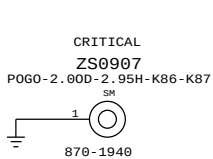


EMI I/O Pogo Pins

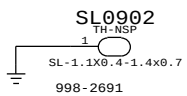
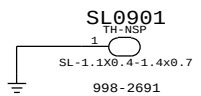
DisplayPort Pogo



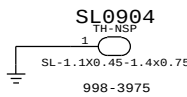
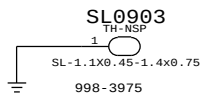
USB/SD Card Pogo



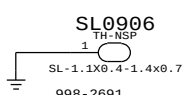
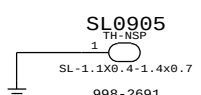
Can Slots



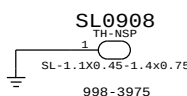
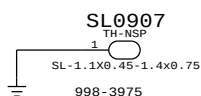
2x MDP connector



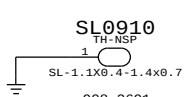
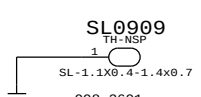
2x USB connector



2x TBT pin diodes

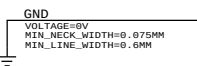


2x TBT chip



2x CPU Vcore

Digital Ground



68 33	DP_TBTSNK0_AUXCH_C_P	==	DPA_IG_AUX_CH_P	17
68 33	DP_TBTSNK0_AUXCH_C_N	==	DPA_IG_AUX_CH_N	17
16	PCIE_CLK100M_ENET_N	==	TRUE	NC_PCIE_CLK100M_ENETN
16	PCIE_CLK100M_ENET_P	==	TRUE	NC_PCIE_CLK100M_ENETP
16	PCIE_EXCARD_D2R_N	==	TRUE	NC_PCIE_EXCARD_D2RN
16	PCIE_EXCARD_D2R_P	==	TRUE	NC_PCIE_EXCARD_D2RP
16	PCIE_EXCARD_R2D_C_N	==	TRUE	NC_PCIE_EXCARD_R2D_CN
16	PCIE_EXCARD_R2D_C_P	==	TRUE	NC_PCIE_EXCARD_R2D_CP
16	PCIE_CLK100M_EXCARD_N	==	TRUE	NC_PCIE_CLK100M_EXCARDN
16	PCIE_CLK100M_EXCARD_P	==	TRUE	NC_PCIE_CLK100M_EXCARDP

68 16	PEG_CLK100M_P	==	MAKE_BASE=TRUE	NC_PEG_CLK100MP
68 16	PEG_CLK100M_N	==	MAKE_BASE=TRUE	NC_PEG_CLK100MN
10	TP_PCH_CLKOUT_DPN	==	MAKE_BASE=TRUE	DPPLL_REF_CLK_N
10	TP_PCH_CLKOUT_DPP	==	MAKE_BASE=TRUE	DPPLL_REF_CLK_P
66 11	MEM_A_CLK_P<1>	==	MAKE_BASE=TRUE	TP_MEM_A_CLKP<1>
66 11	MEM_A_CLK_N<1>	==	MAKE_BASE=TRUE	TP_MEM_A_CLKN<1>
66 11	MEM_B_CLK_P<1>	==	MAKE_BASE=TRUE	TP_MEM_B_CLKP<1>
66 11	MEM_B_CLK_N<1>	==	MAKE_BASE=TRUE	TP_MEM_B_CLKN<1>

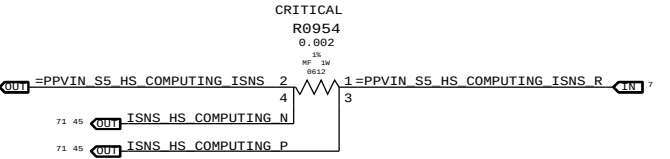
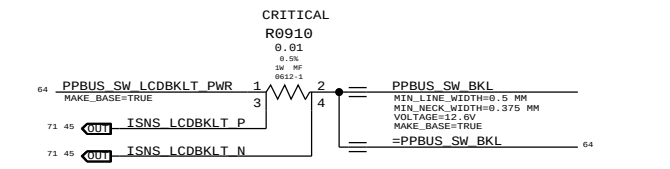
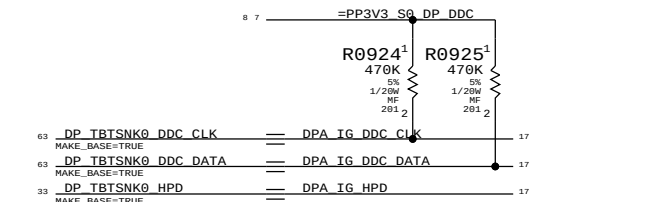
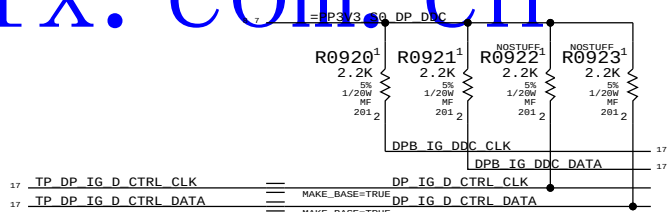
18	USB_EXT_C_P	==	TRUE	NC_USB_EXTCP
18	USB_EXT_C_N	==	TRUE	NC_USB_EXTCN
18	USB3_EXT_C_RX_P	==	TRUE	NC_USB3_EXT_C_RXP
18	USB3_EXT_C_RX_N	==	TRUE	NC_USB3_EXT_C_RXN
18	USB3_EXT_C_TX_P	==	TRUE	NC_USB3_EXT_C_TXP
18	USB3_EXT_C_TX_N	==	TRUE	NC_USB3_EXT_C_TXN
18	USB3_EXT_D_RX_P	==	TRUE	NC_USB3_EXT_D_RXP
18	USB3_EXT_D_RX_N	==	TRUE	NC_USB3_EXT_D_RXN
18	USB3_EXT_D_TX_P	==	TRUE	NC_USB3_EXT_D_TXP
18	USB3_EXT_D_TX_N	==	TRUE	NC_USB3_EXT_D_TXN
18	USB_EXTD_EHCI_P	==	TRUE	NC_USB_EXTD_EHCIP
18	USB_EXTD_EHCI_N	==	TRUE	NC_USB_EXTD_EHCIN

33	TBT_B_R2D_C_P<0>	==	TRUE	NC_TBT_B_R2D_CP<0>
33	TBT_B_R2D_C_N<0>	==	TRUE	NC_TBT_B_R2D_CN<0>
33	TBT_B_R2D_C_P<1>	==	TRUE	NC_TBT_B_R2D_CP<1>
33	TBT_B_R2D_C_N<1>	==	TRUE	NC_TBT_B_R2D_CN<1>
33	TBT_B_D2R_P<0>	==	TRUE	NC_TBT_B_D2RP<0>
33	TBT_B_D2R_N<0>	==	TRUE	NC_TBT_B_D2RN<0>
33	TBT_B_D2R_P<1>	==	TRUE	NC_TBT_B_D2RP<1>
33	TBT_B_D2R_N<1>	==	TRUE	NC_TBT_B_D2RN<1>
33	TBT_B_LSTX	==	TRUE	NC_TBT_B_LSTX

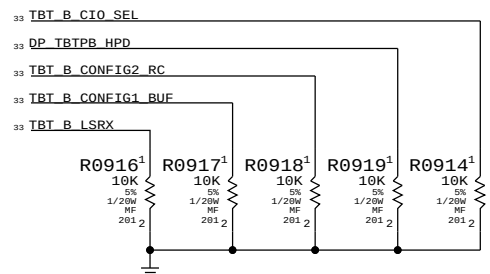
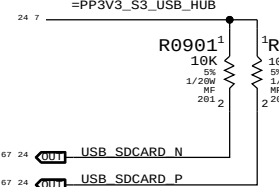
68 33	DP_TBTPB_ML_C_P<1>	==	MAKE_BASE=TRUE	NC_DP_TBTPB_ML_CP<1>
68 33	DP_TBTPB_ML_C_N<1>	==	MAKE_BASE=TRUE	NC_DP_TBTPB_ML_CN<1>
68 33	DP_TBTPB_ML_C_P<3>	==	MAKE_BASE=TRUE	NC_DP_TBTPB_ML_CP<3>
68 33	DP_TBTPB_ML_C_N<3>	==	MAKE_BASE=TRUE	NC_DP_TBTPB_ML_CN<3>

16	PCIE_CLK100M_FW_N	==	MAKE_BASE=TRUE	NC_PCIE_CLK100M_FWN
16	PCIE_CLK100M_FW_P	==	MAKE_BASE=TRUE	NC_PCIE_CLK100M_FWP
9	=PEG_D2R_P<1..0>	==	MAKE_BASE=TRUE	PCIE_SSD_D2R_P<1..0>
9	=PEG_D2R_N<1..0>	==	MAKE_BASE=TRUE	PCIE_SSD_D2R_N<1..0>
9	=PEG_R2D_C_P<1..0>	==	MAKE_BASE=TRUE	PCIE_SSD_R2D_C_P<1..0>
9	=PEG_R2D_C_N<1..0>	==	MAKE_BASE=TRUE	PCIE_SSD_R2D_C_N<1..0>

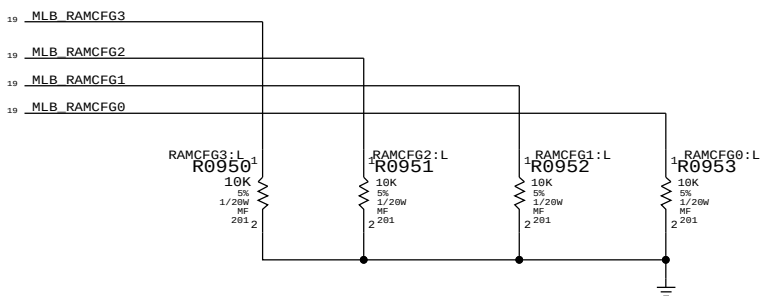
16	PCIE_ENET_D2R_N	==	TRUE	NC_PCIE_ENET_D2RN
16	PCIE_ENET_D2R_P	==	TRUE	NC_PCIE_ENET_D2RP
16	PCIE_ENET_R2D_C_N	==	TRUE	NC_PCIE_ENET_R2D_CN
16	PCIE_ENET_R2D_C_P	==	TRUE	NC_PCIE_ENET_R2D_CP
16	PCIE_FW_D2R_N	==	TRUE	NC_PCIE_FW_D2RN
16	PCIE_FW_D2R_P	==	TRUE	NC_PCIE_FW_D2RP
16	PCIE_FW_R2D_C_N	==	TRUE	NC_PCIE_FW_R2D_CN
16	PCIE_FW_R2D_C_P	==	TRUE	NC_PCIE_FW_R2D_CP



UNUSED SDCARD USB Aliases



23 19	XDP_DD3_PCH_GPI049_ENET_LOW_PWR_PCH	==	ENET_LOW_PWR_PCH	19
23 16	XDP_DC3_PCH_GPI019_SATARDRV_EN	==	SATARDRV_EN	16



TBT DP Ports

17	_DPB_IG_HP	==	DP_TBTSNK1_HP	33 68
17	TP_DP_IG_C_MLP<3..0>	==	MAKE_BASE=TRUE	DP_TBTSNK1_ML_C_P<3..0>
17	TP_DP_IG_C_MLN<3..0>	==	MAKE_BASE=TRUE	DP_TBTSNK1_ML_C_N<3..0>
17	_DPB_IG_AUX_CH_P	==	MAKE_BASE=TRUE	DP_TBTSNK1_AUXCH_C_P
17	_DPB_IG_AUX_CH_N	==	MAKE_BASE=TRUE	DP_TBTSNK1_AUXCH_C_N
17	TP_DP_IG_D_HP	==	MAKE_BASE=TRUE	DP_IG_D_HP
68 33	DP_TBTPB_AUXCH_C_P	==	MAKE_BASE=TRUE	NC_DP_TBTPB_AUXCH_CP
68 33	DP_TBTPB_AUXCH_C_N	==	MAKE_BASE=TRUE	NC_DP_TBTPB_AUXCH_CN
17	TP_DP_IG_B_MLP<3..0>	==	MAKE_BASE=TRUE	DP_TBTSNK0_ML_C_P<3..0>
17	TP_DP_IG_B_MLN<3..0>	==	MAKE_BASE=TRUE	DP_TBTSNK0_ML_C_N<3..0>

16	NC_PCIE_5_R2D_CP	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_P<0>	33 68
16	NC_PCIE_6_R2D_CP	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_P<1>	33 68
16	NC_PCIE_7_R2D_CP	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_P<2>	33 68
16	NC_PCIE_8_R2D_CP	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_P<3>	33 68
16	NC_PCIE_5_R2D_CN	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_N<0>	33 68
16	NC_PCIE_6_R2D_CN	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_N<1>	33 68
16	NC_PCIE_7_R2D_CN	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_N<2>	33 68
16	NC_PCIE_8_R2D_CN	==	MAKE_BASE=TRUE	PCIE_TBT_R2D_C_N<3>	33 68
16	NC_PCIE_5_D2RP	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_P<0>	33 68
16	NC_PCIE_6_D2RP	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_P<1>	33 68
16	NC_PCIE_7_D2RP	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_P<2>	33 68
16	NC_PCIE_8_D2RP	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_P<3>	33 68
16	NC_PCIE_5_D2RN	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_N<0>	33 68
16	NC_PCIE_6_D2RN	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_N<1>	33 68
16	NC_PCIE_7_D2RN	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_N<2>	33 68
16	NC_PCIE_8_D2RN	==	MAKE_BASE=TRUE	PCIE_TBT_D2R_N<3>	33 68

LVDS Aliases

6	TP_LVDS_IG_B_CLKP	==	LVDS_IG_B_CLK_P	
6	TP_LVDS_IG_B_CLKN	==	LVDS_IG_B_CLK_N	
6	NC_LVDS_IG_B_DATAP<0..3>	==	LVDS_IG_B_DATA_P<0..3>	
6	NC_LVDS_IG_B_DATAN<0..3>	==	LVDS_IG_B_DATA_N<0..3>	
6	NC_LVDS_IG_A_DATAP<3>	==	LVDS_IG_A_DATA_P<3>	
6	NC_LVDS_IG_A_DATAN<3>	==	LVDS_IG_A_DATA_N<3>	
64	LCD_BKLT_PWM	==	LVDS_IG_BKLT_PWM	17
64	LCD_IG_PWR_EN	==	LVDS_IG_PANEL_PWR	17
64	LCD_BKLT_EN	==	LVDS_IG_BKLT_ON	17

SATA Aliases

Unused SATA ODD Signals

16	SATA_ODD_R2D_C_P	==	NC_SATA_ODD_R2DCP	
16	SATA_ODD_R2D_C_N	==	NC_SATA_ODD_R2DCN	
16	SATA_ODD_D2R_P	==	NC_SATA_ODD_D2RP	
16	SATA_ODD_D2R_N	==	NC_SATA_ODD_D2RN	

SMC Aliases

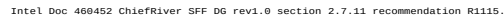
Unused SMC Signals

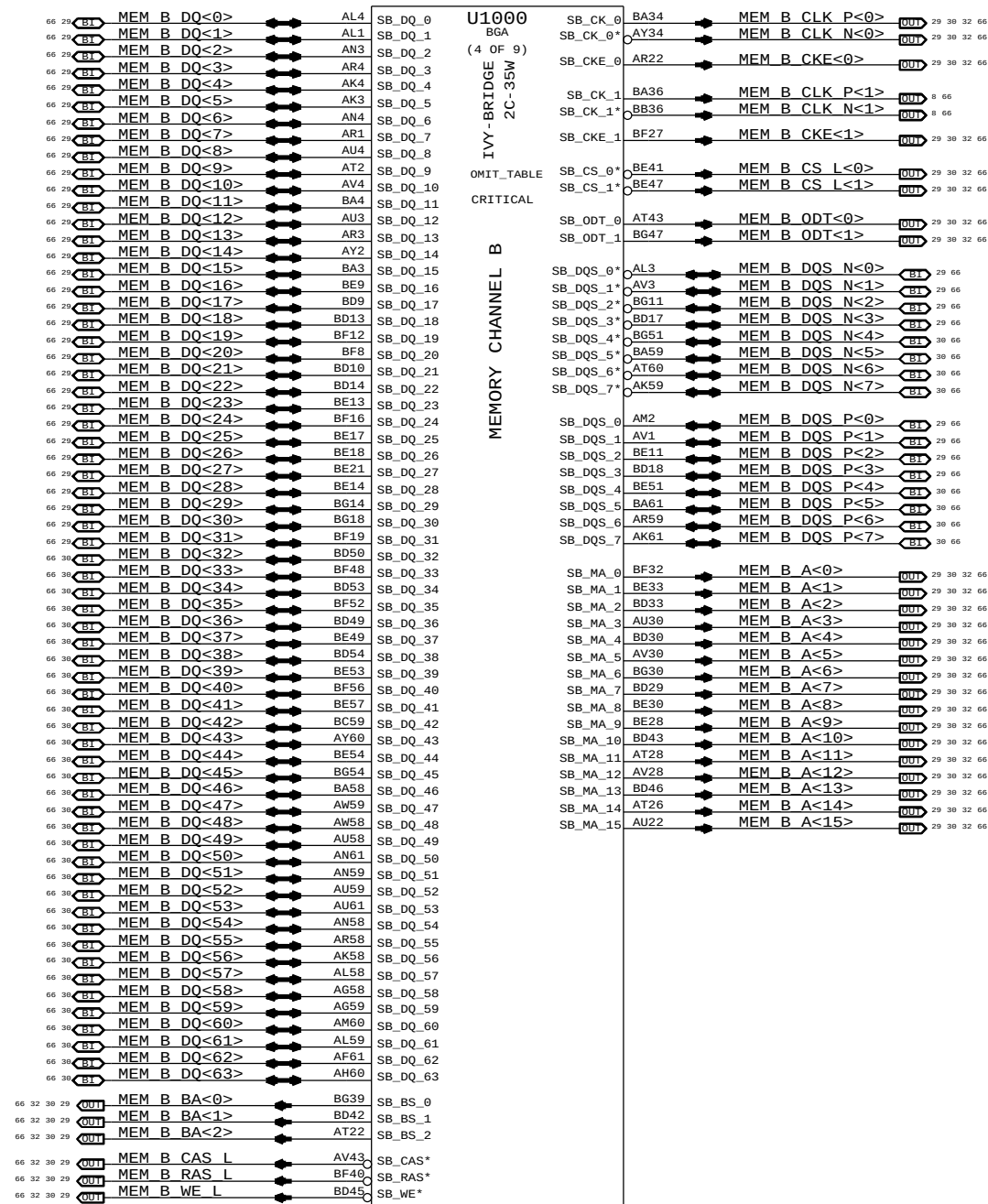
48	SMC_SYS_LED	==	NC_SMC_SYS_LED	
48	IR_RX_OUT_RC	==	NC_IR_RX_OUT_RC	

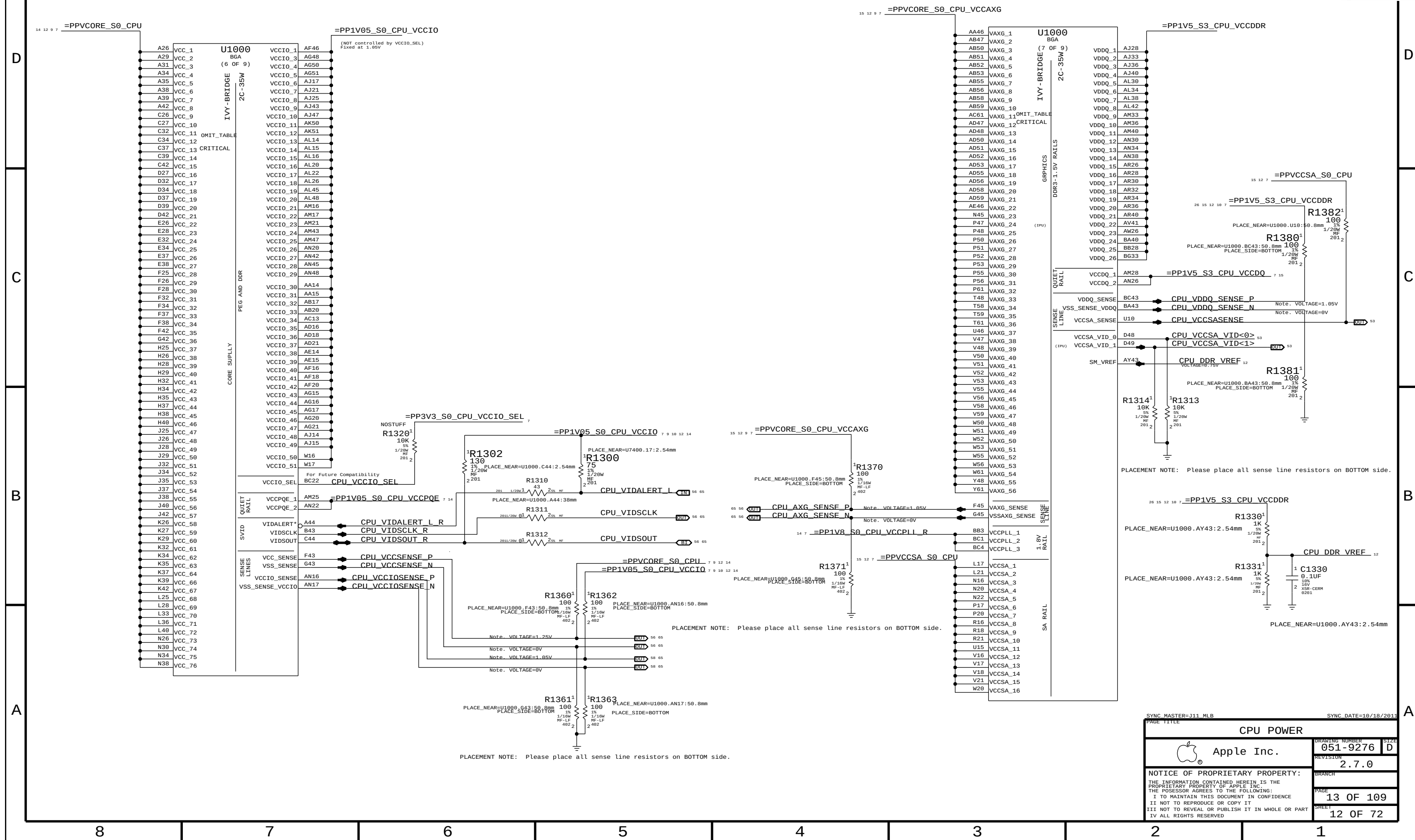
Unused PG00D signal

TP_P1V5S3RS0_RAMP_DONE	==	P1V5S3RS0_RAMP_DONE	INT 59
TP_BDRREG_PG00D	==	DDRREG_PG00D	INT 55

SYNC MASTER=K91_MLB		SYNC DATE=85/15/2016	
PAGE TITLE			
Signal Aliases		DRAWING NUMBER	SIZE
Apple Inc.		051-9276	D
REVISION		2.7.0	
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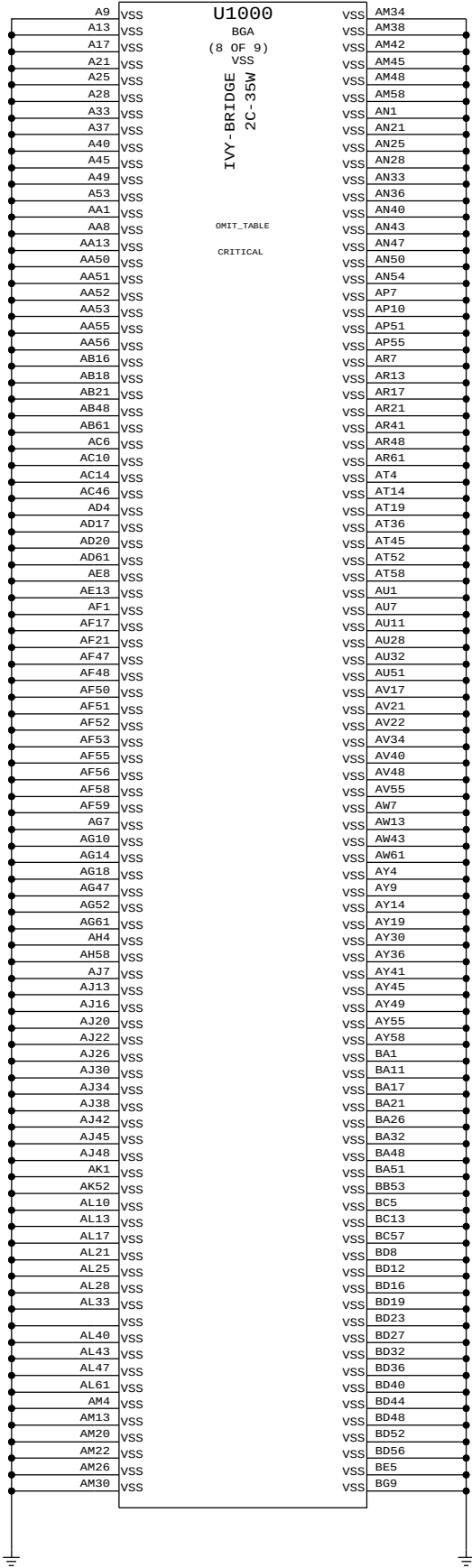
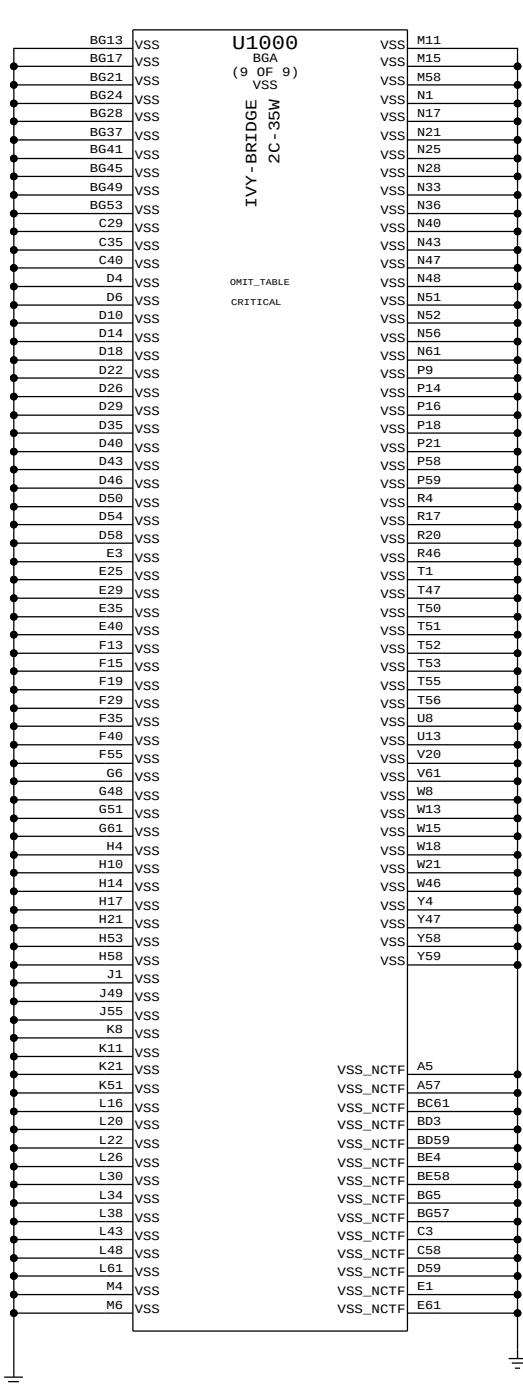
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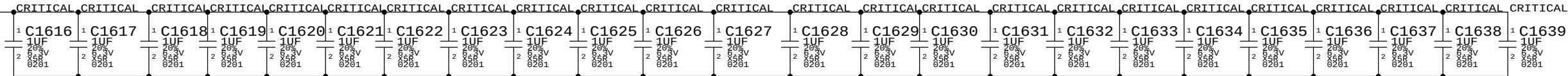
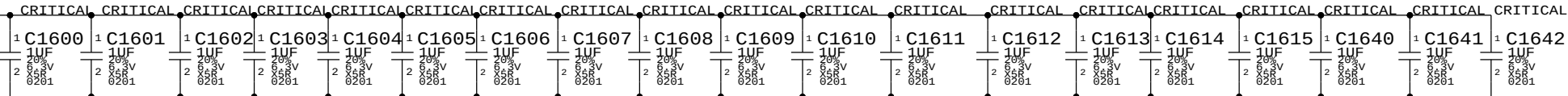


Processor Load Line : -2.9 mOhms

CPU VCORE DECOUPLING

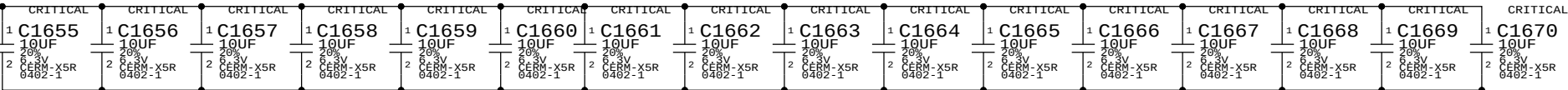
Intel recommendation (Table 7-1): 16x 2.2uF, 12x 22uF, 3x 330uF

12 9 7 =PPVCORE_S0_CPU



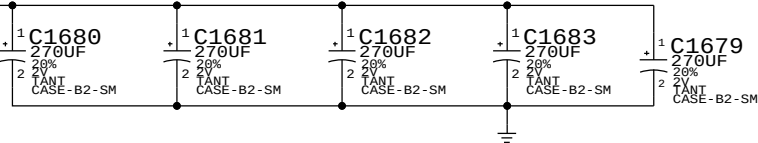
PLACEMENT_NOTE (C1655-C1666):

Place close to U1000 on top side.



PLACEMENT_NOTE (C1667-C1679):

PLACEMENT_NOTE (C1640-C1645):

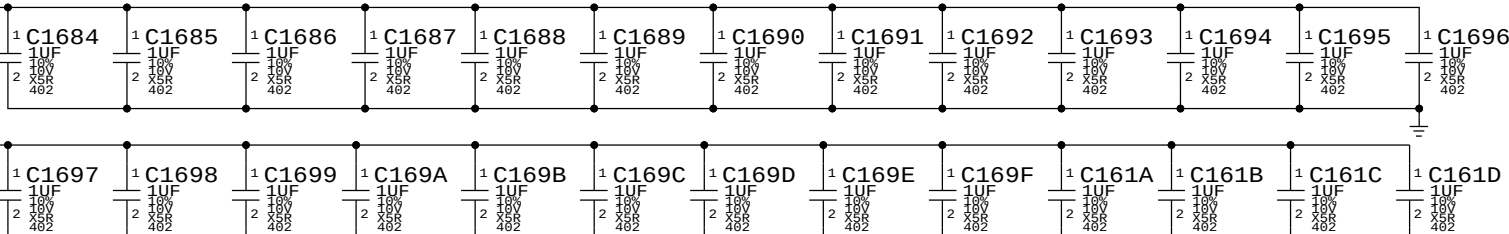


CPU VCCIO/VCCPQ DECOUPLING

Intel recommendation (Section 6.5): 26x 1uF, 19x 10uF, 2x 330uF

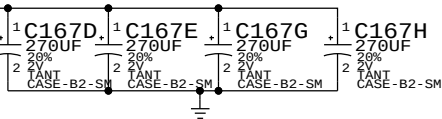
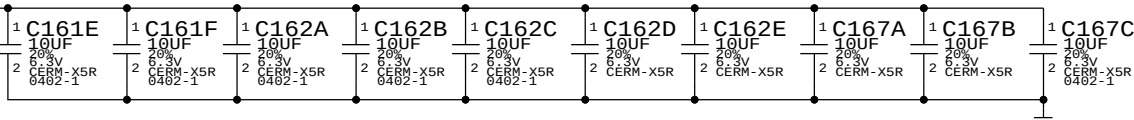
PLACEMENT_NOTE (C1684-C1697):

Place on bottom side of U1000

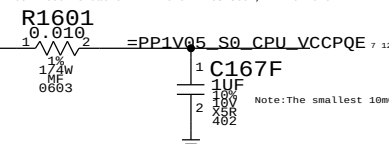


PLACEMENT_NOTE (C1672-C1681):

Place near U1000 on bottom side



Intel recommendation: 1x 10mOhm resistor, 1x 1uF 0402



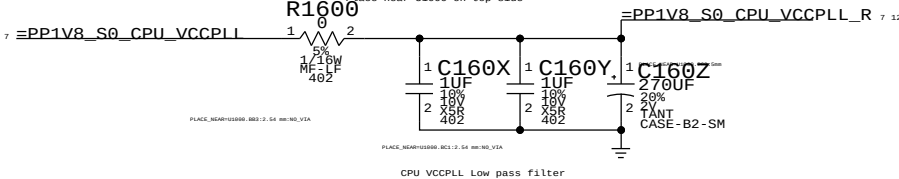
Note: The smallest 10mOhm available in the library are 0805s

CPU VCCPLL DECOUPLING

Intel recommendation (section 6.4): 2x 1uF, 1x 330uF

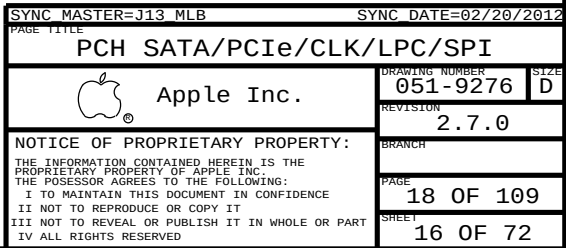
PLACEMENT_NOTE (C1646-C1671):

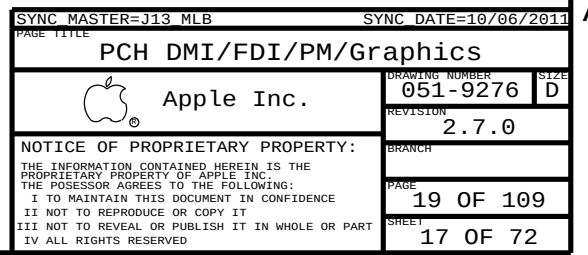
Place near U1000 on top side



CPU VCCPLL Low pass filter

CPU DECOUPLING-I		
Apple Inc.	DRAWING NUMBER	051-9276
	REVISION	2.7.0
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	PAGE	16 OF 109
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OMIT TABLE

U1800
PCH-PPT-MB-SFF-ES1
BGA
QP8D-MM915462
(5 OF 10)

NC X	BH24	TP1
NC X	BK24	TP2
NC X	BH20	TP3
NC X	BK16	TP4
NC X	BH16	TP5
NC X	AN42	TP6
NC X	AN40	TP7
NC X	AR40	TP8
NC X	AR42	TP9
NC X	D20	TP10
NC X	M30	TP11
NC X	E3	TP12
NC X	AM4	TP13
NC X	AT4	TP14
NC X	AT2	TP15
NC X	AD10	TP16
NC X	B24	TP17
NC X	D24	TP18
NC X	AD44	TP19
NC X	AD46	TP20
NC X	BJ48	TP21
NC X	BL7	TP22
NC X	W40	TP23
NC X	K30	TP24

RSVD

AU6	X NC
AU8	X NC
AW1	X NC
AW3	X NC
AY2	X NC
AY4	X NC
AY6	X NC
AY8	X NC
BA1	X NC
BA3	X NC
BB6	X NC
BC1	X NC
BC3	X NC
BD2	X NC
BD4	X NC
BE1	X NC
BE3	X NC
BE6	X NC
BF6	X NC
BF7	X NC
BG1	X NC
BG3	X NC
BH3	X NC
BH4	X NC
BJ4	X NC
BJ5	X NC
BJ7	X NC
BK6	X NC
BL5	X NC

67 38	IN	USB3_EXTB_RX_N	BJ25	USB3RN1
67 38	IN	USB3_EXTB_RX_N	BJ27	USB3RN2
67 38	IN	USB3_EXTB_RX_N	BJ31	USB3RN3
67 38	IN	USB3_EXTB_RX_N	BJ29	USB3RN4
67 38	IN	USB3_EXTB_RX_P	BL25	USB3RP1
67 38	IN	USB3_EXTB_RX_P	BL27	USB3RP2
67 38	IN	USB3_EXTB_RX_P	BL31	USB3RP3
67 38	IN	USB3_EXTB_RX_P	BL29	USB3RP4
67 38	OUT	USB3_EXTB_TX_N	BF26	USB3TN1
67 38	OUT	USB3_EXTB_TX_N	BF28	USB3TN2
67 38	OUT	USB3_EXTB_TX_N	BF28	USB3TN3
67 38	OUT	USB3_EXTB_TX_N	BF30	USB3TN4
67 38	OUT	USB3_EXTB_TX_P	BD26	USB3TP1
67 38	OUT	USB3_EXTB_TX_P	AY28	USB3TP2
67 38	OUT	USB3_EXTB_TX_P	BD28	USB3TP3
67 38	OUT	USB3_EXTB_TX_P	BD30	USB3TP4

USB

USBP0N	F24	USB_EXTB_N	BT	38 67	Ext A (XHCI/EHCI)
USBP0P	H24	USB_EXTB_P	BT	38 67	
USBP1N	C25	USB_EXTB_XHCI_N	BT	24 67	Ext B (XHCI)
USBP1P	A25	USB_EXTB_XHCI_P	BT	24 67	
USBP2N	C27	USB_EXTB_N	BT	8	Ext C (XHCI/EHCI)
USBP2P	A27	USB_EXTB_P	BT	8	
USBP3N	H28	USB_EXTD_XHCI_N	BT	24 67	Ext D (XHCI) (Mobiles: Trackpad?)
USBP3P	F28	USB_EXTD_XHCI_P	BT	24 67	
USBP4N	M26	TP_USB_4N			Unused
USBP4P	K26	TP_USB_4P			
		TP_USB_SDN			
USBP5N	D28	TP_USB_SDP			RSVD: SD
USBP5P	B28	TP_USB_SDP			
USBP6N	H26	TP_USB_WLANN			RSVD: WiFi
USBP6P	F26	TP_USB_WLANP			
USBP7N	D32	USB_HUB_UP_N	BT	24 67	USB Hub (All LS/FS Devices)
USBP7P	B32	USB_HUB_UP_P	BT	24 67	
USBP8N	M28	USB_CAMERA_N	BT	6 39 67	Camera
USBP8P	K28	USB_CAMERA_P	BT	6 39 67	
USBP9N	C29	USB_EXTB_EHCI_N	BT	24 67	Ext B (EHCI)
USBP9P	A29	USB_EXTB_EHCI_P	BT	24 67	
USBP10N	C31	USB_EXTD_EHCI_N	BT	8	Ext D (EHCI)
USBP10P	A31	USB_EXTD_EHCI_P	BT	8	
USBP11N	H33	TP_USB_BT_HSN			RSVD: BT (HS)
USBP11P	F33	TP_USB_BT_HSP			
USBP12N	H30	TP_USB_12N			Unused
USBP12P	F30	TP_USB_12P			
USBP13N	M33	TP_USB_13N			Unused
USBP13P	K33	TP_USB_13P			

USBRBIAS*
USBRBIAS

OC0*/GPIO59	C17	XDP_DA0_PCH_GPIO59_USB_EXTB_OC_L	BT	18 23
OC1*/GPIO40	A17	XDP_DA1_PCH_GPIO40_USB_EXTB_OC_L	BT	18 23
OC2*/GPIO41	A13	XDP_DA2_PCH_GPIO41_USB_EXTB_OC_L	BT	18 23
OC3*/GPIO42	D16	XDP_DA3_PCH_GPIO42_USB_EXTB_OC_L	BT	18 23
OC4*/GPIO43	A11	XDP_DB0_PCH_GPIO43_USB_EXTB_OC_EHCI_L	BT	23
OC5*/GPIO109	B16	XDP_DB1_PCH_GPIO109_USB_EXTD_OC_EHCI_L	BT	23
OC6*/GPIO1010	C23	XDP_DB2_PCH_GPIO1010_AP_PWR_EN	BT	23
OC7*/GPIO1014	H15	XDP_DB3_PCH_GPIO1014_SDCONN_STATE_CHANGE	BT	18 23

PLACE_NEAR=U1800.A33:2.54mm
R2070
22.6
1%
1/20W
MF
201

35 25 19 18 17 16 7	=PP3V3_S0_PCH_GPIO	R2010	10K	1	2	5%	1/20W	MF	201	PCI_INTA_L	D49	PIRQA*
		R2011	10K	1	2	5%	1/20W	MF	201	PCI_INTB_L	C48	PIRQB*
		R2012	10K	1	2	5%	1/20W	MF	201	PCI_INTC_L	C47	PIRQC*
		R2013	10K	1	2	5%	1/20W	MF	201	PCI_INTD_L	C45	PIRQD*
									</			

SYNC_MASTER=J13 MLB		SYNC_DATE=09/22/2011	
PAGE TITLE			
PCH PCI/USB/TP/RSVD			
Apple Inc.		DRAWING NUMBER	051-9276
		REVISION	2.7.0
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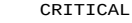
chip-down memory should pull all 4 RAMCFG GPIOs high.
memory should add pull-downs on another page and set straps per software.

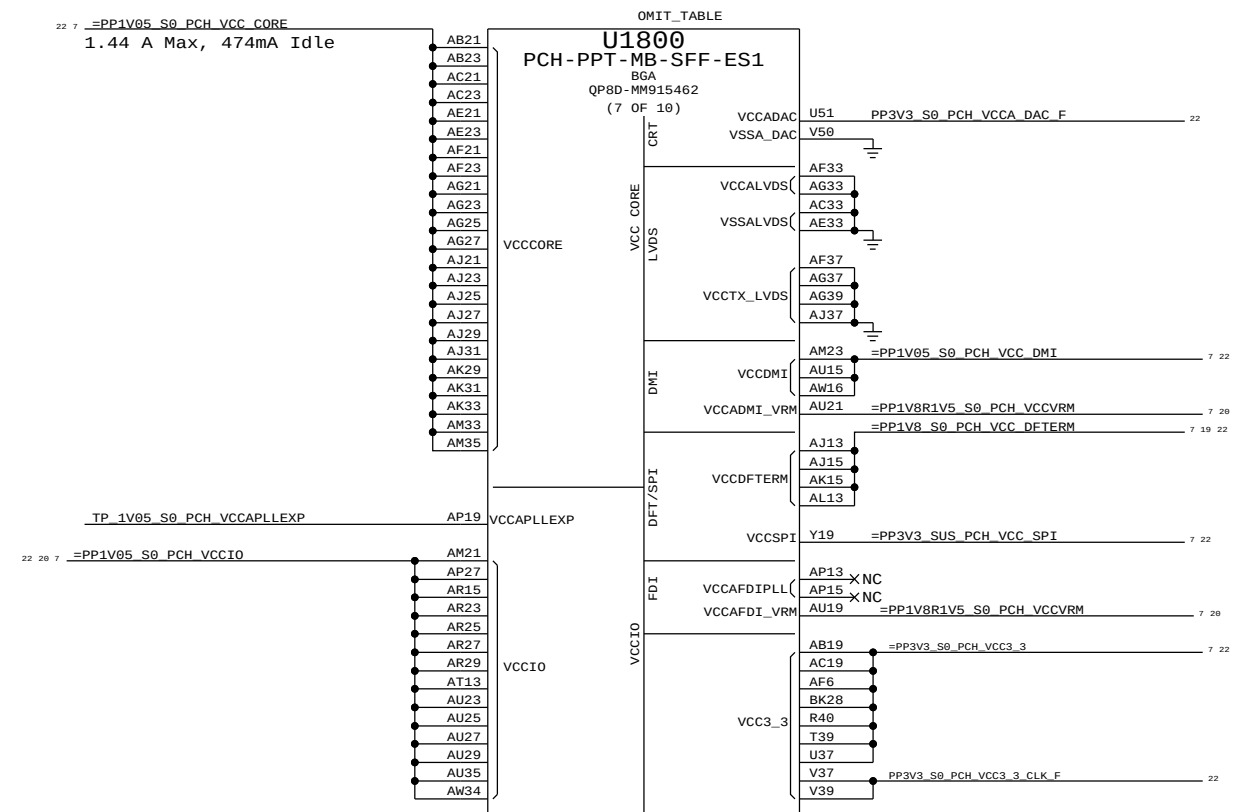
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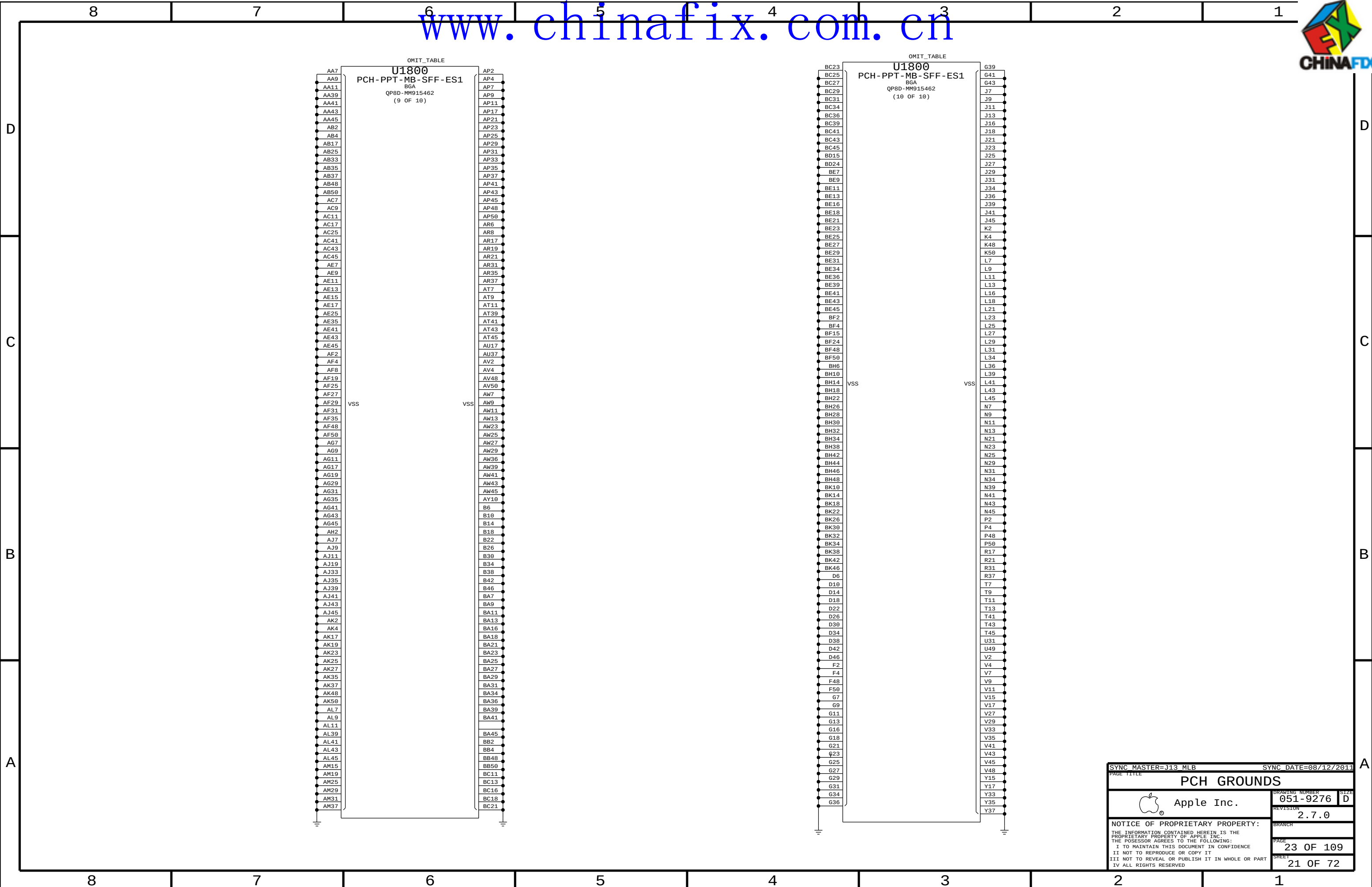
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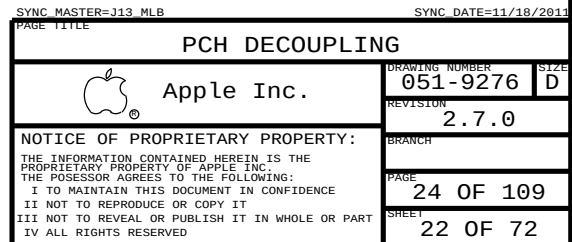
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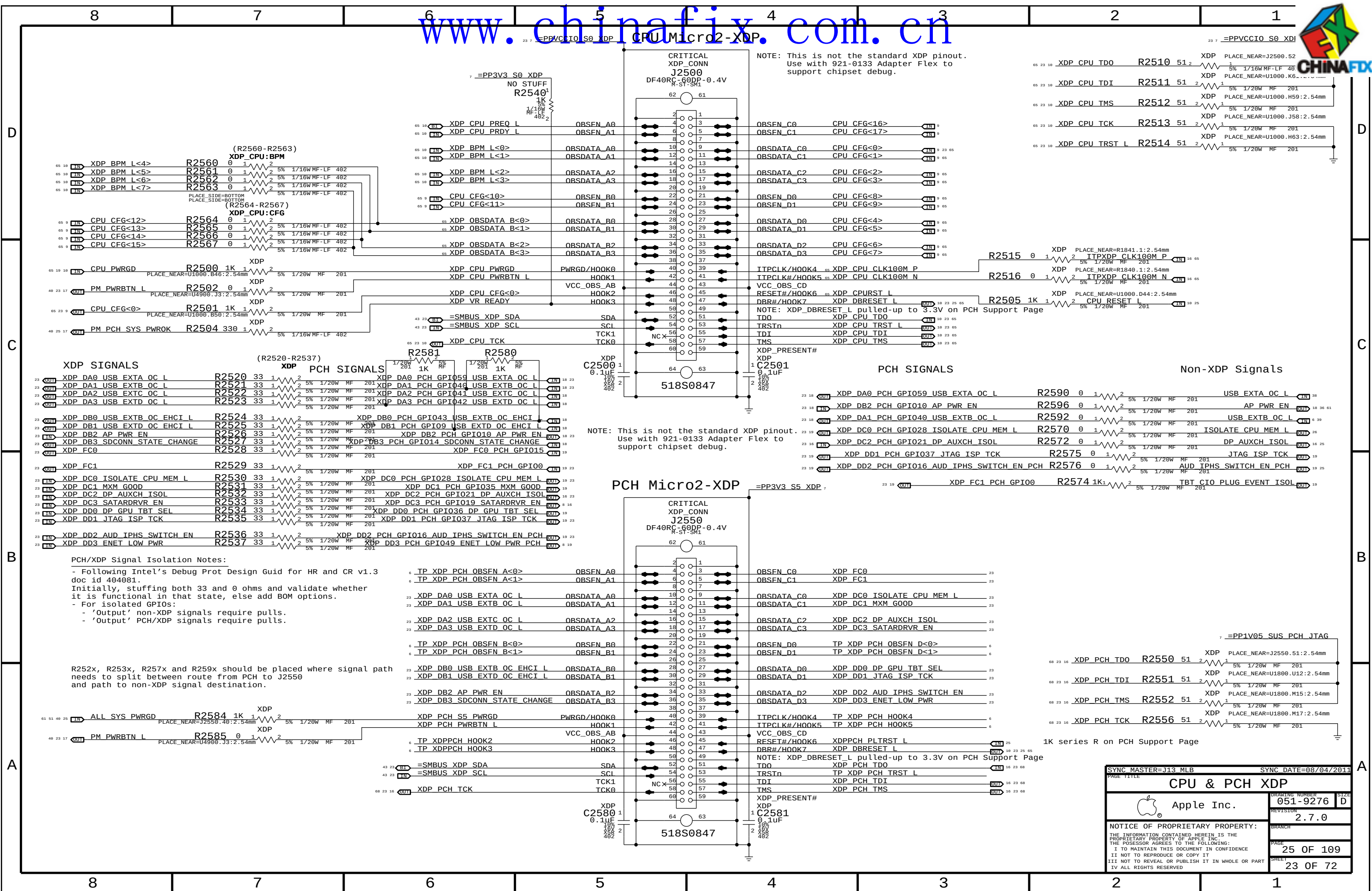
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SYNC MASTER=J13 MLB		SYNC DATE=08/12/2011	
PAGE TITLE		PCH GROUNDS	
		DRAWING NUMBER	051-9276
		REVISION	2.7.0
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		SHEET	21 OF 72



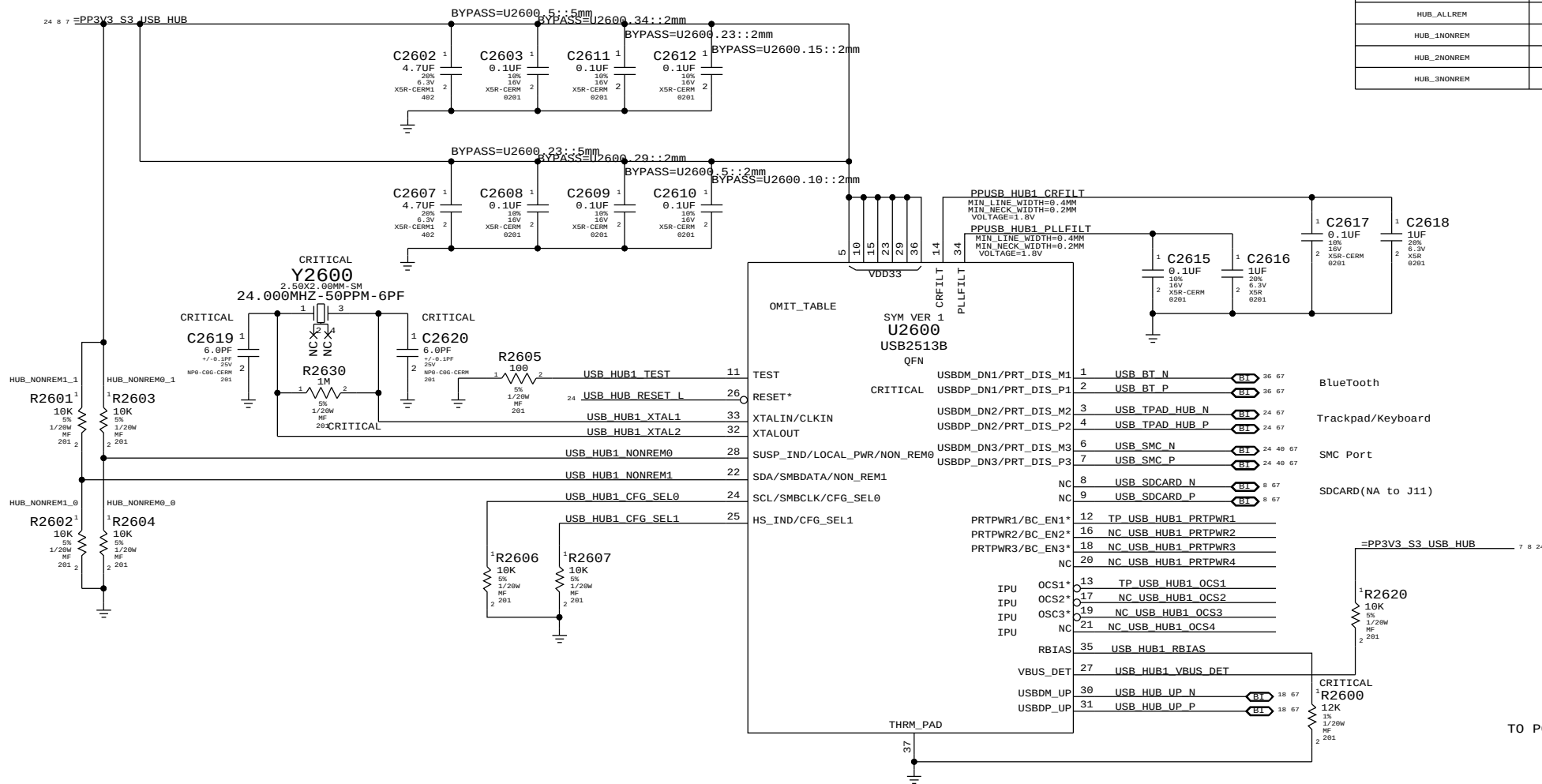


BOM GROUP		BOM OPTIONS	
HUB_ALLREM		HUB_NONREM1_0, HUB_NONREM0_0	
HUB_1NONREM		HUB_NONREM1_0, HUB_NONREM0_1	
HUB_2NONREM		HUB_NONREM1_1, HUB_NONREM0_0	
HUB_3NONREM		HUB_NONREM1_1, HUB_NONREM0_1	

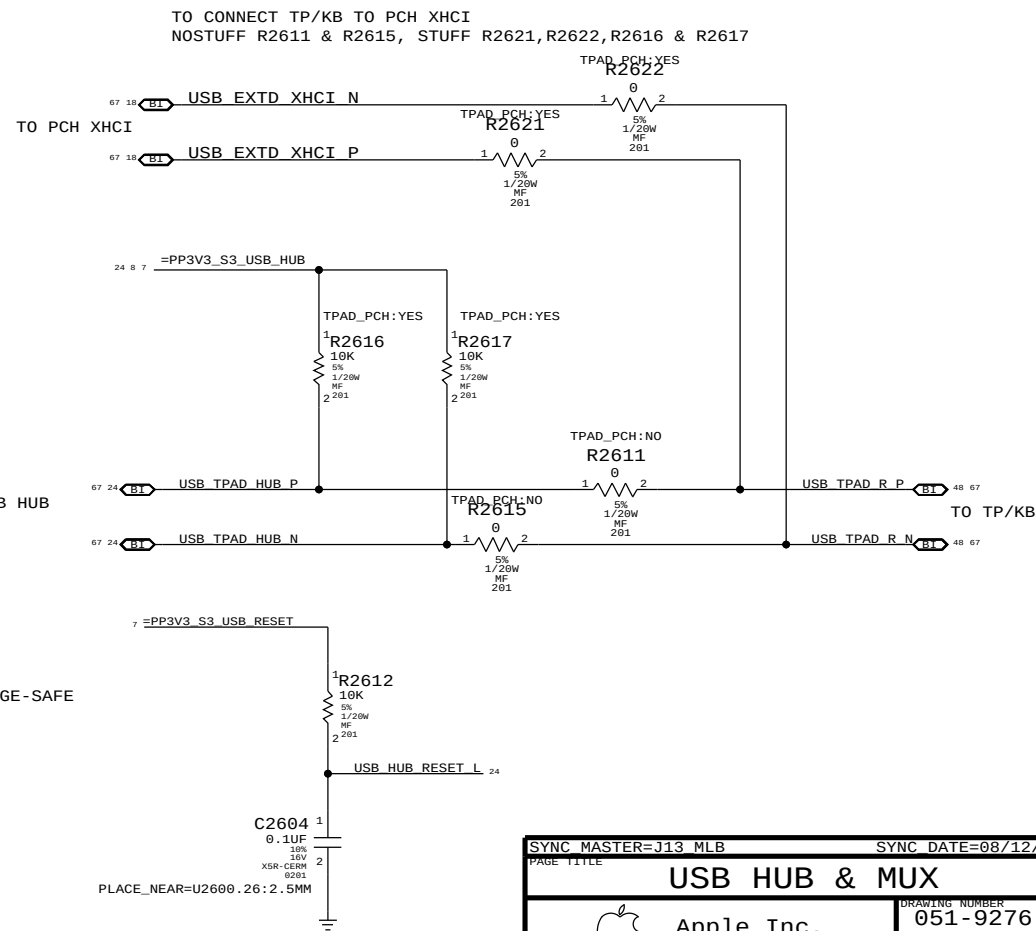
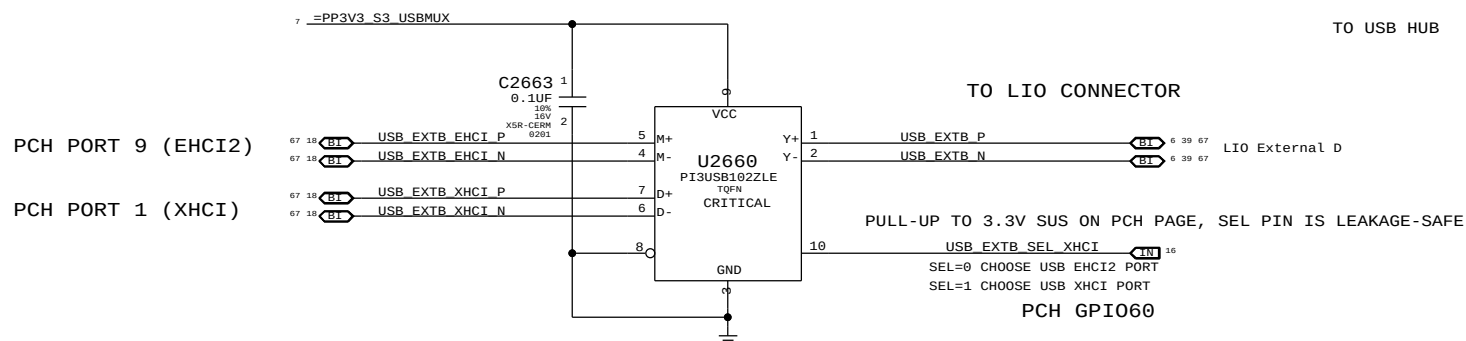
NON_REM1	NON_REM0	DESCRIPTION
0	0	All ports are removable
0	1	Port 1 is non removable
1	0	Port 1 and 2 are non removable
1	1	Port 1, 2, and 3 are non removable

BOM TABLE

PART#	QTY	DESCRIPTION	REFERENCE DESIGNATOR(S)	CRITICAL	BOM OPTION
33859983	1	IC, USB2512B, USB 2.0 HUB CNTRL, 36-QFN	U2600	CRITICAL	USBHUB2512B
33859923	1	IC, USB2513B, USB 2.0 HUB CNTRL, 36-QFN	U2600	CRITICAL	USBHUB2513B



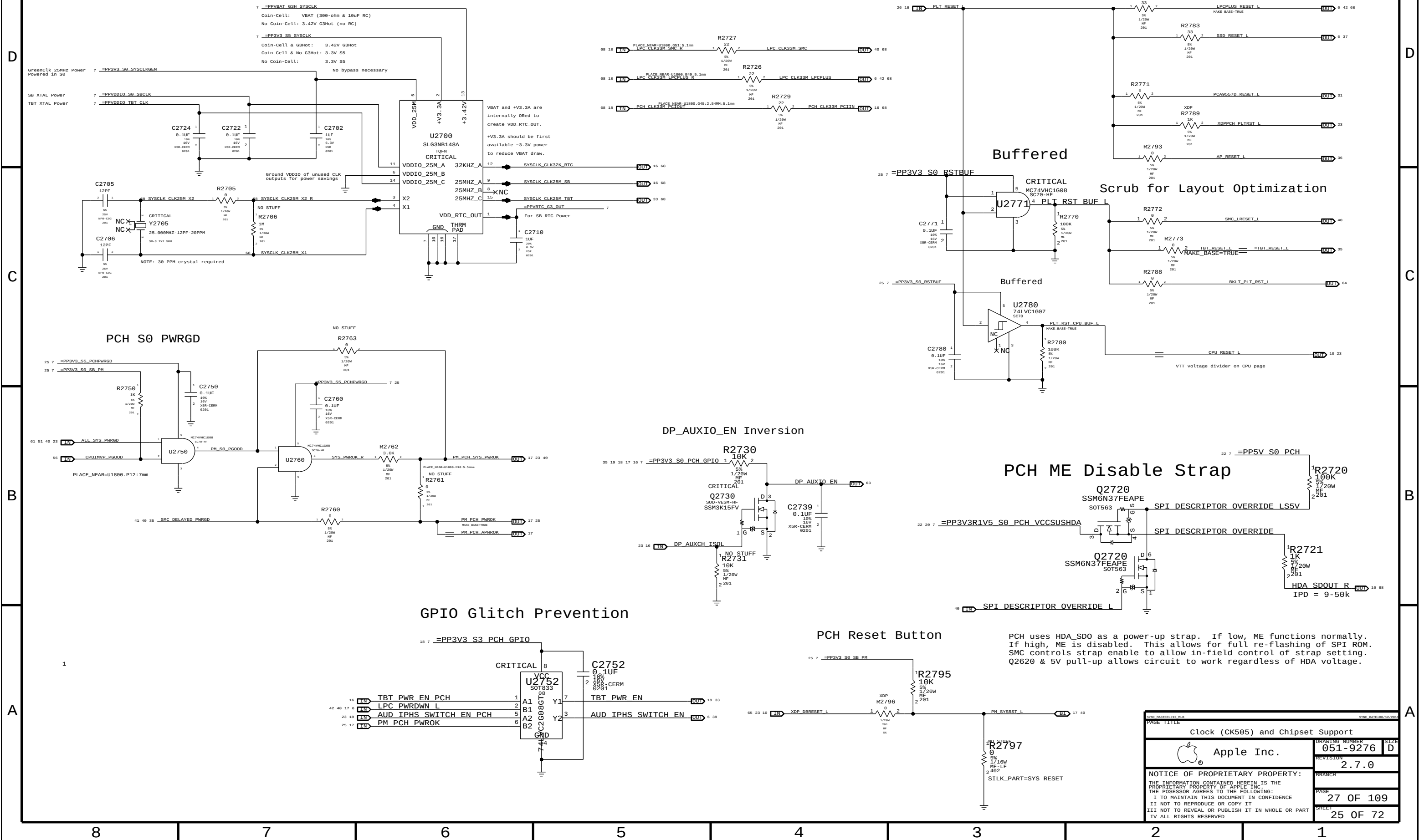
USB XHCI/EHCI2 PORT MUX FOR EXT B



SYNC MASTER=J13 MLB		SYNC DATE=08/12/2011	
PAGE TITLE		USB HUB & MUX	
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System RTC Power Source & 32kHz / 25MHz Clock Generator

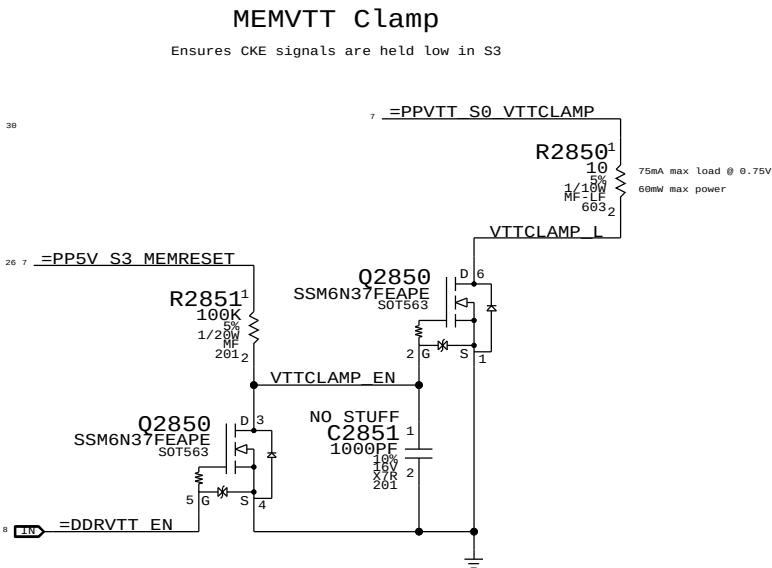
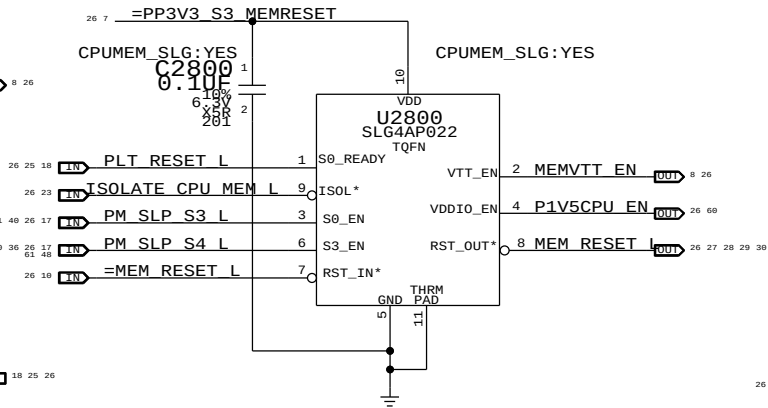
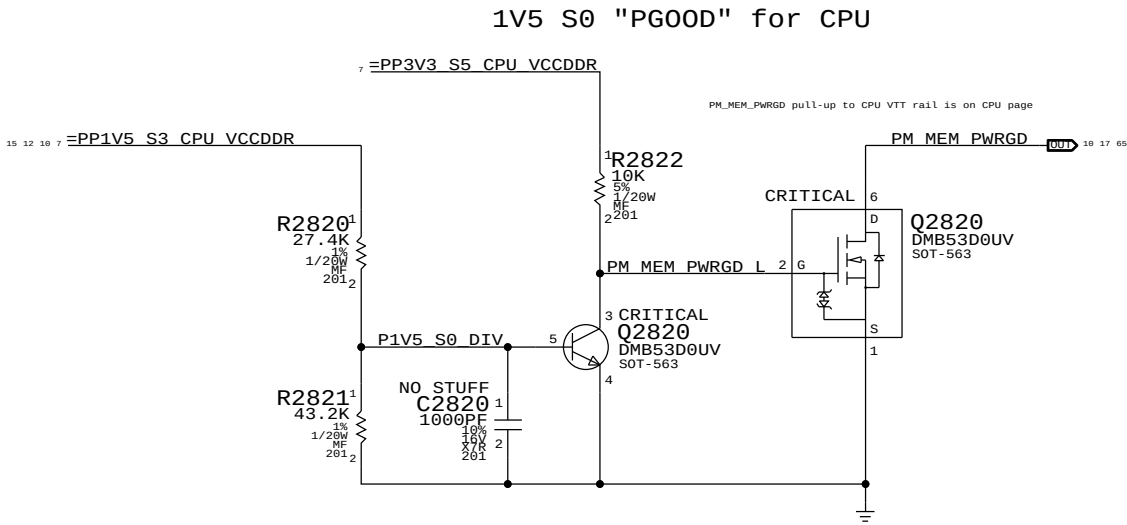
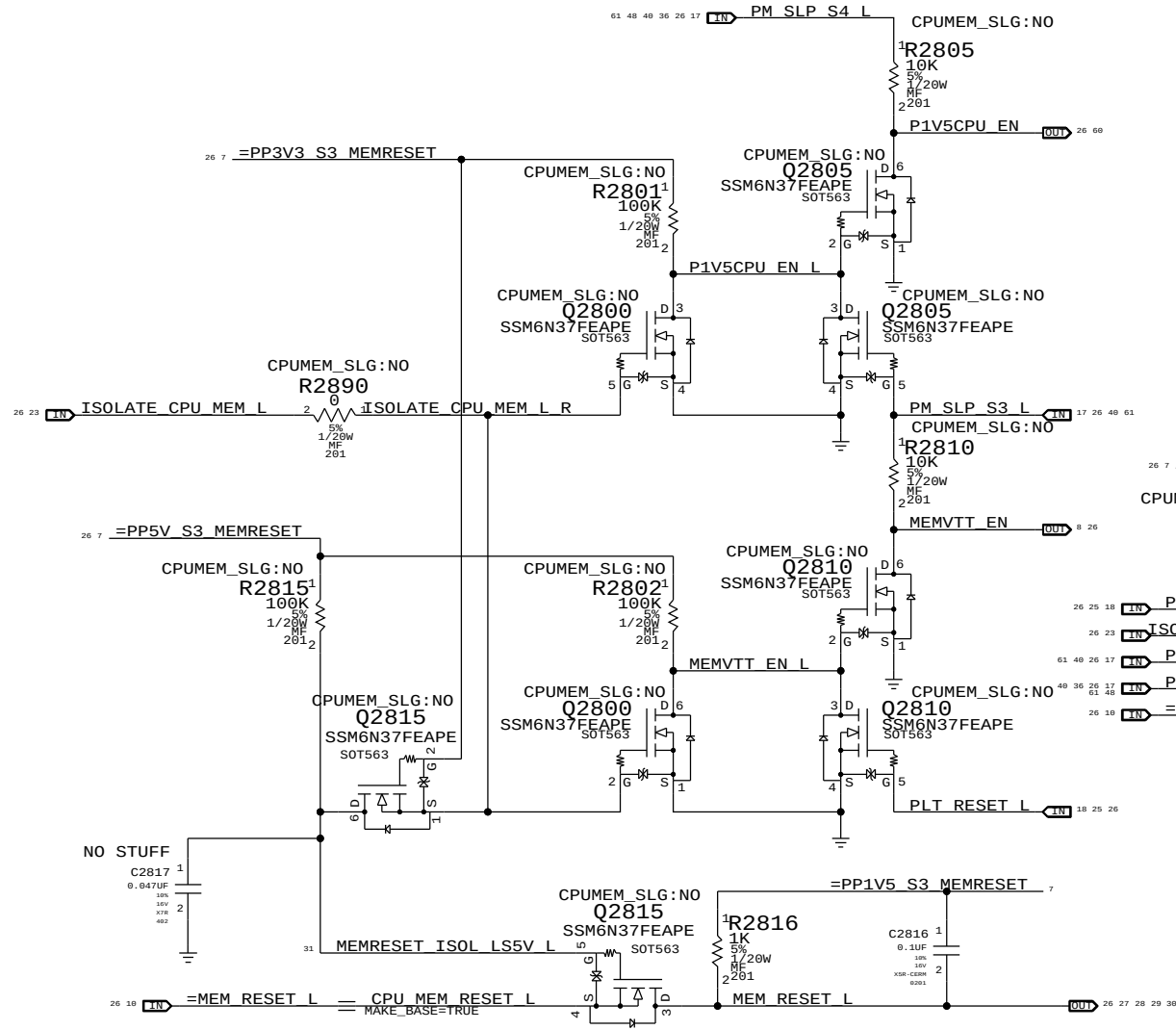
Platform Reset Connections
Unbuffered



The circuit below handles CPU and VTT power during S0->S3->S0 transitions, as well as isolating the CPU's SM_DRAMRST# output from the S0-DIMMs when necessary.

ISOLATE_CPU_MEM_L GPIO state during S3->S0 transitions determines behavior of signals.
WHEN HIGH: CPU 1.5V remains powered in S3, VTT follows S0 rails, MEM_RESET_L not isolated.
WHEN LOW: CPU 1.5V follows S0 rails, VTT ensures clean CKE transition, MEM_RESET_L isolated.

P1V5CPU_EN = (ISOLATE_CPU_MEM_L + PM_SLP_S3_L) * PM_SLP_S4_L
MEMVTT_EN = (ISOLATE_CPU_MEM_L + PLT_RST_L) * PM_SLP_S3_L
MEM_RESET_L = !ISOLATE_CPU_MEM_L + CPU_MEM_RESET_L

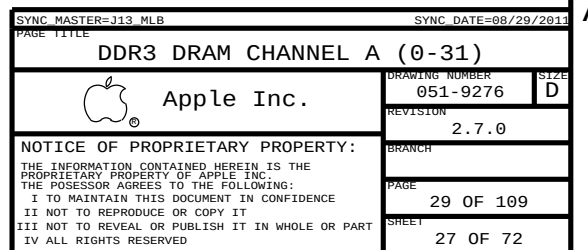
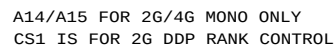


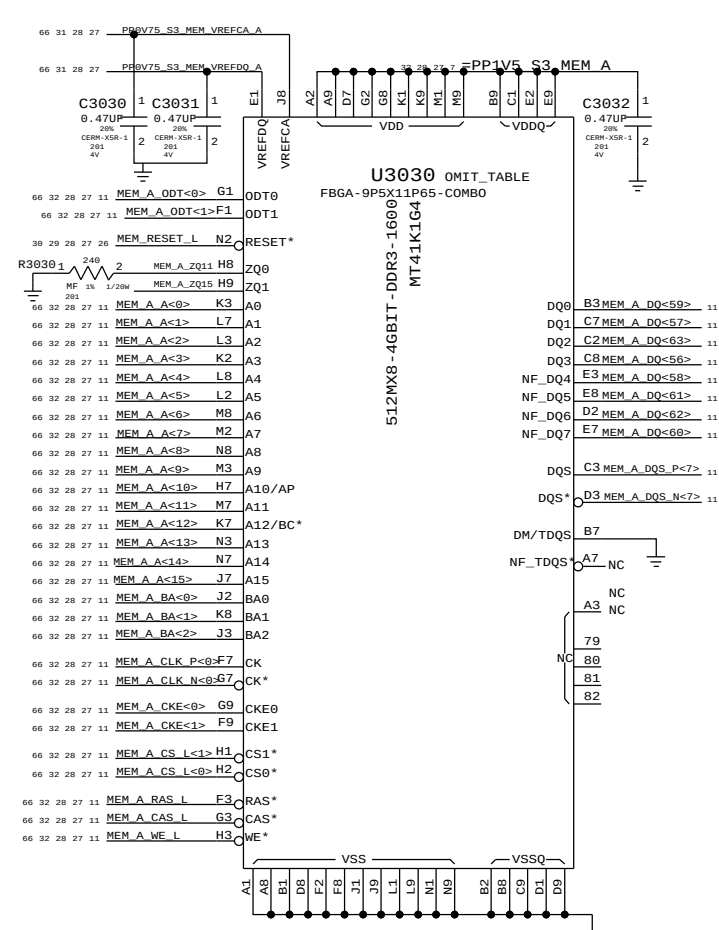
Step	ISOLATE_CPU_MEM_L	PLT_RESET_L	PM_SLP_S3_L	PM_SLP_S4_L	CPUMEM_RESET_L	MEM_RESET_L	MEMVTT_EN	P1V5CPU_EN
S0	0	1	1	1	1	CPUMEM_RESET_L	1	1
1	0	1	1	1	1	1	1	1
2	0	0	1	1	1	1	0	1
3	0	0	0	1	X	1	0	0
S3	4	0	0	1	1	X	0	1
5	0	1	1	1	0 (*)	1	1	1
6	0	1	1	1	1	1	1	1
S0	7	1	1	1	1	CPUMEM_RESET_L	1	1

(*) CPU_MEM_RESET_L asserts due to loss of PM_MEM_PWRGD, must wait for software to clear before deasserting ISOLATE_CPU_MEM_L GPIO.

NOTE: In the event of a S3->S5 transition ISOLATE_CPU_MEM_L will still be asserted on next S5->S0 transition. Rails will power-up as if from S3, but MEM_RESET_L will not properly assert. Software must deassert ISOLATE_CPU_MEM_L and then generate a valid reset cycle on CPU_MEM_RESET_L.

SYNC_MASTER-313_MLB		SYNC_DATE=11/19/2011	
PAGE TITLE			
CPU Memory S3 Support			
 Apple Inc.		DRAWING NUMBER	051-9276
		SIZE	D
		REVISION	2.7.0
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PAGE TITLE			
DDR3 DRAM CHANNEL A (32-63)			
 Apple Inc.		DRAWING NUMBER 051-9276	
		SIZE D	
		REVISION 2.7.0	
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		PAGE 30 OF 109	
		SHEET 28 OF 72	

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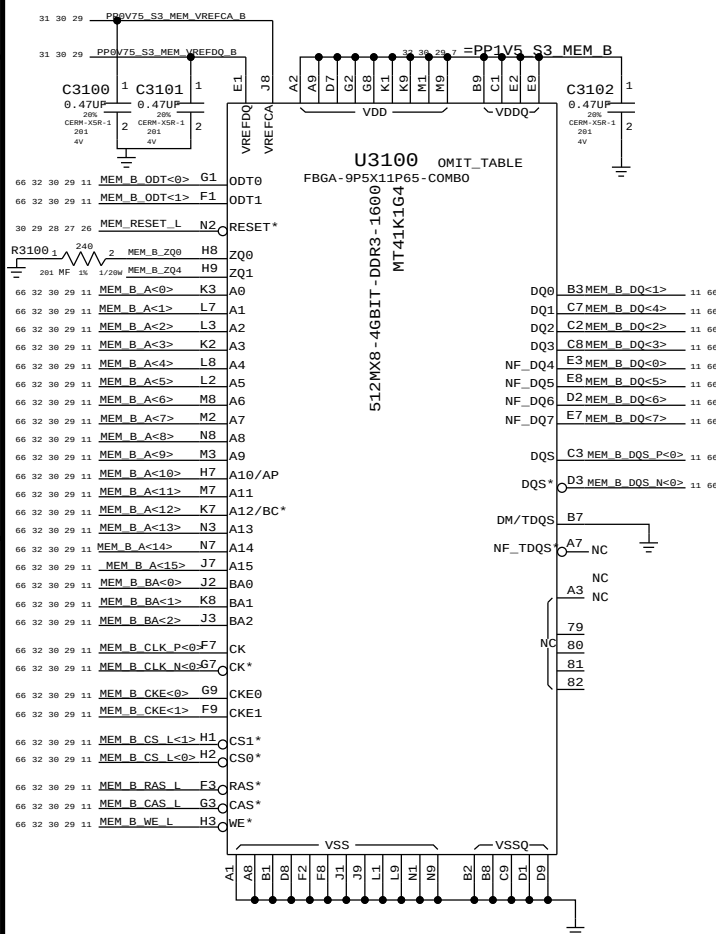
A

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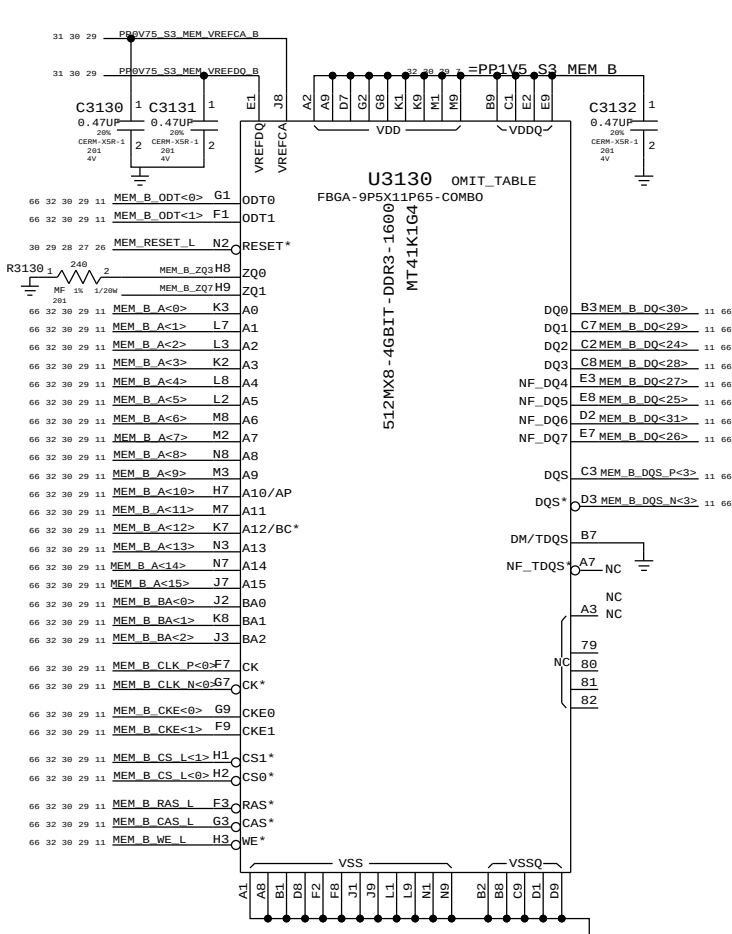
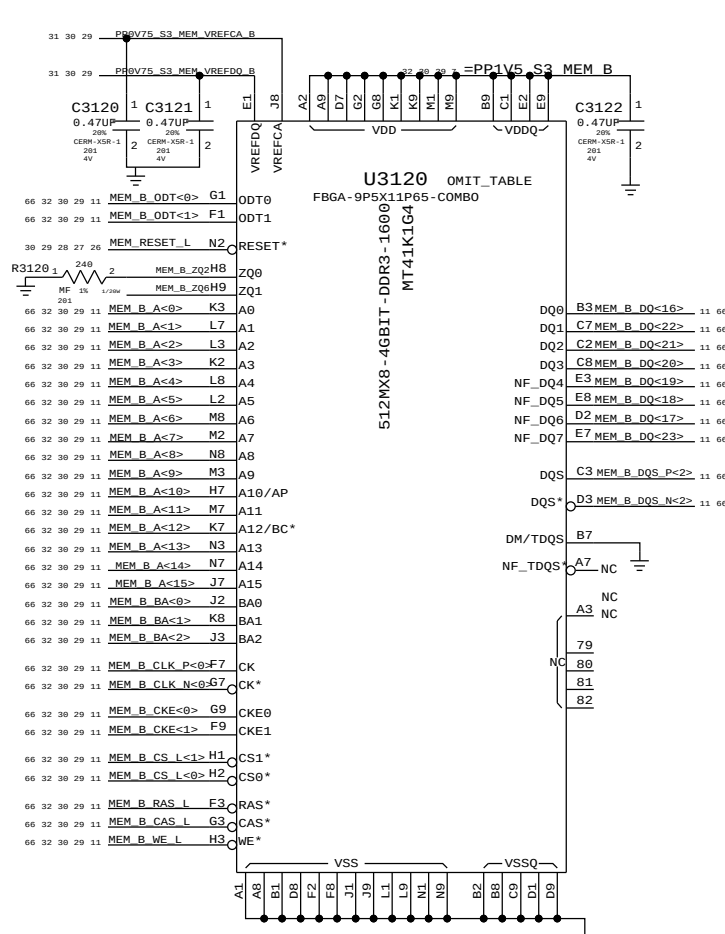
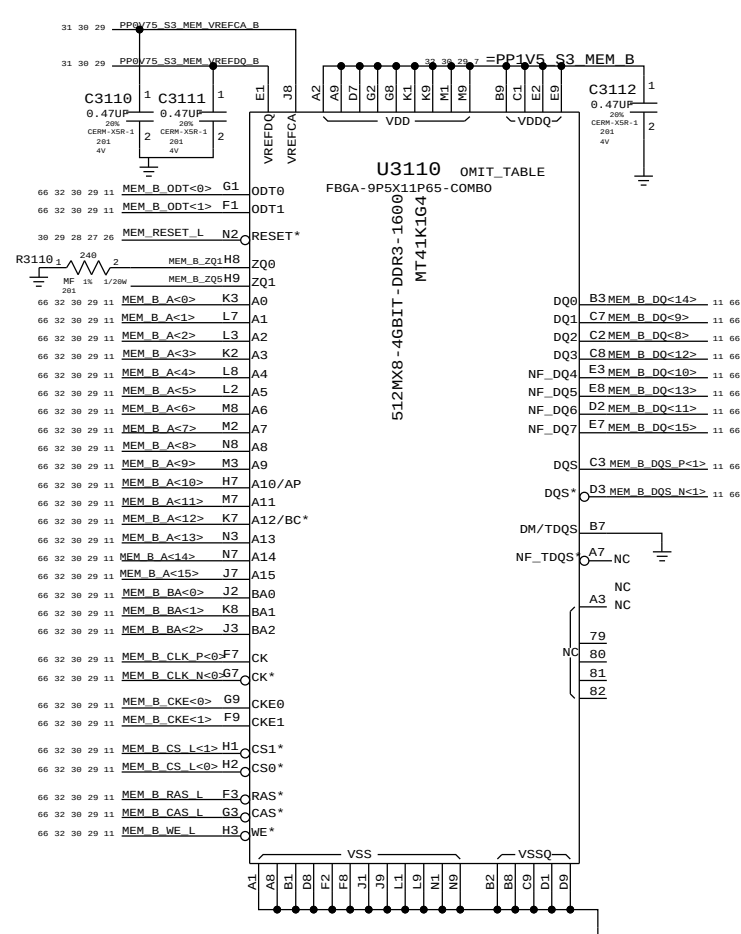
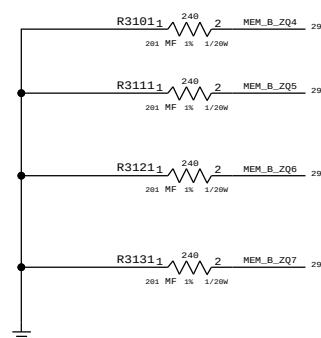
C

B

A



A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL



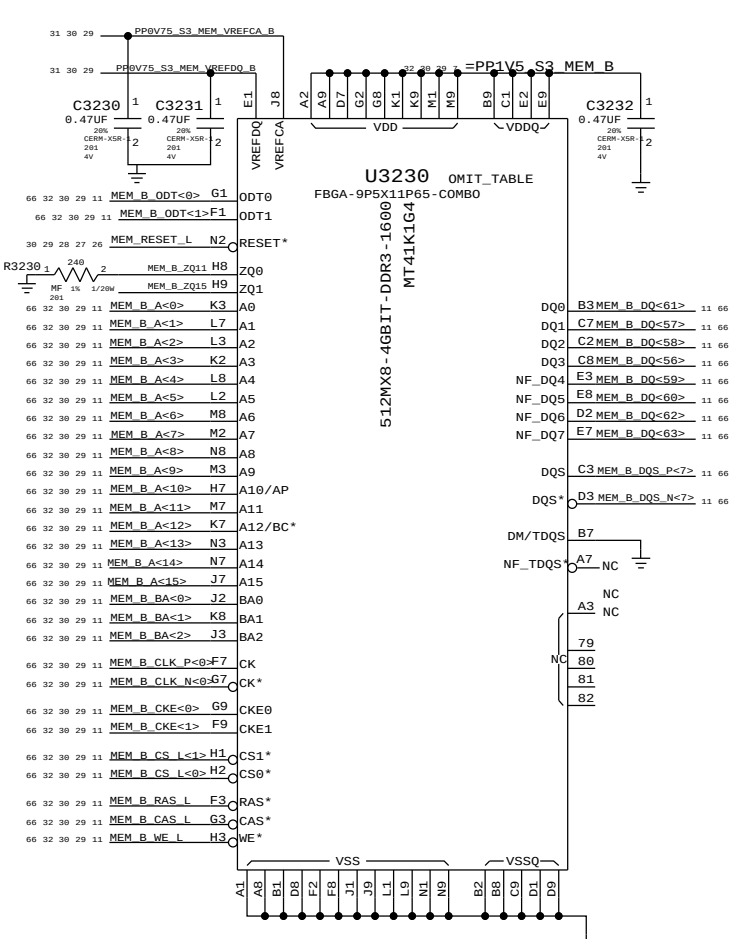
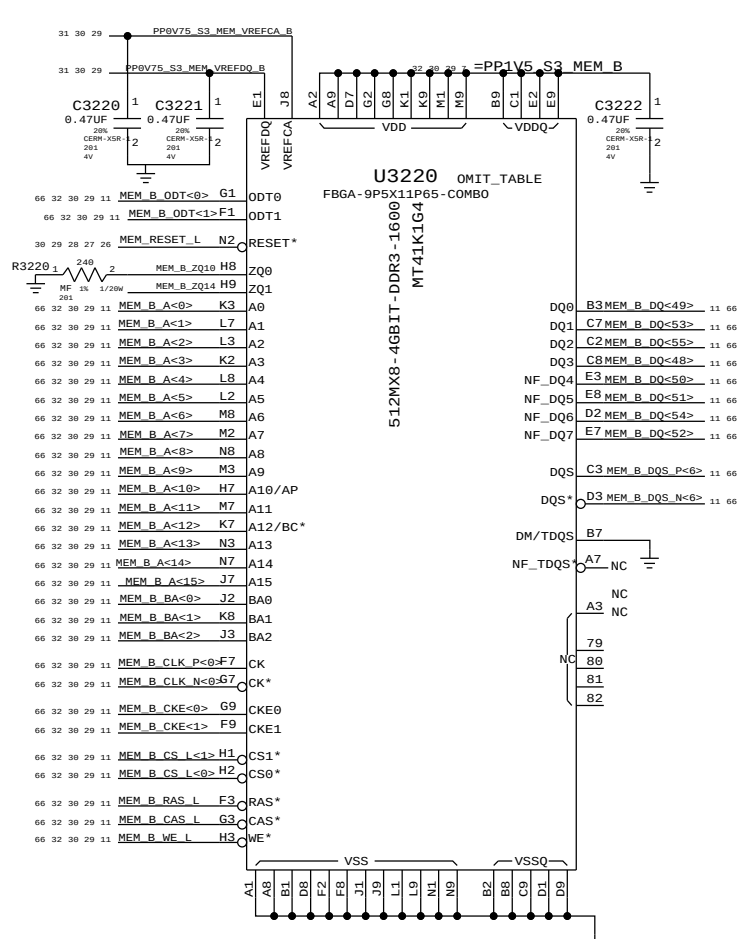
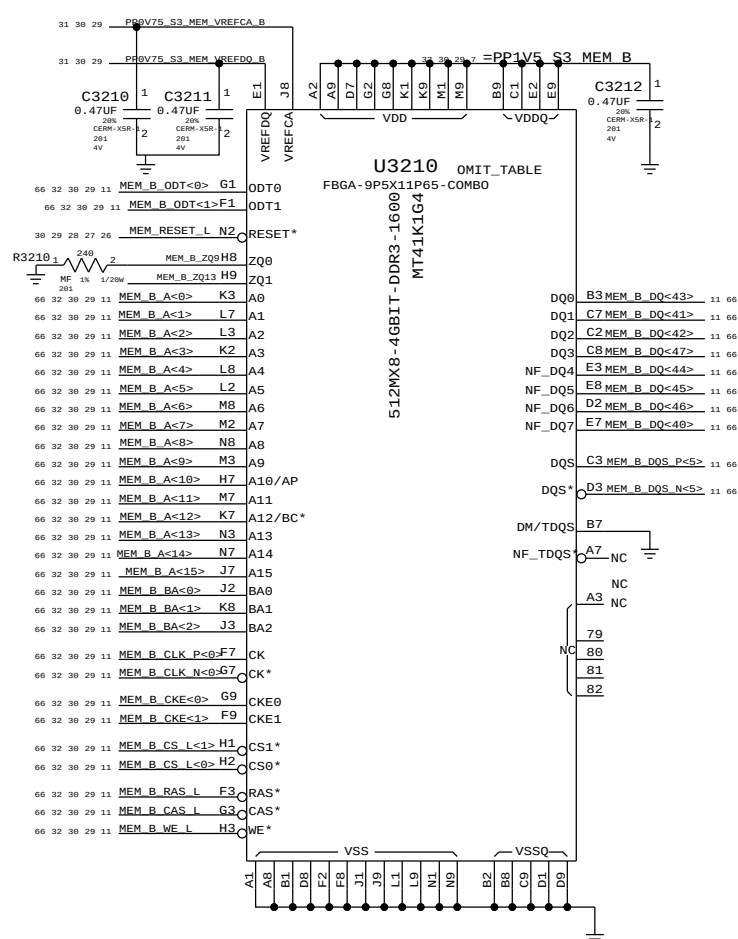
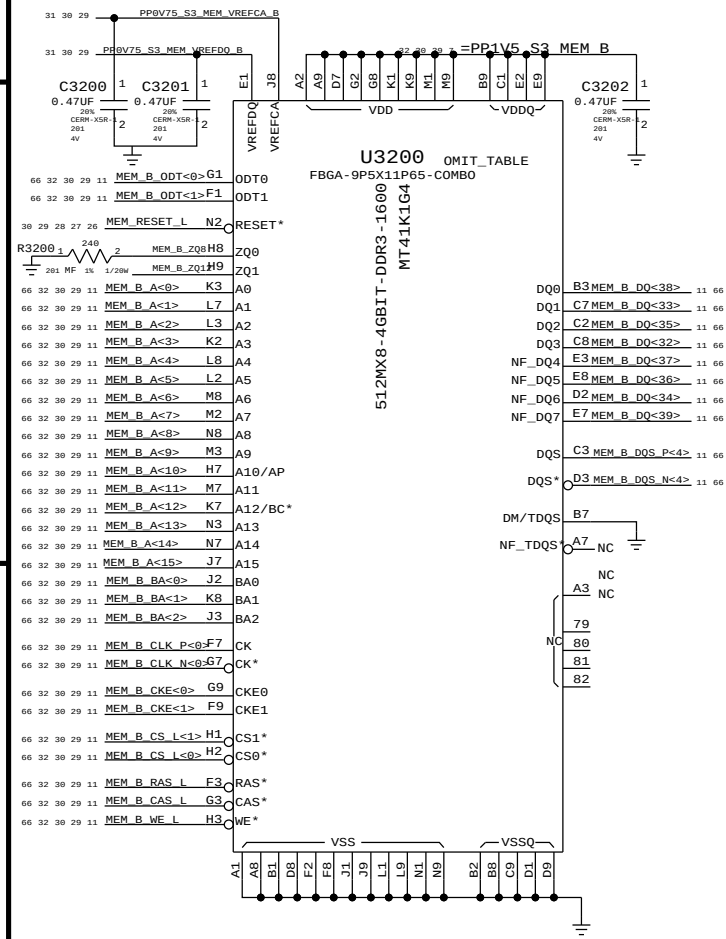
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DDR3 DRAM CHANNEL B (0-31)			
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		REVISION	2.7.0
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D

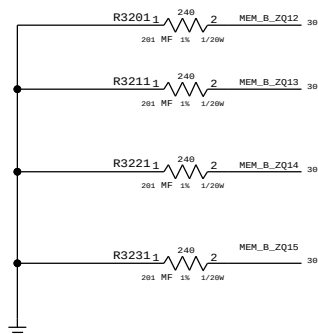
C

B

A



A14/A15 FOR 2G/4G MONO ONLY
CS1 IS FOR 2G DDP RANK CONTROL



SYNC MASTER=J13_M8B		SYNC DATE=08/29/2011	
PAGE TITLE			
DDR3 DRAM CHANNEL B (32-63)			
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	REVISION	2.7.0	
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D

C

B

A

NOTE: Must not enable more than two S0-DIMM margining buffers at once or Vref source may be overloaded.

D

C

B

A

D

C

B

A

Page Notes

Power aliases required by this page:

- =PP3V3_S3_VREFMRGN
- =PPVTT_S3_DDR_BUF
- =PPDDR_S3_MEMVREF

Signal aliases required by this page:

- =I2C_VREFDACS_SCL
- =I2C_VREFDACS_SDA
- =I2C_PCA9557D_SCL
- =I2C_PCA9557D_SDA

BOM options provided by this page:

DDRREF_DAC - Stuffs Apple margining circuit.

VREFDQ:LDO - LDO outputs sent to DQ inputs.

VREFDQ:LDO_DAC - Margined LDO outputs sent to DQ inputs.

VREFDQ:M1_M3 - CPU margined DDR voltage divider sent to DQ inputs.

VREFDQ:M1_DAC - DAC margined DDR voltage divider sent to DQ inputs.

VREFCA:LDO - LDO outputs sent to CA inputs.

VREFCA:LDO_DAC - DAC margined LDO outputs sent to CA inputs.

Addr=0x30(WR)/0x31(RD)

NOTE: Margining will be disabled across all soft-resets and sleep/wake cycles.

NOTE: CPU DAC output step sizes:
DDR3 (1.5V) 7.70mV per step
DDR3L (1.35V) 6.99mV per step

Required zero ohm resistors when no Vref margining circuit stuffed

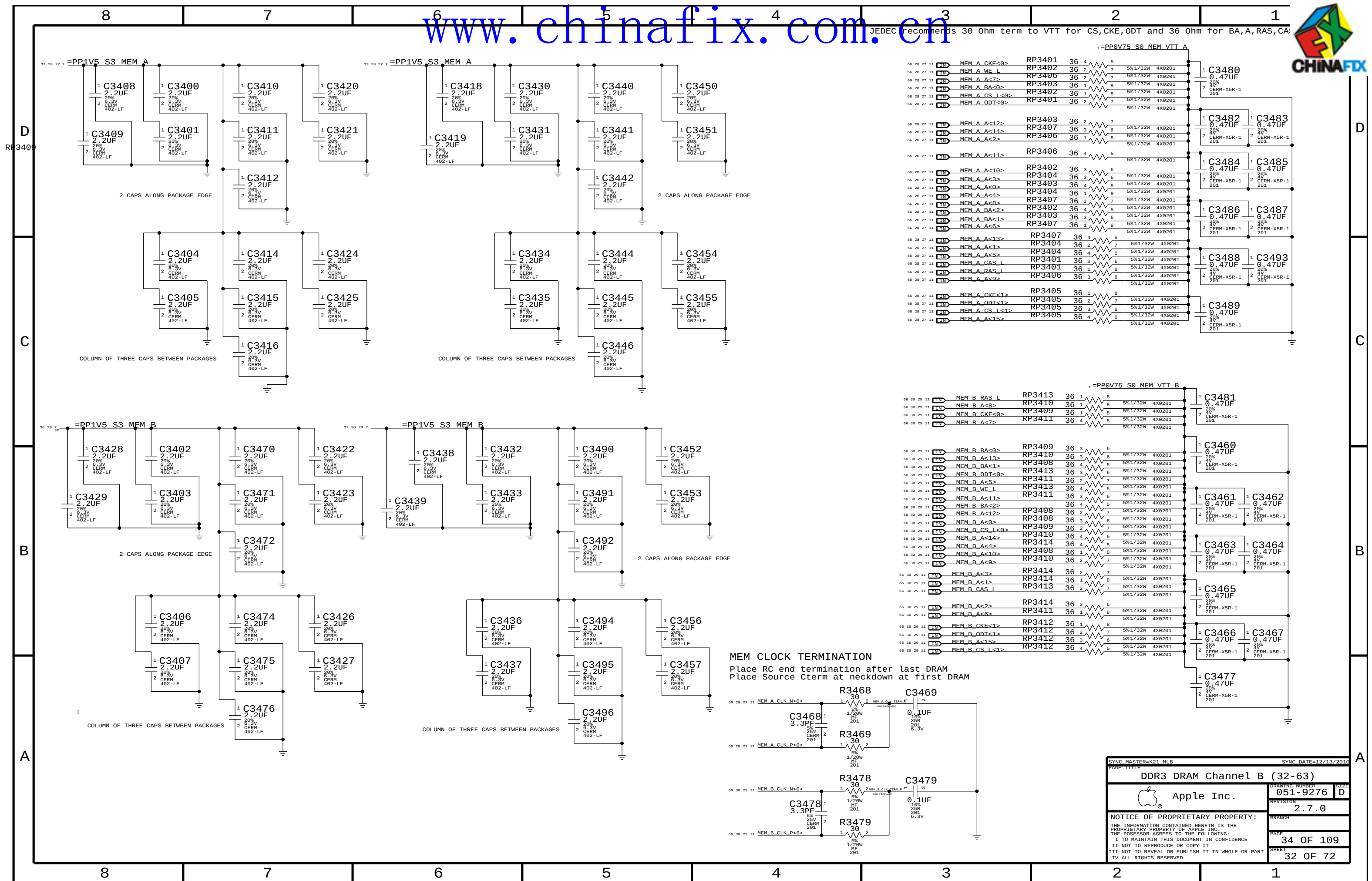
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
117S9002	2	RES, MF, 1/20W, 0.0 OHM, 5, 0201, SMD	R3303, R3306		VREFDQ:LDO
117S9002	2	RES, MF, 1/20W, 0.0 OHM, 5, 0201, SMD	R3309, R3305		VREFCA:LDO

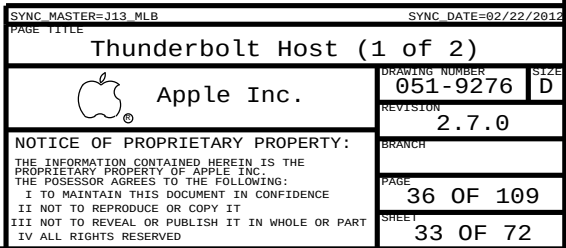
PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
118S9012	4	RES, MF, 1K0HM, 1, 1/20W, 0201	R3321, R3322, R3311, R3312		VREFDQ:M1_DAC
118S9003	1	RES, MF, 3320HM, 1, 1/20W, 0201	R3304		VREFDQ:M3_DAC

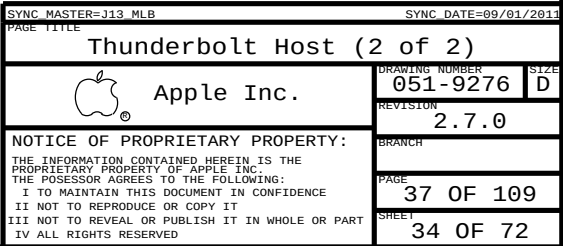
	MEM A VREF DQ	MEM B VREF DQ	MEM A VREF CA	MEM B VREF CA	MEM VREG	GPU Frame Buffer (1.8V, 70% Vref)
DAC Channel:	A	B	C	C	D	D
PCA9557D Pin:	1	2	3	4	5	6
Nominal value		0.75V (DAC: 0x3A)			1.5V (DAC: 0x3A)	1.267V (DAC: 0x8B)
Margined target:		0.300V - 1.200V (+/- 450mV)			1.000V - 2.000V (+/- 500mV)	1.056V - 1.442V (+/- 180mV)
DAC range:		0.000V - 1.501V (0x00 - 0x74)			0.000V - 3.000V (0x00 - 0x74)	0.000V - 3.300V (0x00 - 0xFF)
VRef current:		+3.4mA - -3.4mA (- = sourced)			+61uA - -61uA (- = sourced)	+6.0mA - -5.0mA (- = sourced)
DAC step size:		7.69mV / step @ output			8.59mV / step @ output	1.51mV / step @ output

PAGE TITLE		PAGE DATE=11/19/2013	
FSB/DDR3/FRAMEBUF Vref Margining			
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JEDEC recommends 30 Ohm term to VTT for CS,CKE,ODT and 36 Ohm for BA,A,RAS,CA'







Page Notes

Power aliases required by this page:

- =PPVIN_SW_TBTBST (8-13V Boost Input)
- =PP18V_TBT_REG (18V Boost Output)
- =PP3V3_TBT_P3V3TBTFTET (3.3V FET Input)
- =PP3V3_TBT_FET (3.3V FET Output)
- =PP3V3_S0_TBT_PWRCTL
- =PP1V05_TBT_P1V05TBTFTET (1.05V FET Input)
- =PP1V05_TBT_FET (1.05V FET Output)

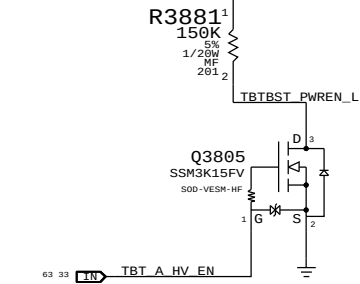
Signal aliases required by this page:

- =TBT_CLKREQ_L
- =TBT_RESET_L

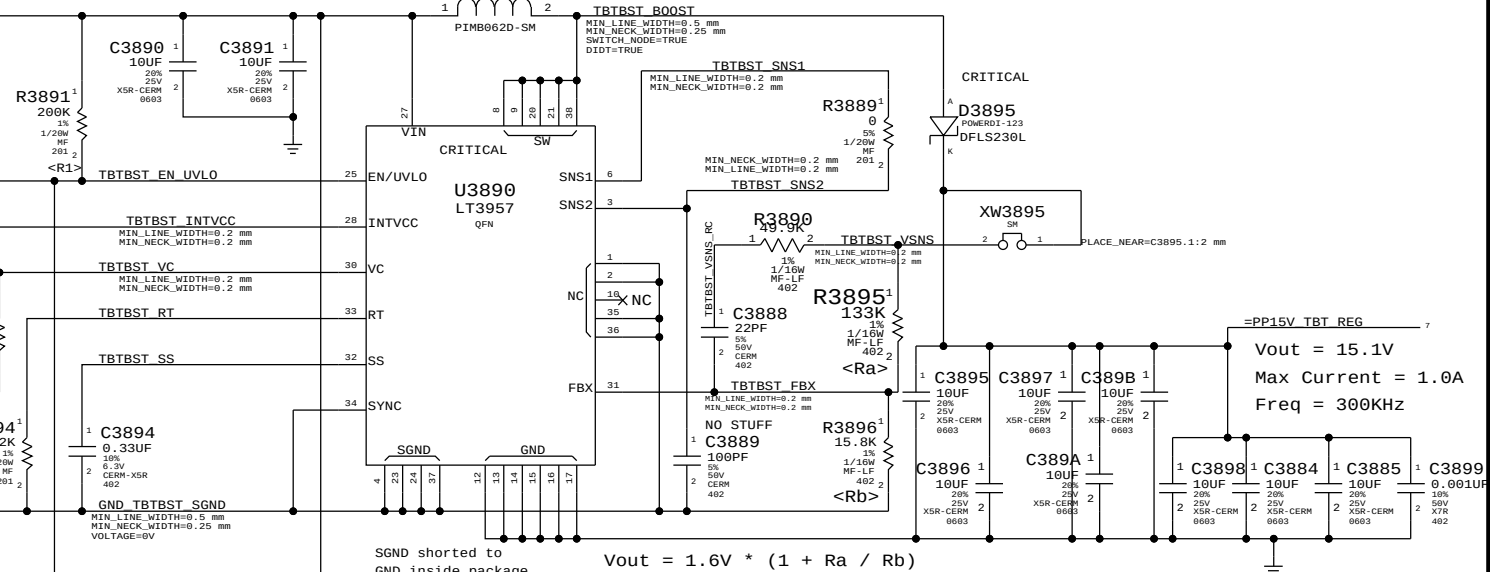
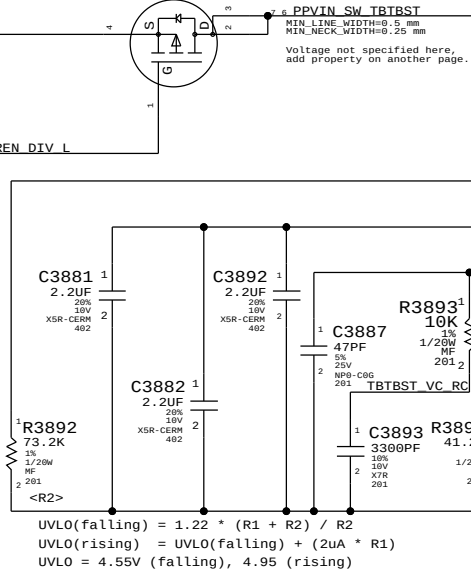
BOM options provided by this page:

TBTBST:Y - Stuffs 18V boost circuitry.

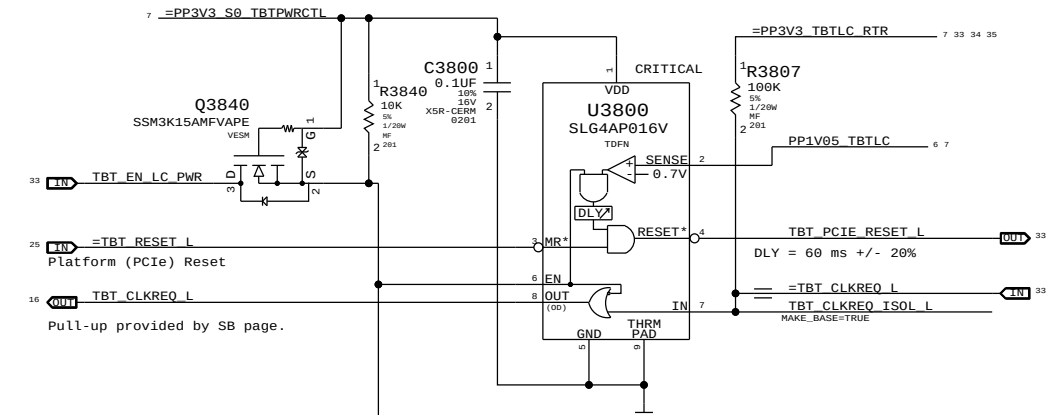
=PPVIN_SW_TBTBST
8-13V Input
Changes required
for 2S.



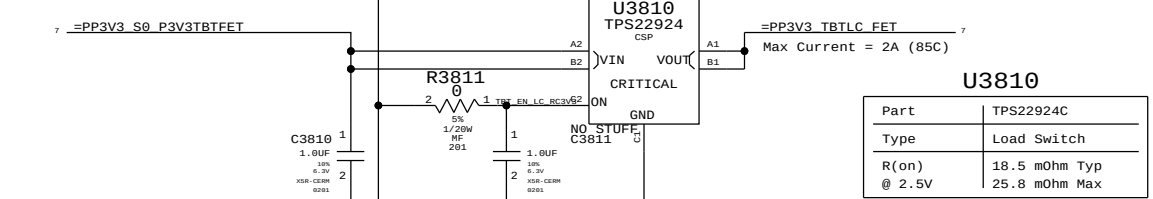
CRITICAL
Q3880
SI8409DB
BGA
Vds(max): -30V
Vgs(max): +/-12V
Vgs(th): -1.4V
Rds(on): 46mOhm @ 4.5V Vgs
Id(max): 3.7A @ 70C



Supervisor & CLKREQ# Isolation

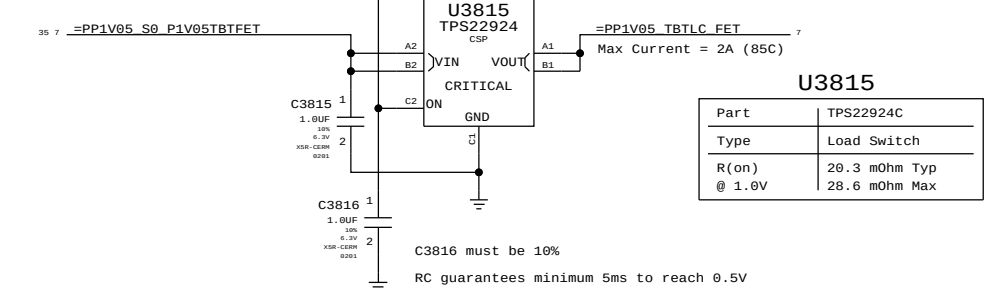


3.3V TBT "LC" Switch



U3810	
Part	TPS22924C
Type	Load Switch
R(on)	18.5 mOhm Typ 25.8 mOhm Max
@ 2.5V	

1.05V TBT "LC" Switch

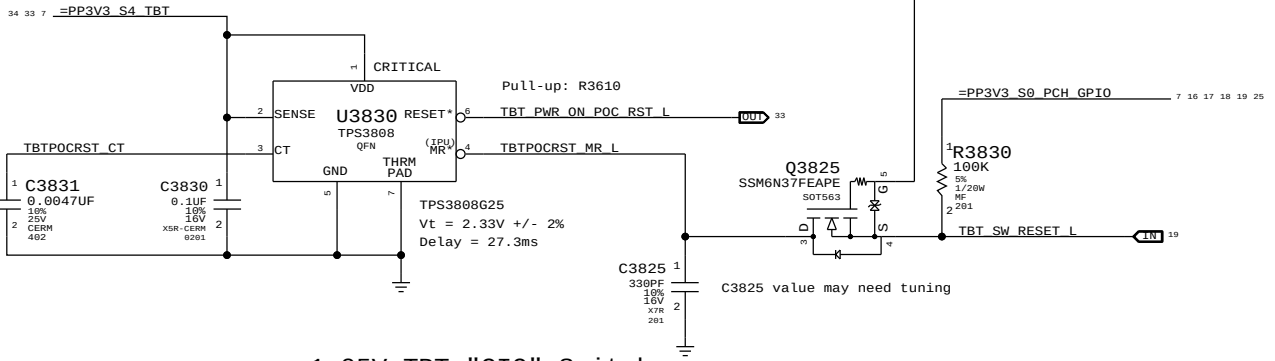


U3815	
Part	TPS22924C
Type	Load Switch
R(on)	20.3 mOhm Typ 28.6 mOhm Max
@ 1.0V	

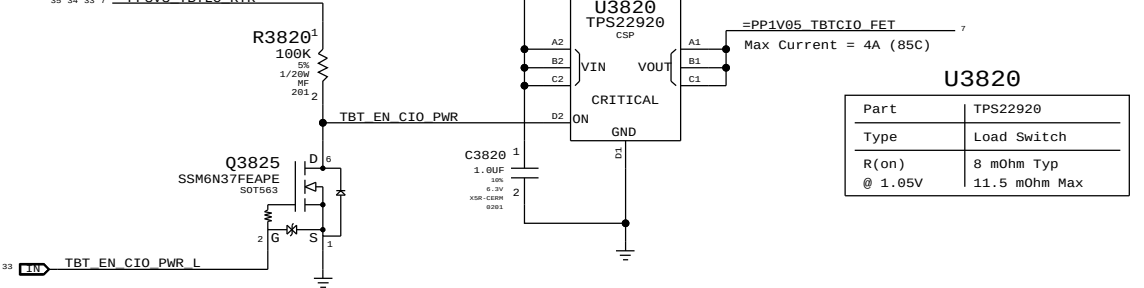
C3816 must be 10%
RC guarantees minimum 5ms to reach 0.5V

TBT "POC" Power-up Reset

Intel investigating whether RC is sufficient.

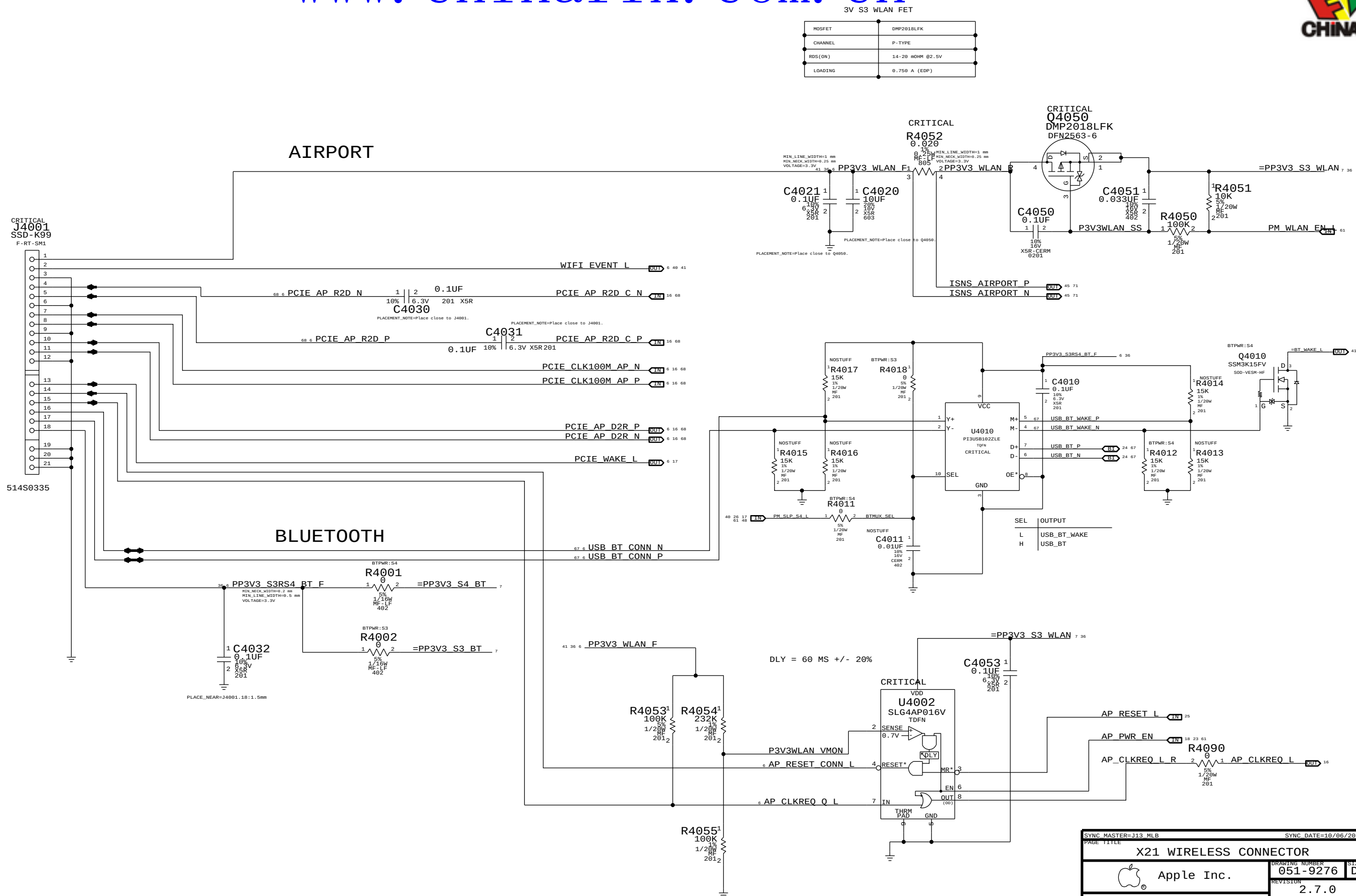


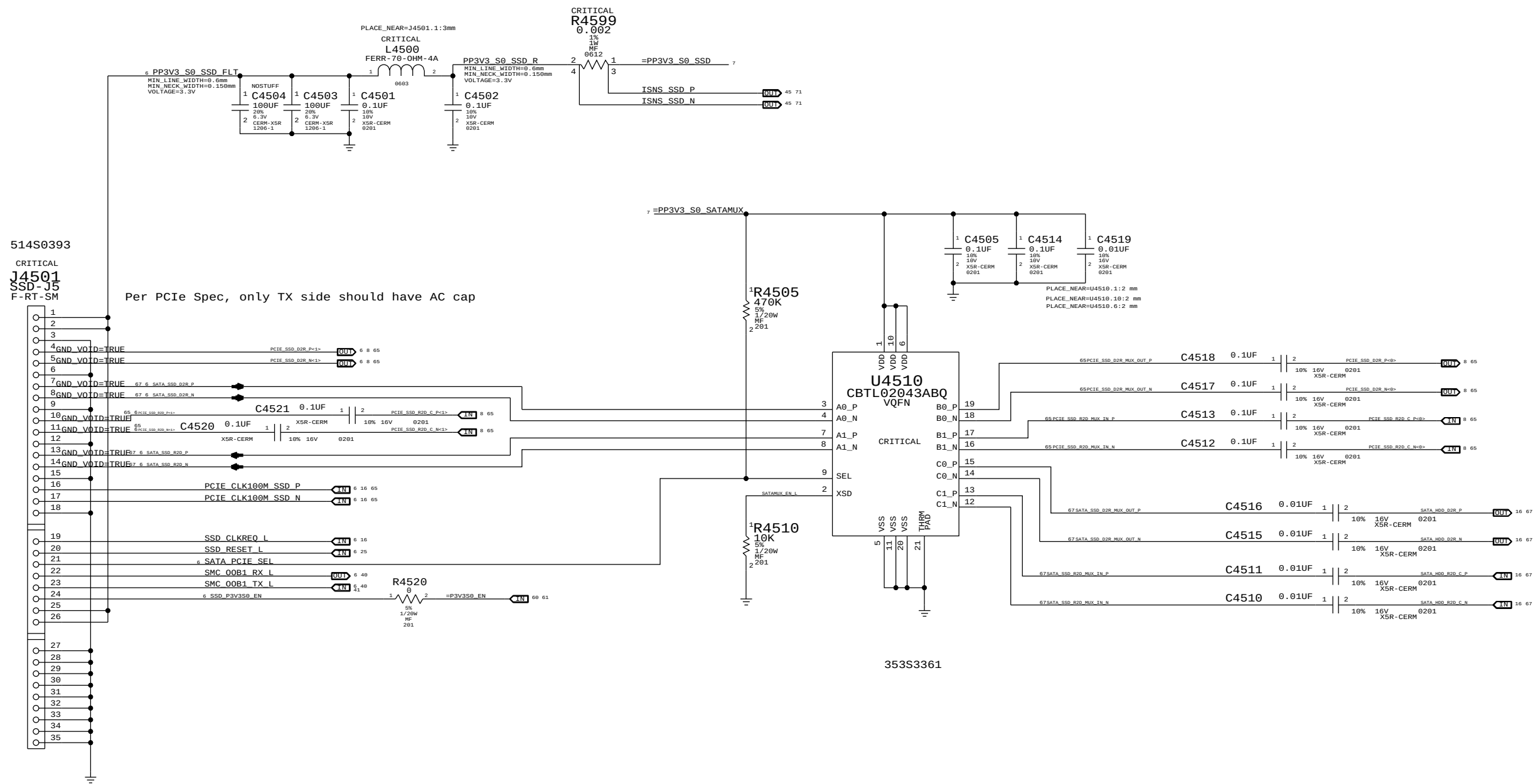
1.05V TBT "CIO" Switch



U3820	
Part	TPS22920
Type	Load Switch
R(on)	8 mOhm Typ 11.5 mOhm Max
@ 1.05V	

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TBT Power Support		DRAWING NUMBER	
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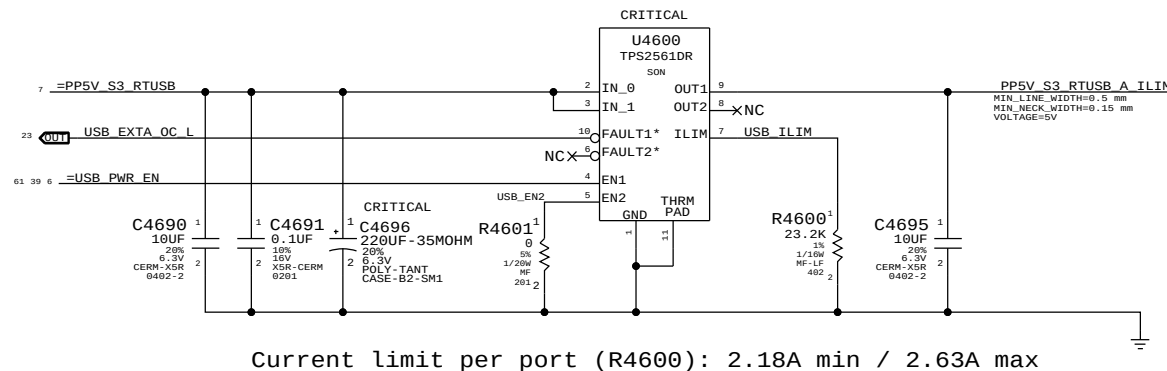




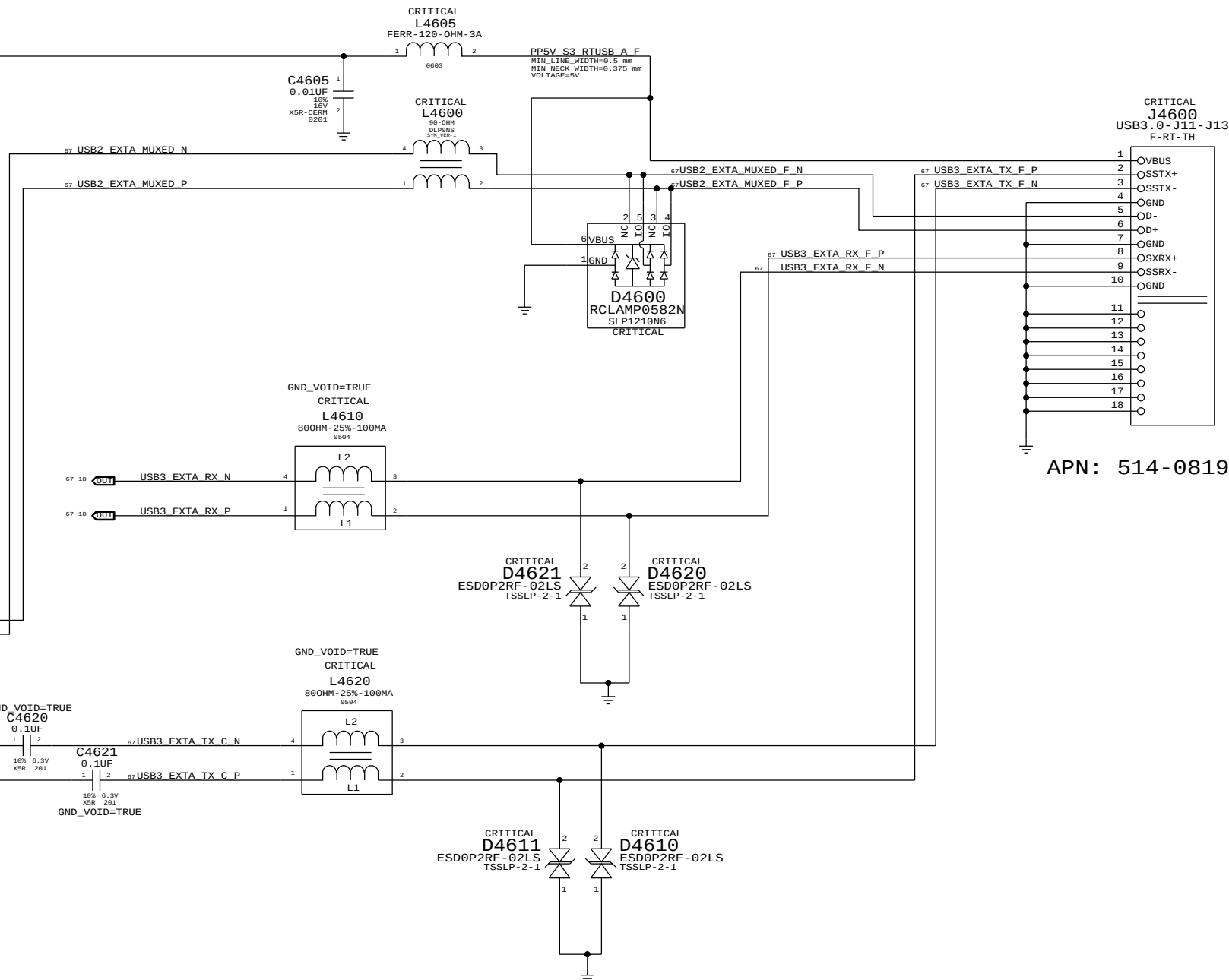
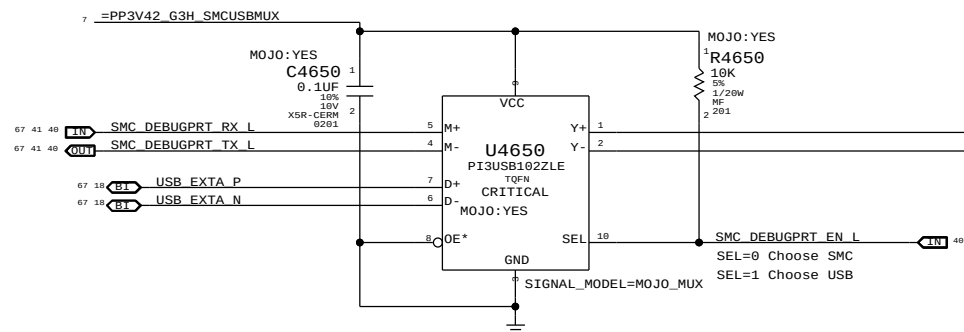
PCIE/SATA GUMSTICK2 CONNECTOR

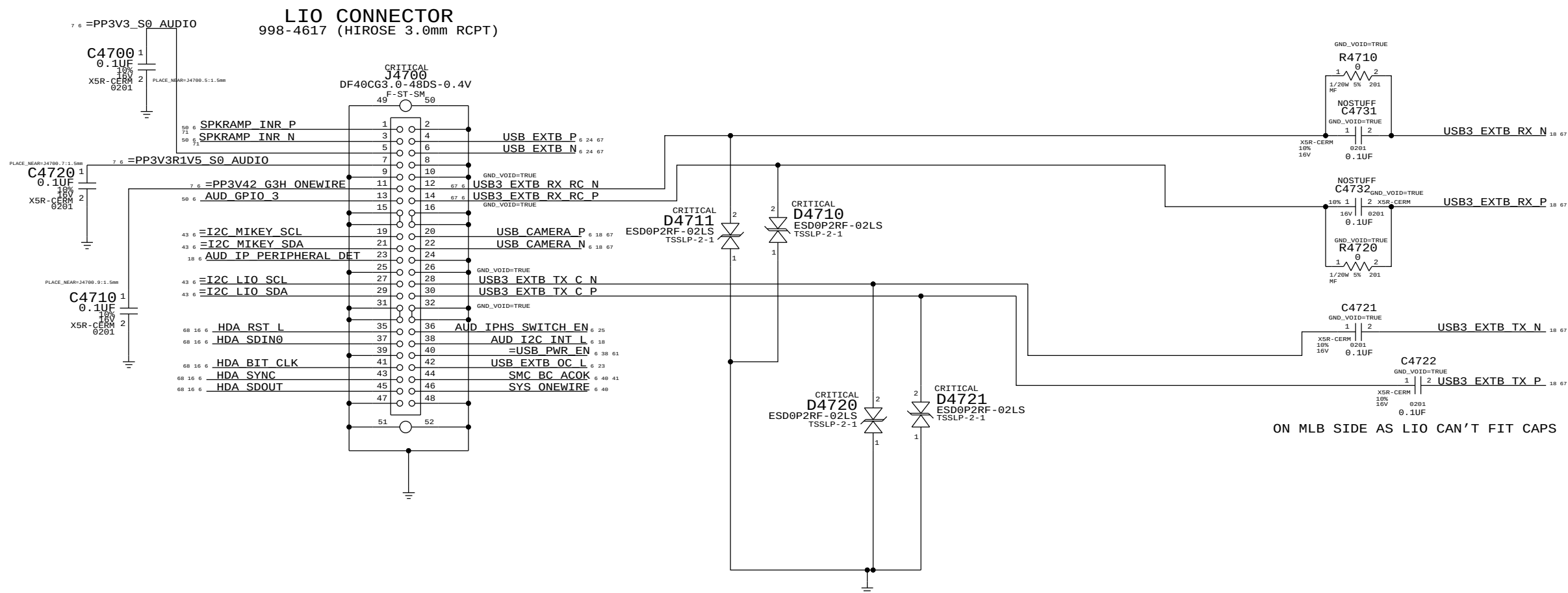
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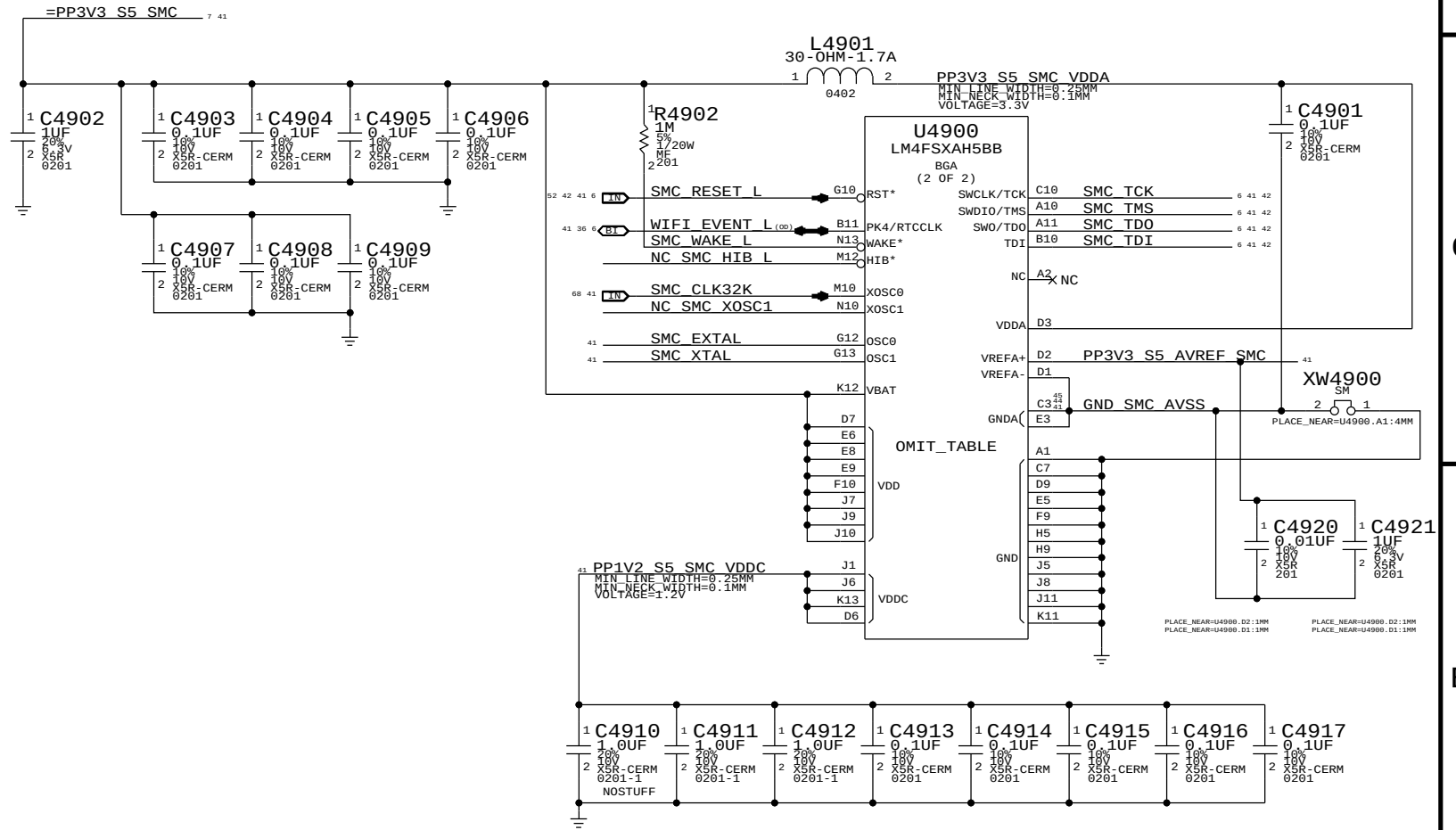
USB Port Power Switch



Mojo SMC Debug Mux



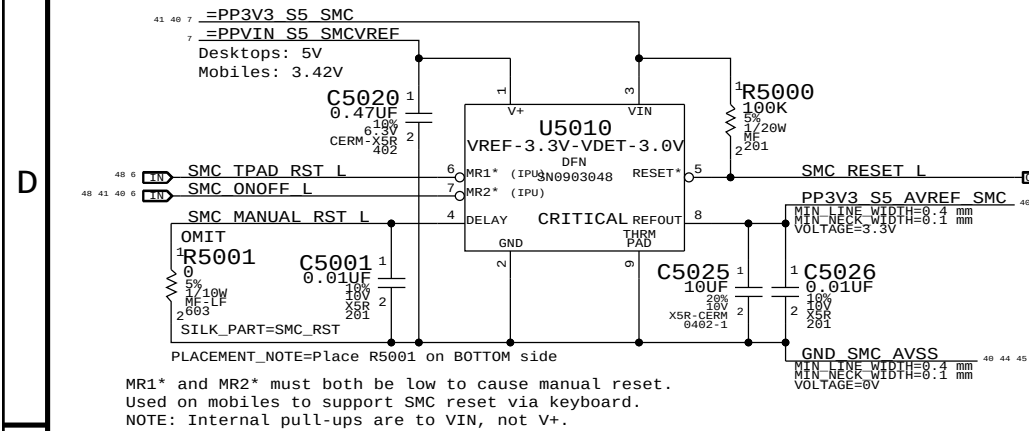




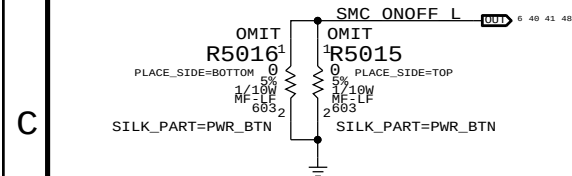
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SMC Reset "Button", Supervisor & AVREF Supply



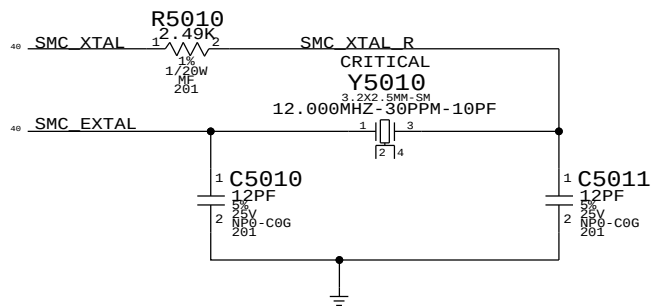
Debug Power "Buttons"



SMC Crystal Circuit

SMC USB Clock require these crystal values: 5, 6, 8, 10, 12, 16, 18, 20, 24, 25 MHz

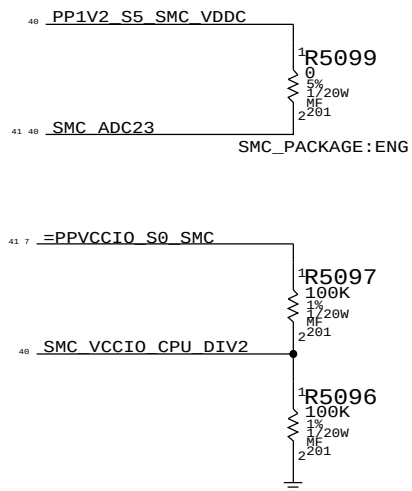
Note:
ADC10 and ADC11 are shared
with comparators on Stack Board.



SMC ADC0	MAKE_BASE=TRUE	SMC CPU ISENSE	MAKE_BASE=TRUE
SMC ADC1	MAKE_BASE=TRUE	SMC VCCSA VSENSE	MAKE_BASE=TRUE
SMC ADC2	MAKE_BASE=TRUE	SMC DCIN VSENSE	MAKE_BASE=TRUE
SMC ADC3	MAKE_BASE=TRUE	SMC DCIN ISENSE	MAKE_BASE=TRUE
SMC ADC4	MAKE_BASE=TRUE	SMC PBUS VSENSE	MAKE_BASE=TRUE
SMC ADC5	MAKE_BASE=TRUE	SMC HDD ISENSE	MAKE_BASE=TRUE
SMC ADC6	MAKE_BASE=TRUE	SMC BMON ISENSE	MAKE_BASE=TRUE
SMC ADC7	MAKE_BASE=TRUE	SMC HS COMPUTING ISENSE	MAKE_BASE=TRUE
SMC ADC8	MAKE_BASE=TRUE	SMC OTHER HI ISENSE	MAKE_BASE=TRUE
SMC ADC9	MAKE_BASE=TRUE	SMC 1V5S3 ISENSE	MAKE_BASE=TRUE
SMC ADC10	MAKE_BASE=TRUE	SMC CPUVCC10 ISENSE	MAKE_BASE=TRUE
SMC ADC11	MAKE_BASE=TRUE	SMC GFX VSENSE	MAKE_BASE=TRUE
SMC ADC12	MAKE_BASE=TRUE	SMC CPU SA ISENSE	MAKE_BASE=TRUE
SMC ADC13	MAKE_BASE=TRUE	SMC 3V3S0 ISENSE	MAKE_BASE=TRUE
SMC ADC14	MAKE_BASE=TRUE	SMC WLAN ISENSE	MAKE_BASE=TRUE
SMC ADC15	MAKE_BASE=TRUE	SMC LCDBKLT ISENSE	MAKE_BASE=TRUE
SMC ADC16	MAKE_BASE=TRUE	NC SMC ADC17	MAKE_BASE=TRUE
SMC ADC17	MAKE_BASE=TRUE	SMC SMC ADC20	MAKE_BASE=TRUE
SMC ADC18	MAKE_BASE=TRUE	NC SMC ADC21	MAKE_BASE=TRUE
SMC ADC19	MAKE_BASE=TRUE	NC SMC ADC22	MAKE_BASE=TRUE
SMC ADC20	MAKE_BASE=TRUE	SMC ADC23	MAKE_BASE=TRUE
SMC ADC21	MAKE_BASE=TRUE	NC SMC GFX OVERTEMP	MAKE_BASE=TRUE
SMC ADC22	MAKE_BASE=TRUE	NC SMC GFX THROTTLE L	MAKE_BASE=TRUE
SMC ADC23	MAKE_BASE=TRUE	NC SMC FAN 1 CTL	MAKE_BASE=TRUE
SMC GFX OVERTEMP	MAKE_BASE=TRUE	NC SMC FAN 1 TACH	MAKE_BASE=TRUE
SMC GFX THROTTLE L	MAKE_BASE=TRUE	ENET ASF GPIO	MAKE_BASE=TRUE
SMC FAN 1 CTL	MAKE_BASE=TRUE	NC SMC MPM5 LED PWR	MAKE_BASE=TRUE
SMC FAN 1 TACH	MAKE_BASE=TRUE	NC SMC MPM5 LED CHG	MAKE_BASE=TRUE
ENET ASF GPIO	MAKE_BASE=TRUE	NC SYS TDM ONEWIRE	MAKE_BASE=TRUE
SMC MPM5 LED PWR	MAKE_BASE=TRUE		
SMC MPM5 LED CHG	MAKE_BASE=TRUE		
SYS TDM ONEWIRE	MAKE_BASE=TRUE		

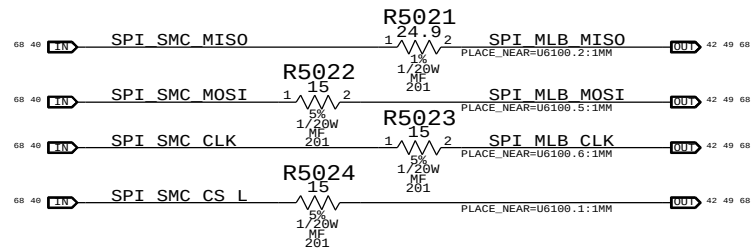
SMC12 Eng Pkg Support

Eng Package requires 1.2V ON SMC_ADC23 pin.

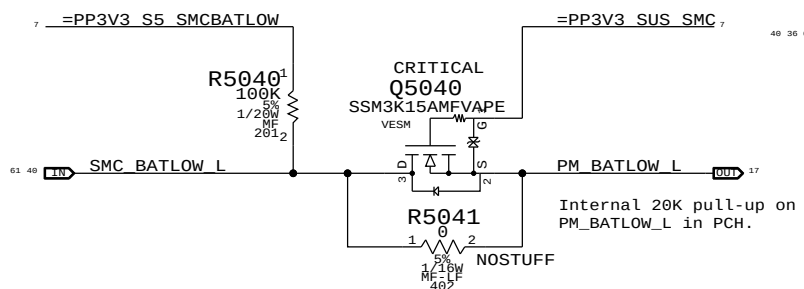


SMC12 SPI Support

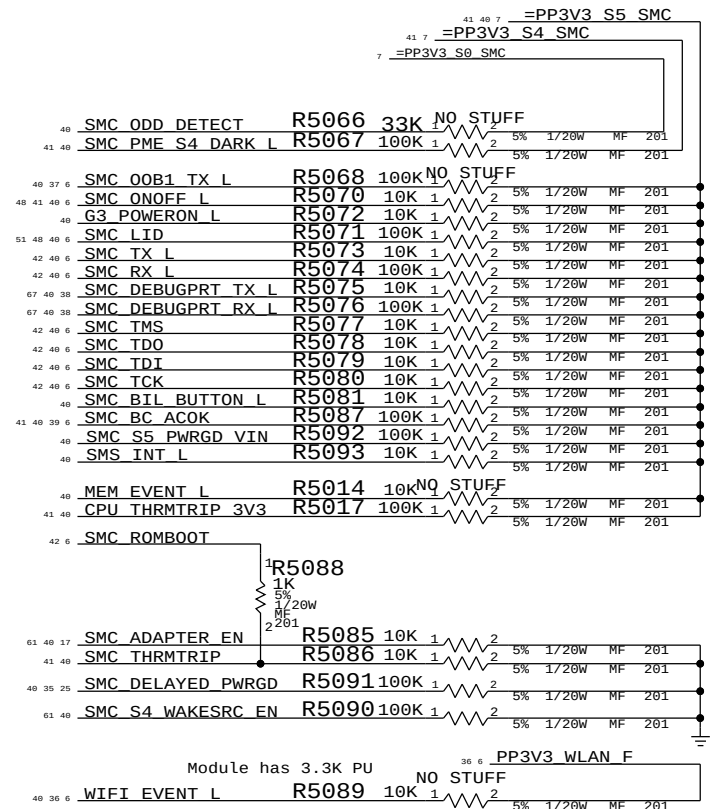
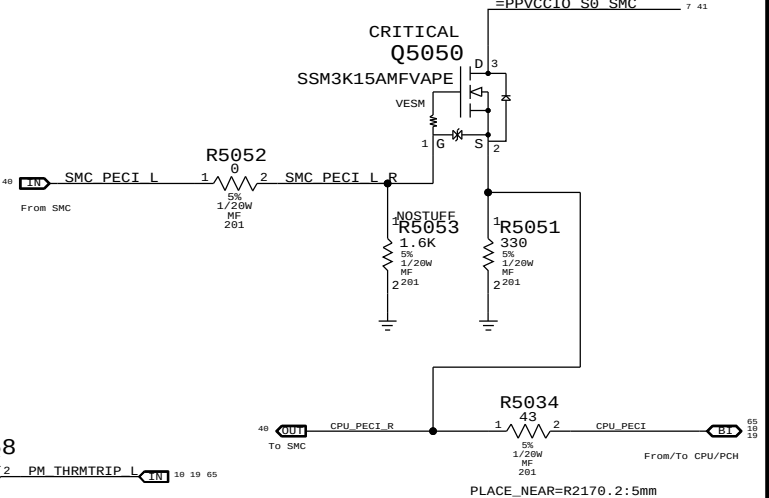
Series resistors are no stuffed until the topology of 2 SPI Masters are verified.



BATLOW# Isolation

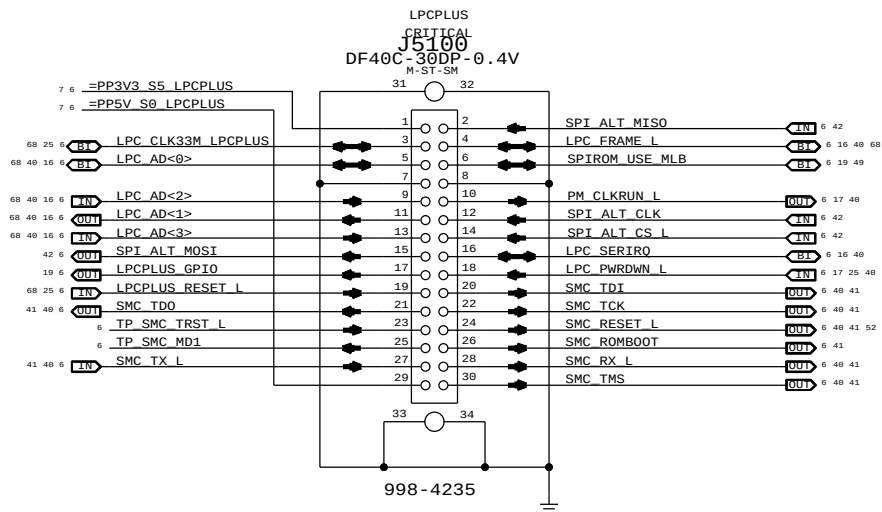


SMC12 PECl Support

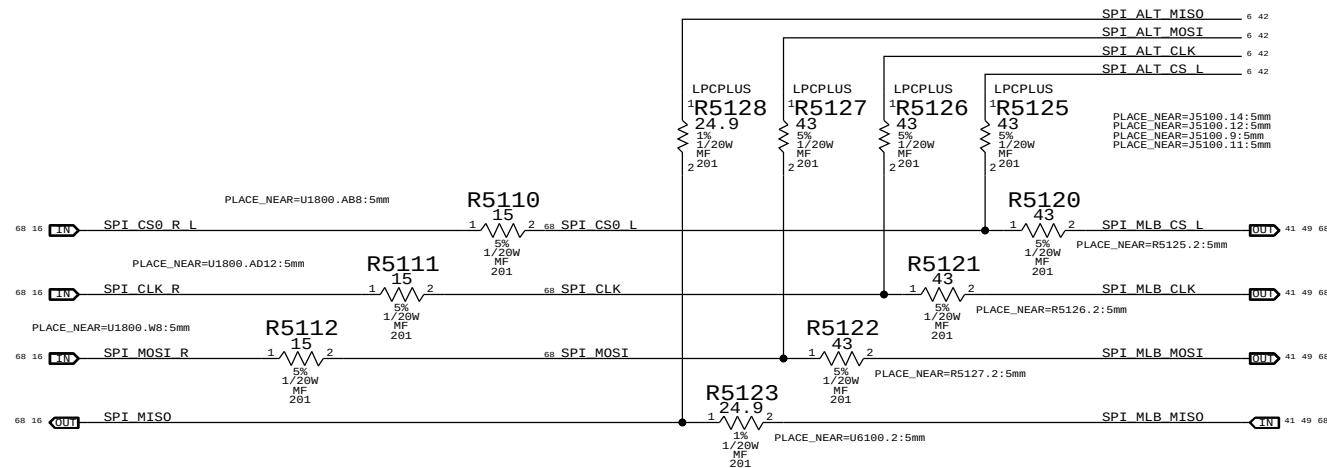


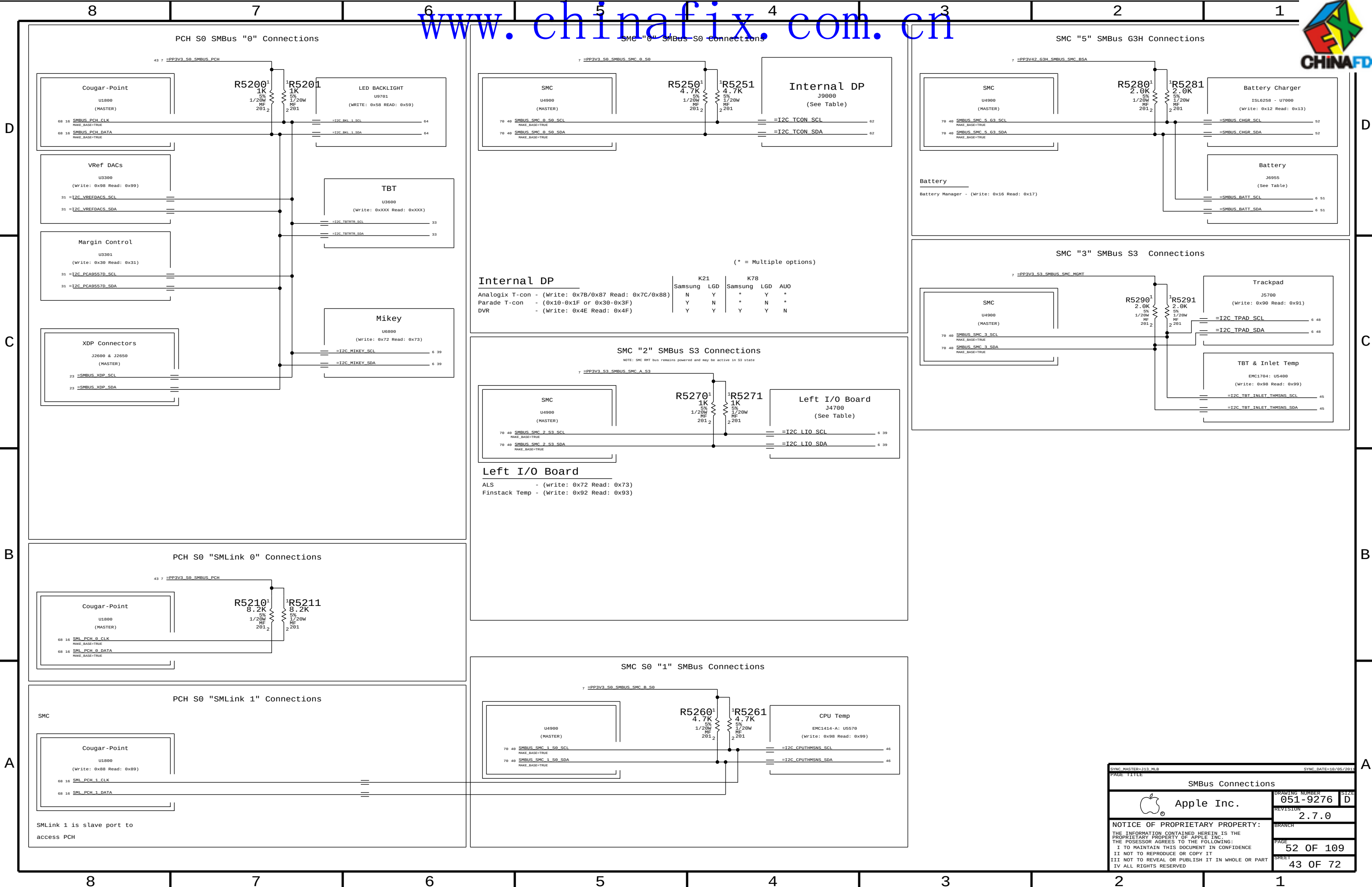
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LPC+SPI Connector

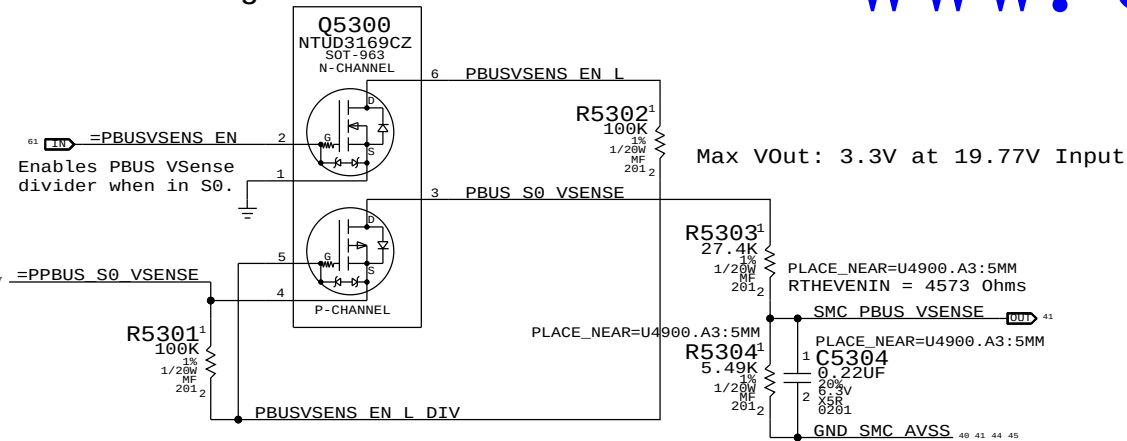


SPI Bus Series Termination

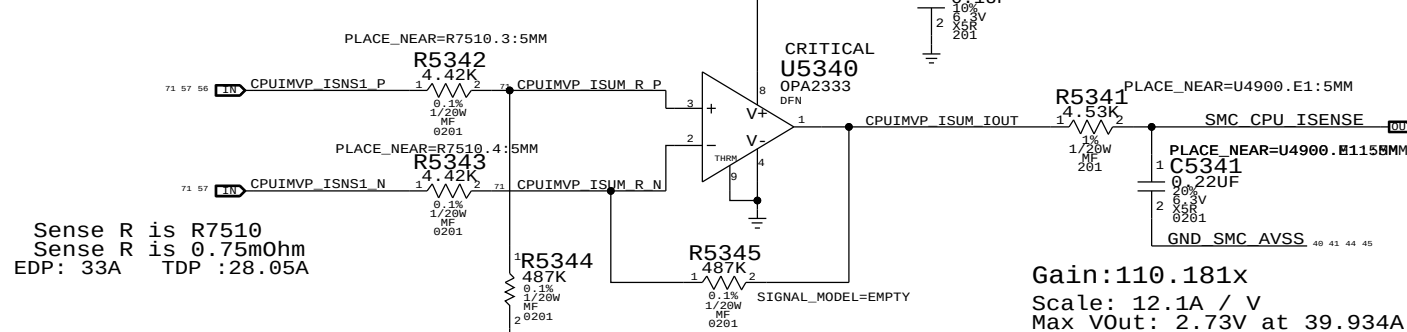




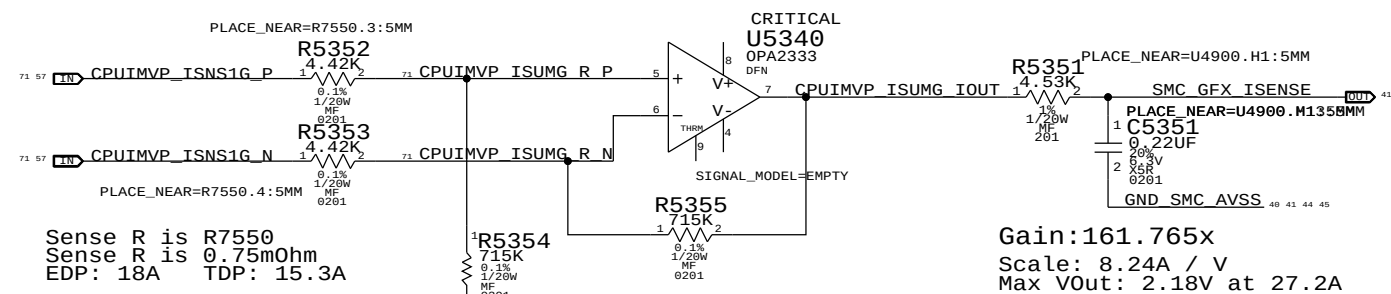
PBUS Voltage Sense Enable & Filter



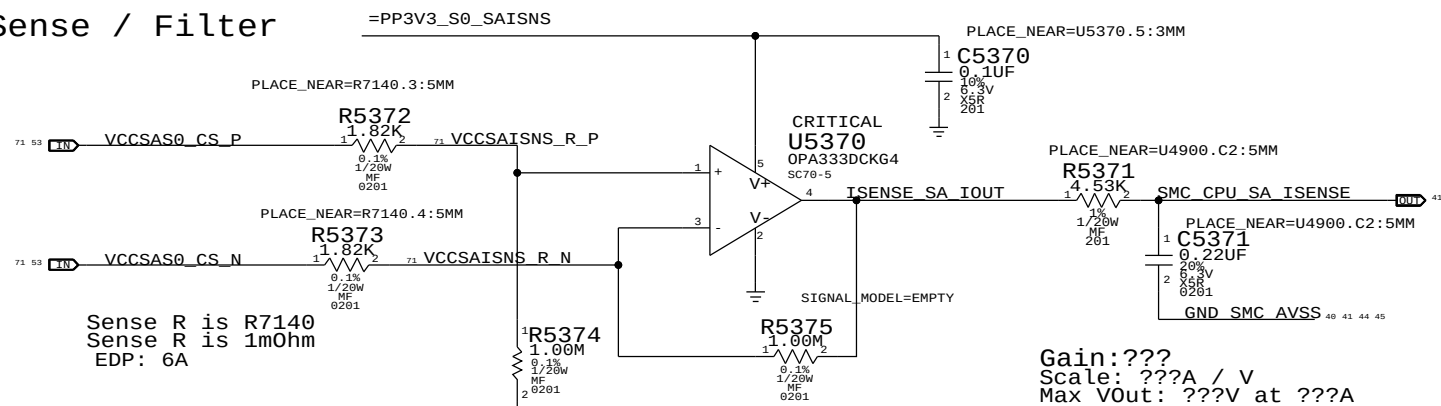
CPU VCore Load Side Current Sense / Filter



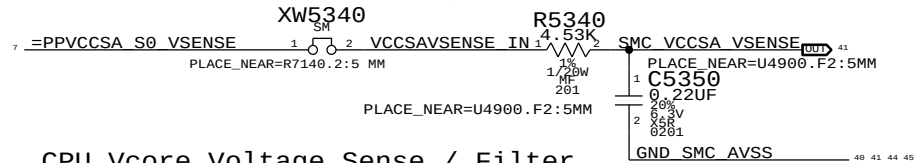
GFX/IG VCore Load Side Current Sense / Filter



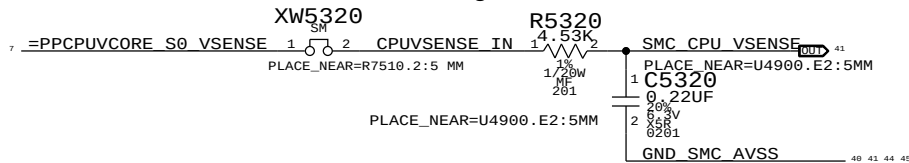
CPU SA Current Sense / Filter



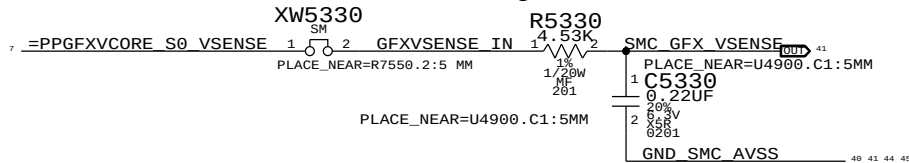
VCCSA Voltage Sense / Filter



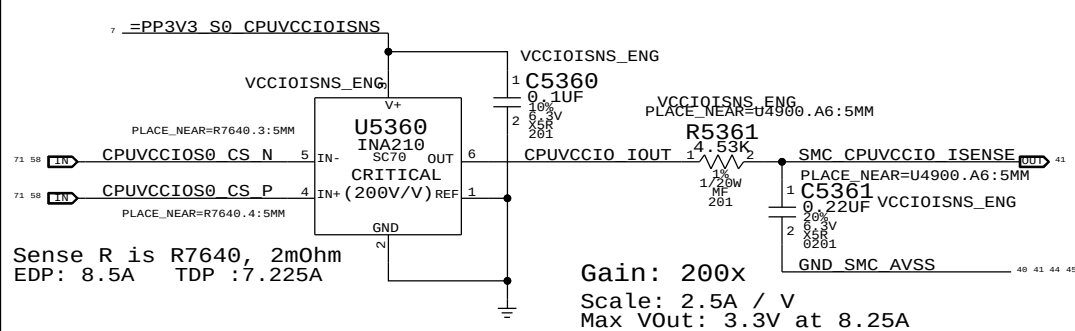
CPU Vcore Voltage Sense / Filter



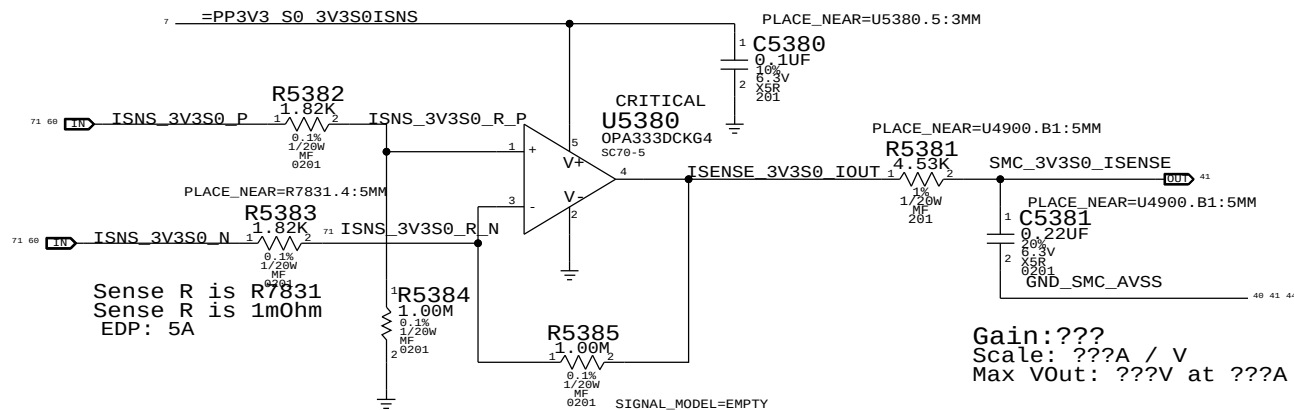
GFX/IG Vcore Voltage Sense / Filter



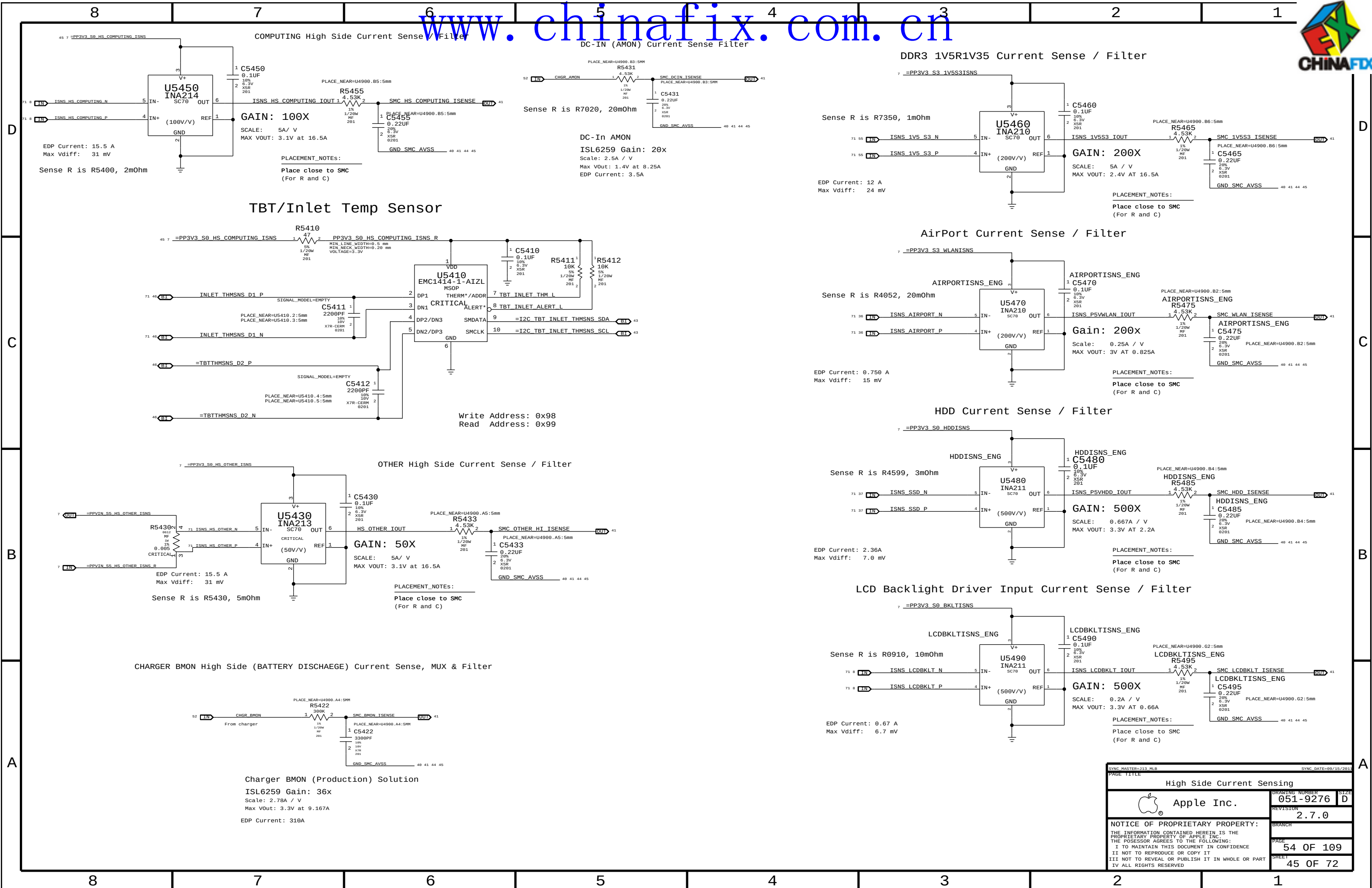
CPU 1.05V VCCIO Current Sense / Filter



3.3V S0 FET Current Sense / Filter

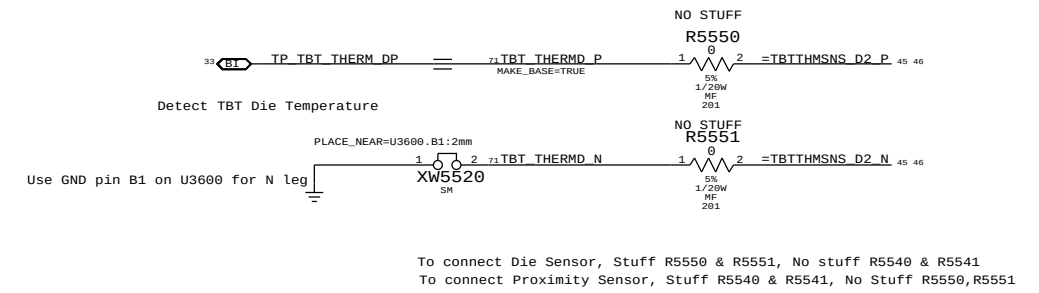


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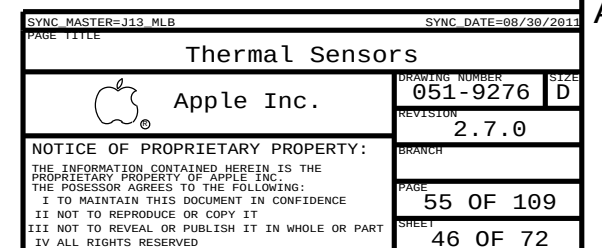
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TBT Die



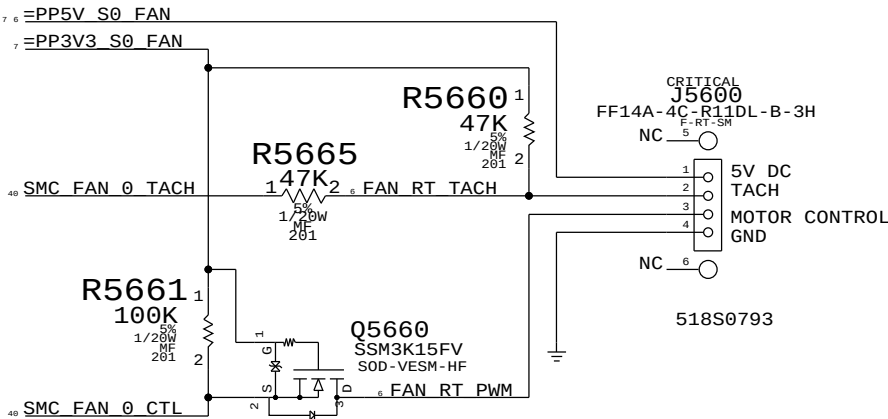
Replacing caps with 100K PD on ISENSE SMC inputs

Replacing caps with 100K PD on ISENSE SMC inputs



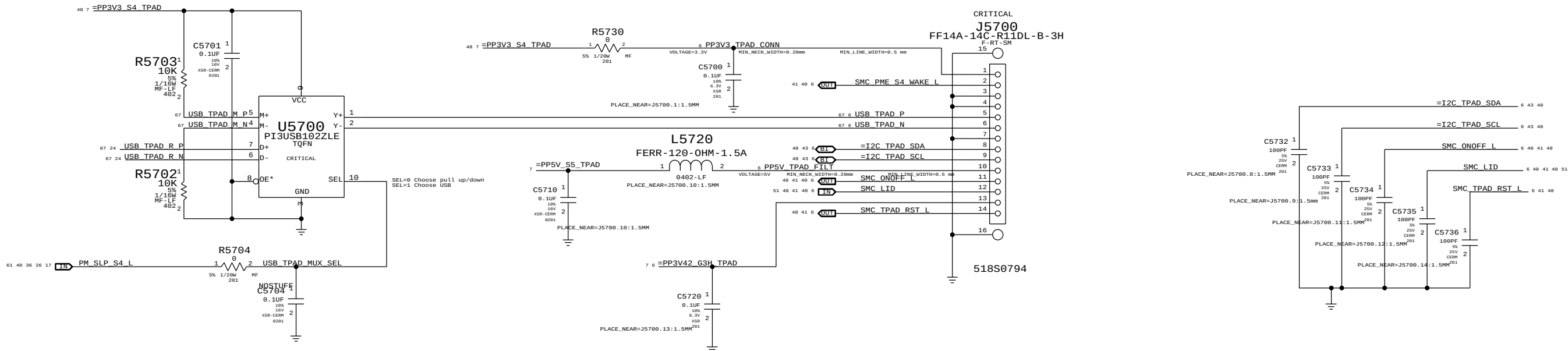


FAN CONNECTOR

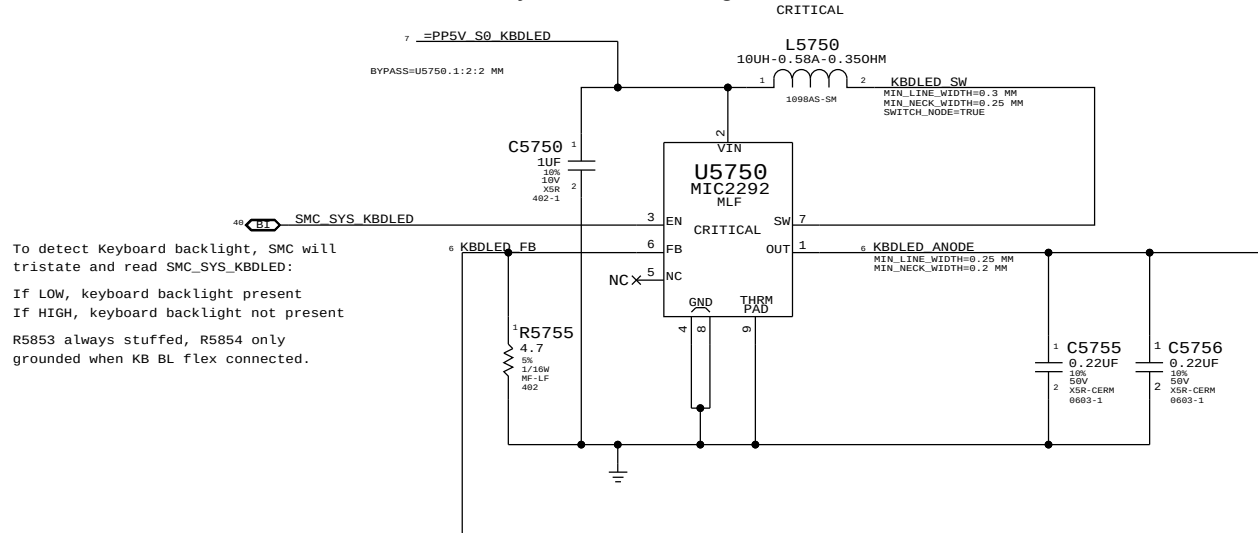


SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE		Fan	
		DRAWING NUMBER	051-9276
		REVISION	2.7.0
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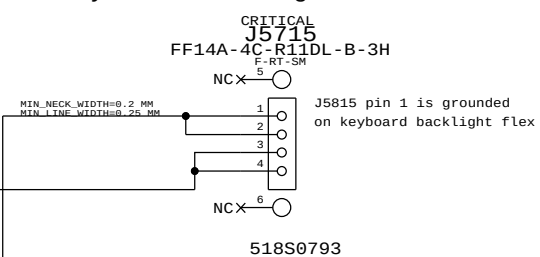
IPD Flex Connector



Keyboard Backlight Driver & Detection



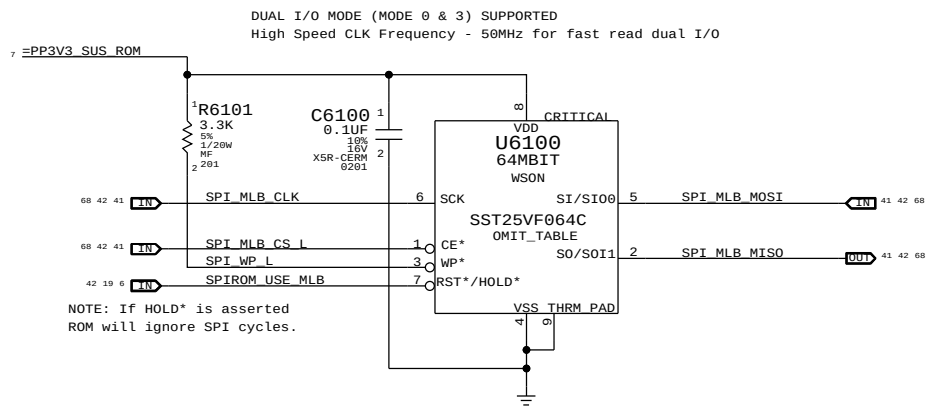
Keyboard Backlight Connector



SYNC MASTER=K21 MLB		SYNC DATE=12/13/2010	
PAGE TITLE			
IPD / KBD Backlight			
 Apple Inc.		DRAWING NUMBER	051-9276
		SIZE	D
		REVISION	2.7.0
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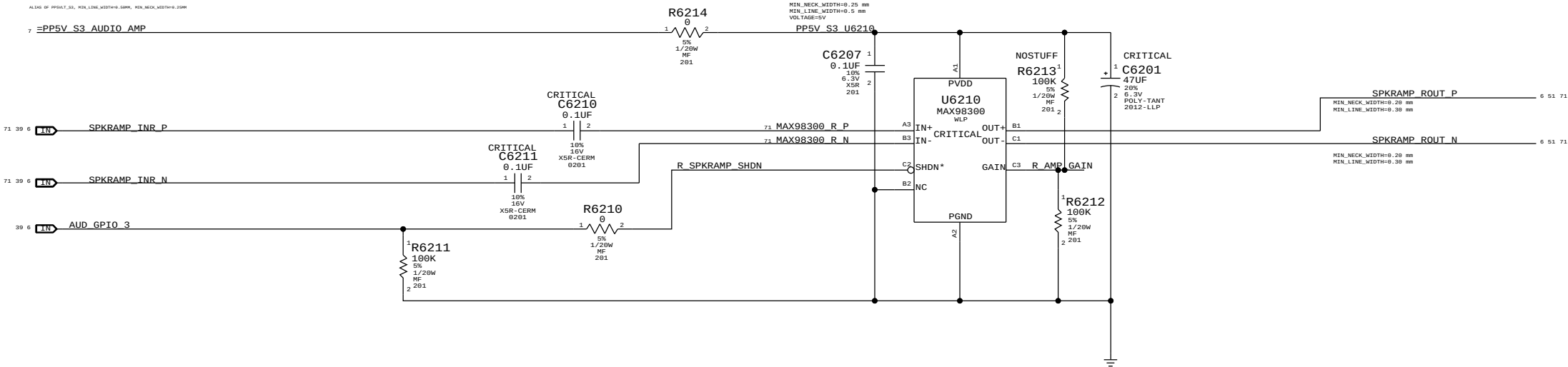
SYNC MASTER=J13_MLB		SYNC DATE=10/13/2011	
PAGE TITLE			
SPI ROM			
	DRAWING NUMBER		SIZE
	051-9276		D
Apple Inc.		REVISION	
		2.7.0	
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SPEAKER AMPLIFIERS

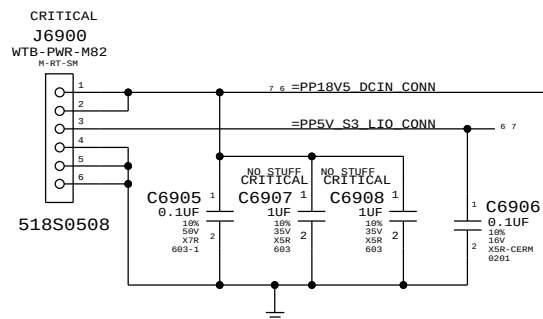
APN:353S2888

SPEAKER LOWPASS 80 HZ < FC < 132 HZ
GAIN 6DB



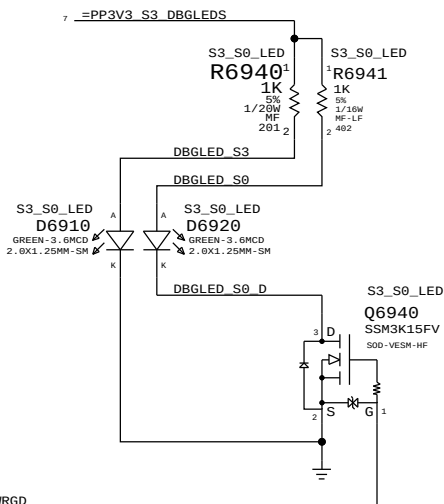


MLB to LIO Power Cable Connector



Debug LEDs

(For development only)



CRITICAL

Q6910

SI5419DU

POWERPAK

MPMS: YES

C6912

0.1uF

100V

XSR-CERM

0402

DCIN_ISOL_GATE

6.8V Zener

CRITICAL

D6912

GDZT2R6.8

GDZ-0201

MPMS: NO

R6911

0

5%

1/20W

MF-LF

201

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

CRITICAL

D6905

BAT30CWFILM

SOT-223

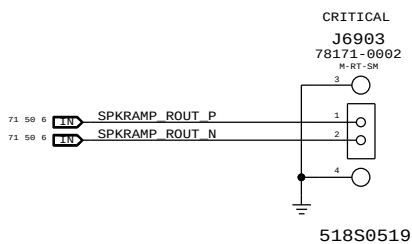
CRITICAL

D6905

BAT30CWFILM

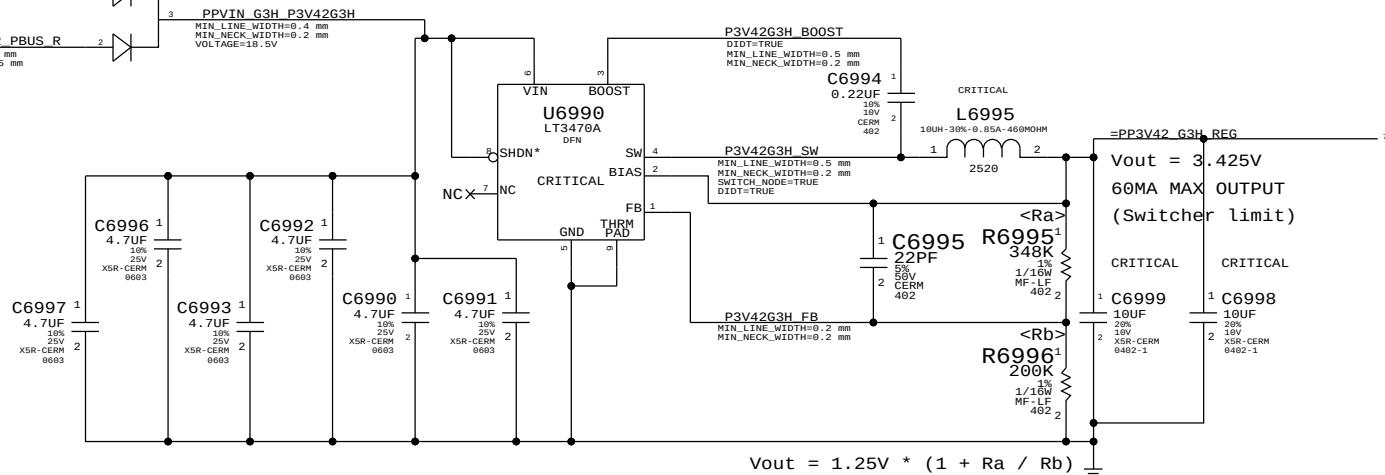
SOT-223

Right Speaker Connector



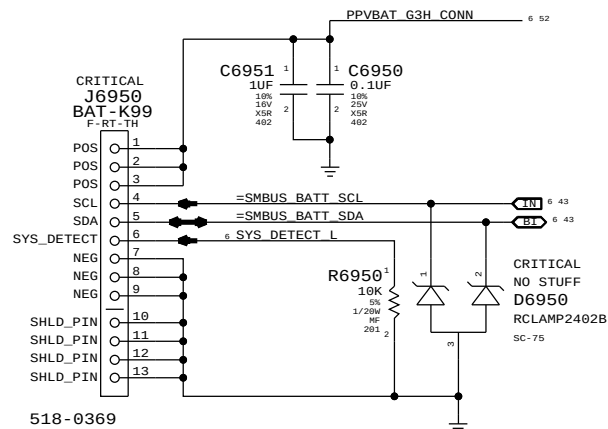
3.425V "G3Hot" Supply

Supply needs to guarantee 3.31V delivered to SMC Vref generator

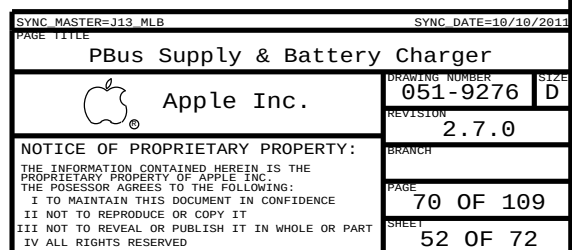
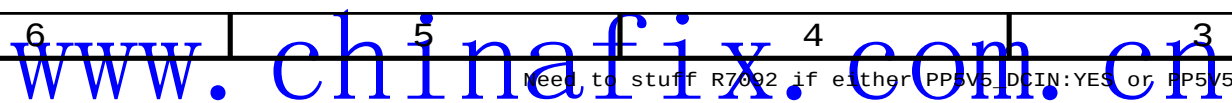


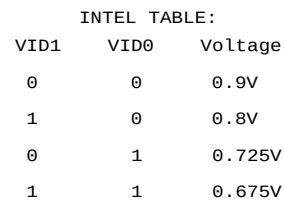
$$V_{out} = 1.25V * (1 + R_a / R_b)$$

K99-Specific Battery Connector



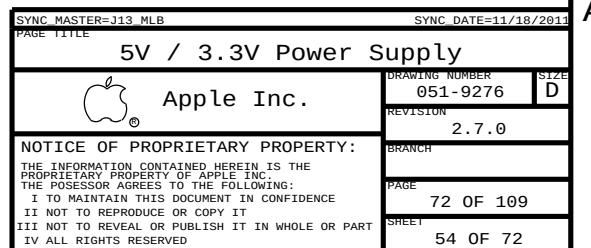
PAGE TITLE		PAGE NUMBER	
DC-In & Battery Connectors		051-9276	
Apple Inc.		2.7.0	
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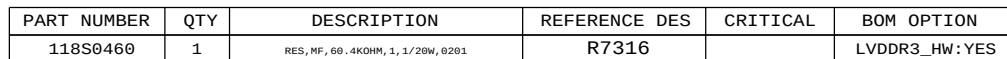




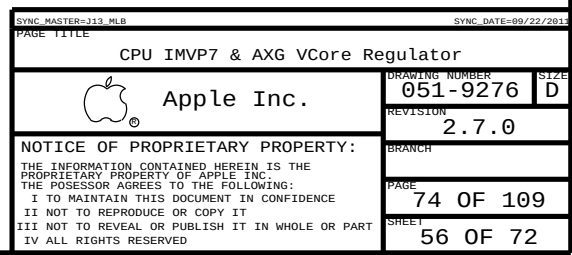
INTEL TABLE:		
VID1	VID0	Voltage
0	0	0.9V
1	0	0.8V
0	1	0.725V
1	1	0.675V

SYNC MASTER-J13 MLB		SYNC DATE=09/01/2013	
PAGE TITLE			
System Agent Supply			
 Apple Inc.		DRAWING NUMBER	051-9276
		SIZE	D
		REVISION	2.7.0
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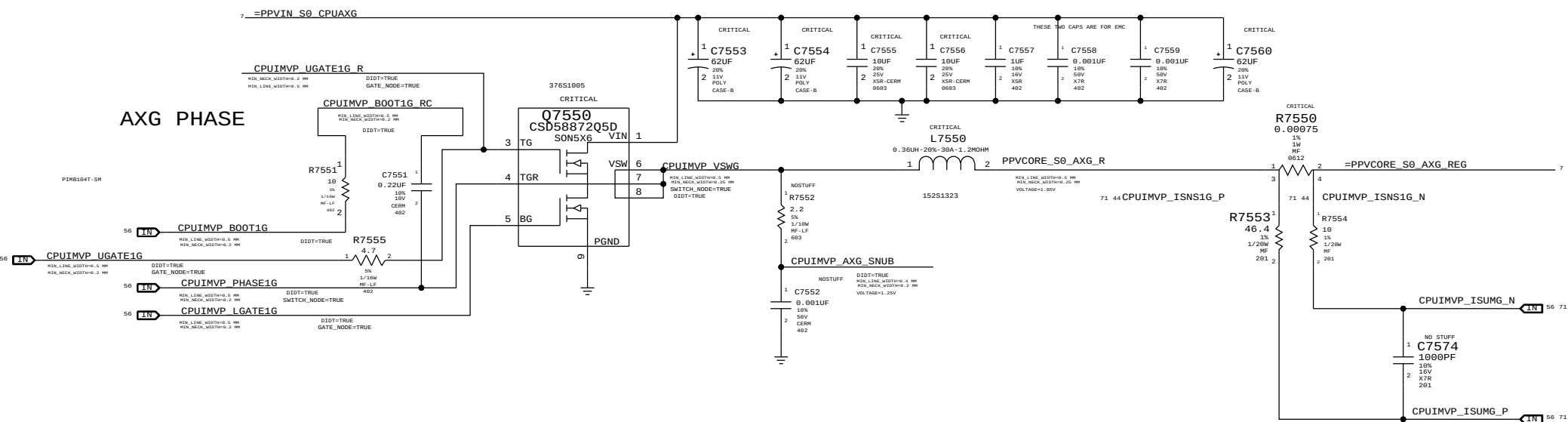
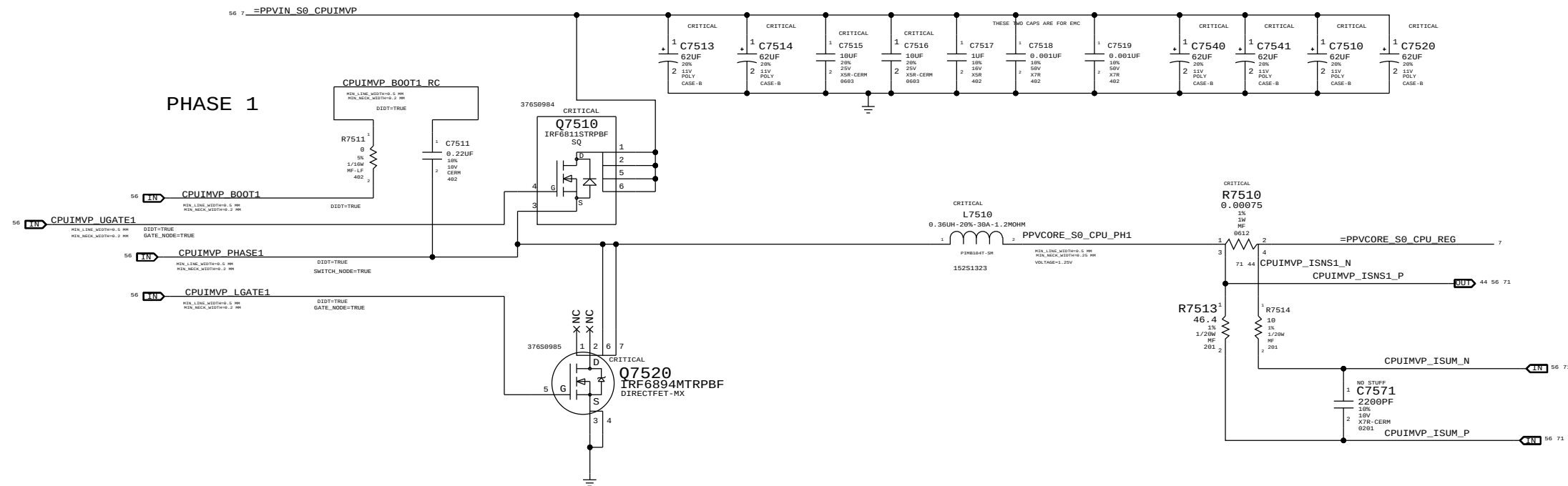




Vout = 1.5V
14.1A max output
(Q7335 limit)
f = 400 kHz



CPU=IV Bridge ULV, AXG=GT2



PAGE TITLE		DRAWING NUMBER		SIZE	
CPU IMVP7 & AXG VCore Output		051-9276		D	
Apple Inc.		REVISION		2.7.0	
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D

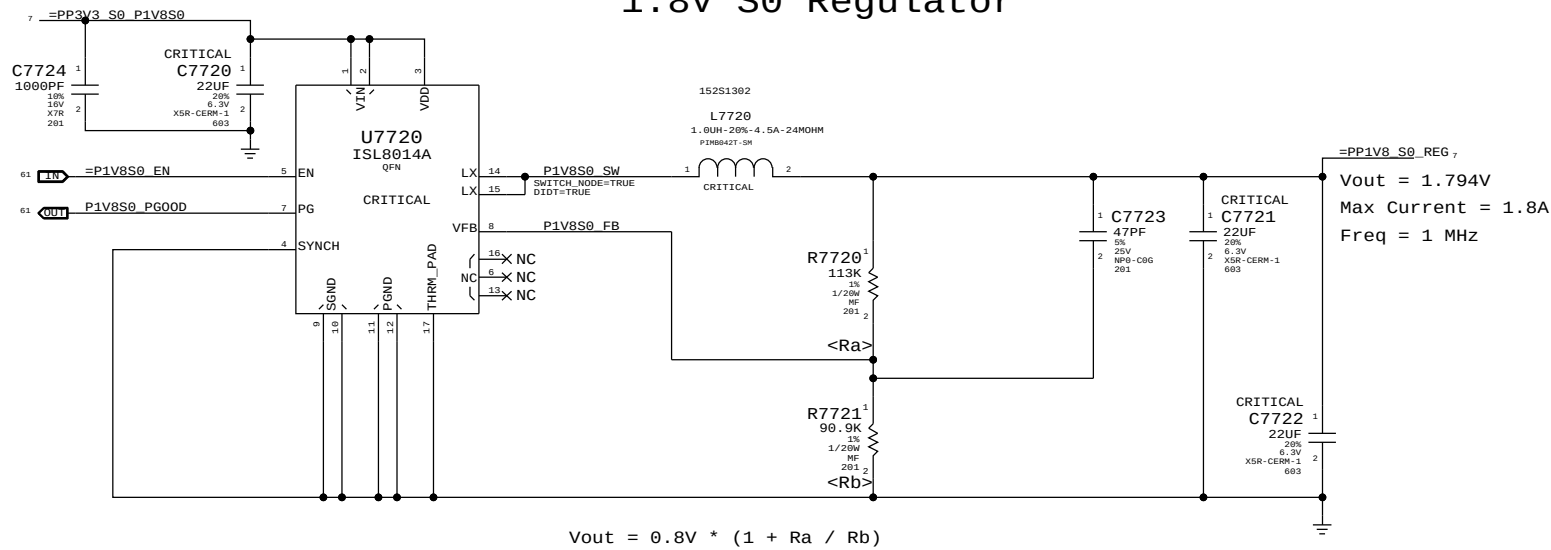


B

A

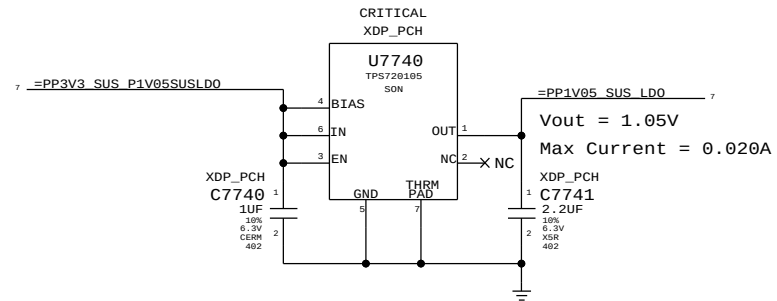
SYNCH MASTER-213 HL-B		SYNCH DATE=09/01/2013	
PAGE TITLE			
CPU VCCIO1 (1.05V) Power Supply			
	Apple Inc.	DRAWING NUMBER	SIZE
		051-9276	D
		REVISION	
		2.7.0	
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1.8V S0 Regulator

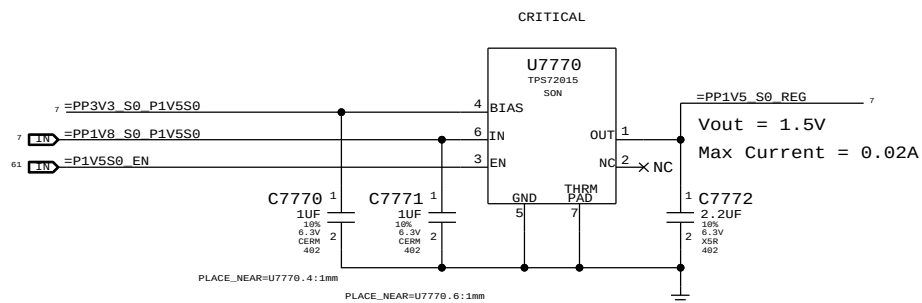


1.05V SUS LDO

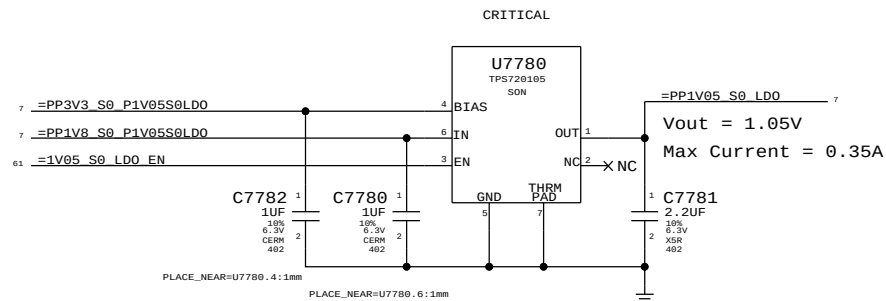
Cougar Point requires JTAG pull-ups to be powered at 1.05V when SUS suspend well is active. Pull-ups (3) must be 51 ohms to support XDP (not required in production). 70mA is required to support pull-ups. Alternative is strong voltage dividers (200/100) to 3.3V S5, which burns 100mW in all S-states.

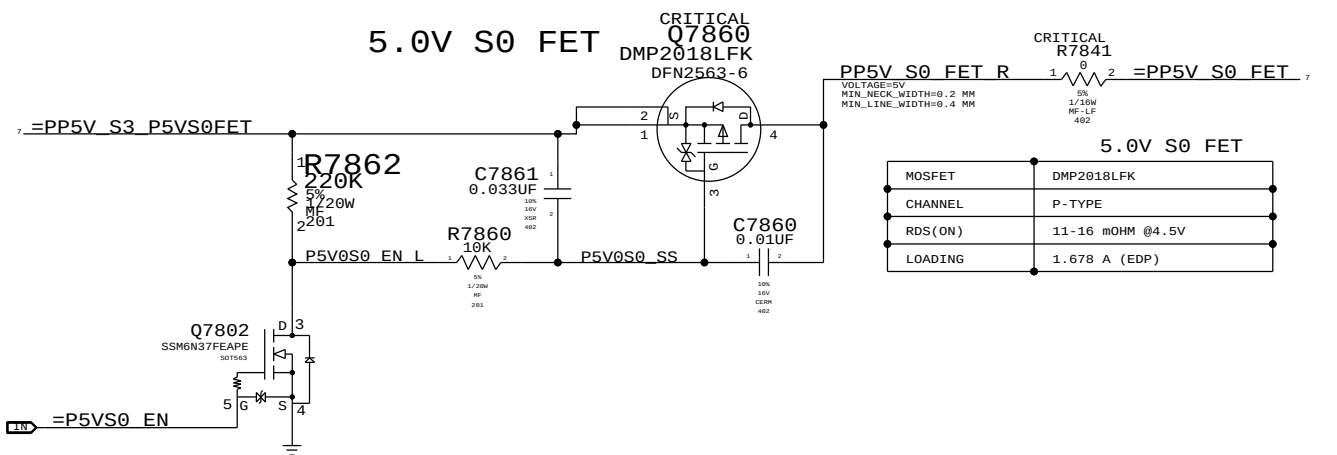
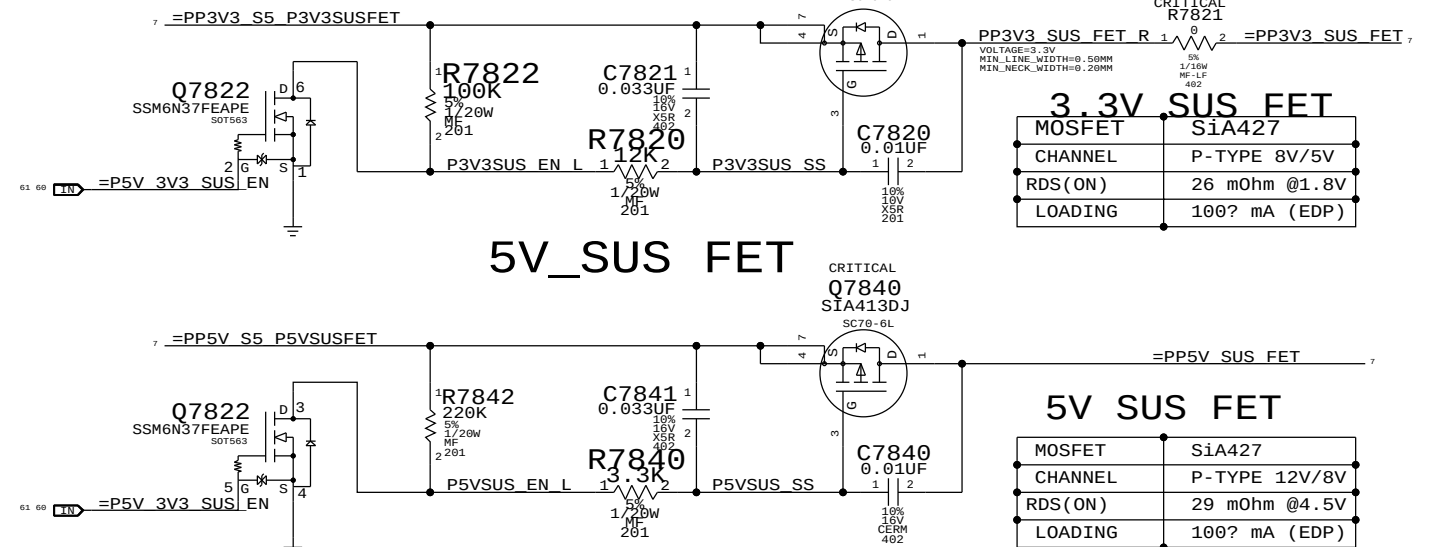
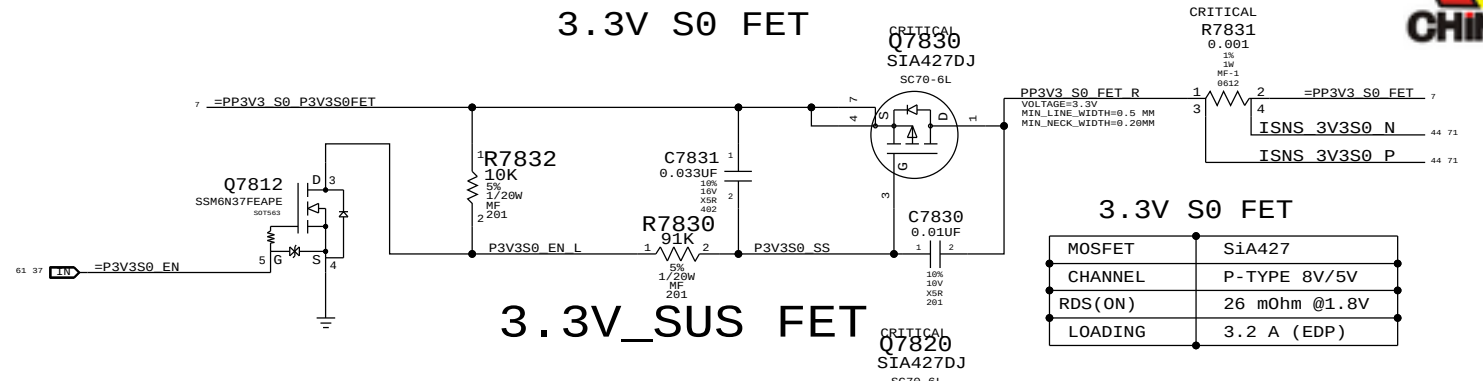


1.5V S0 LDO

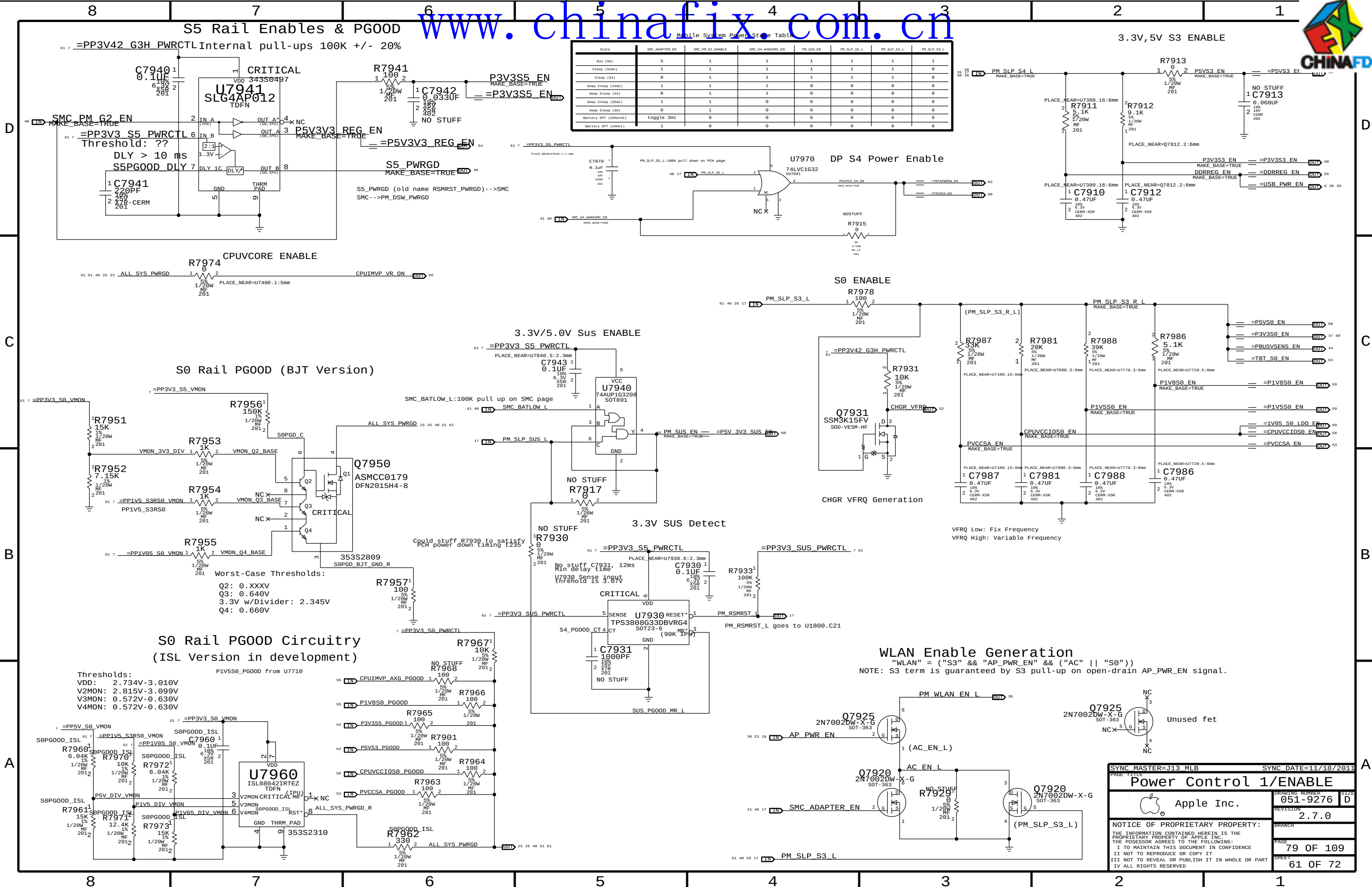


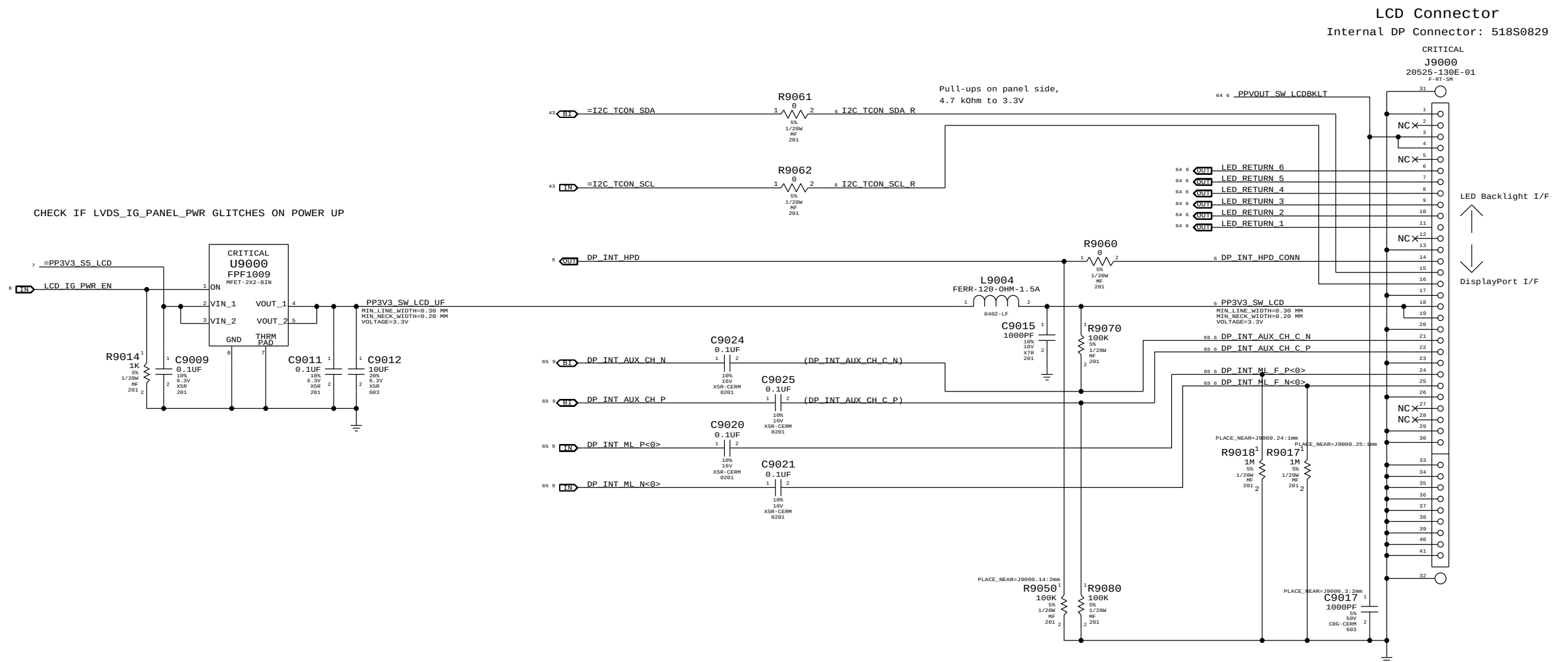
1.05V S0 LDO

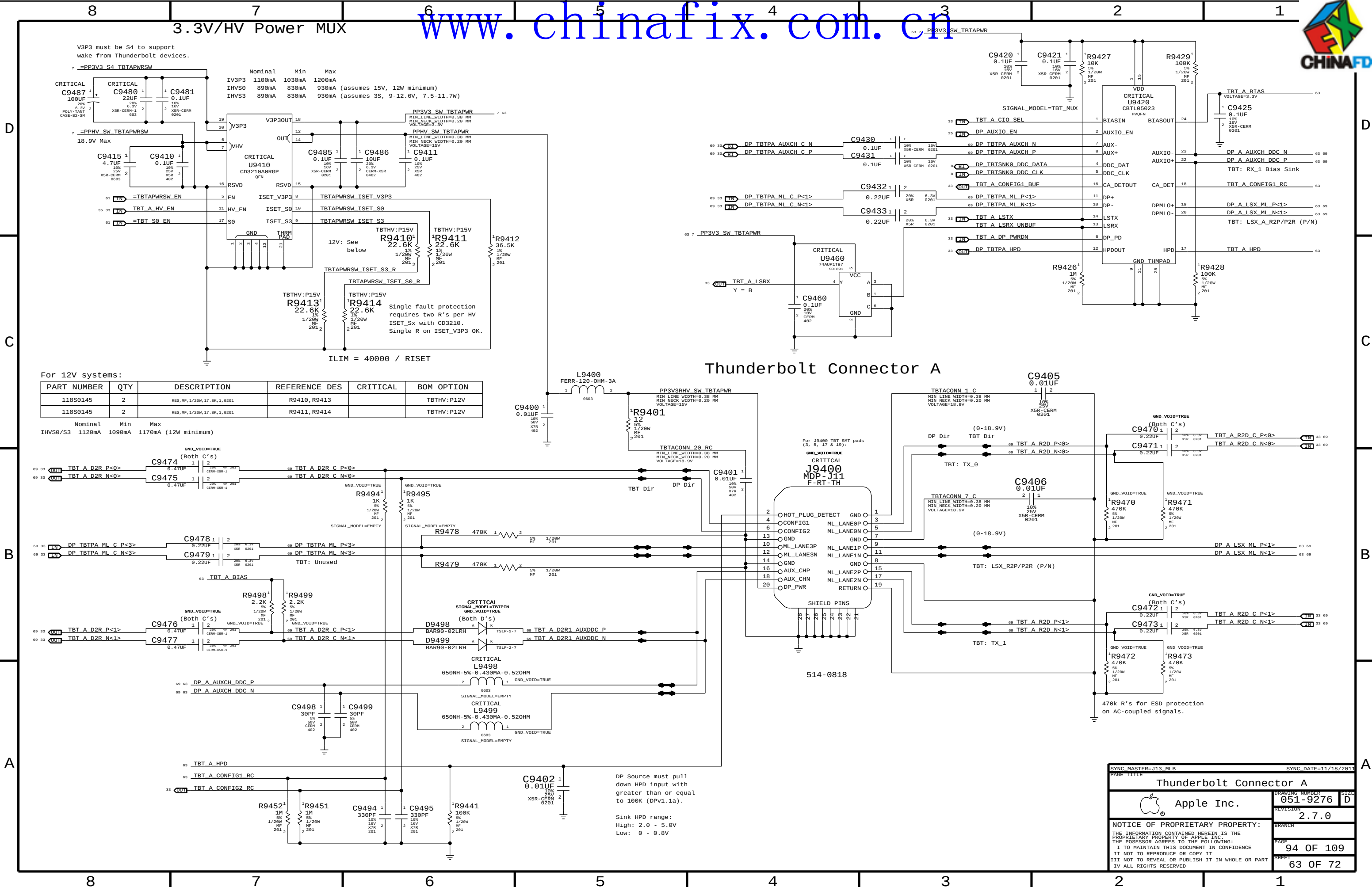




MOSFET	DMP2018LFK
CHANNEL	P - TYPE
RDS(ON)	11-16 mOHM @4.5V
LOADING	1.678 A (EDP)







SYNC MASTER=J13 MLB SYNC DATE=11/18/2011

PAGE TITLE

Thunderbolt Connector A

Apple Inc.

DRAWING NUMBER 051-9276 SIZE D

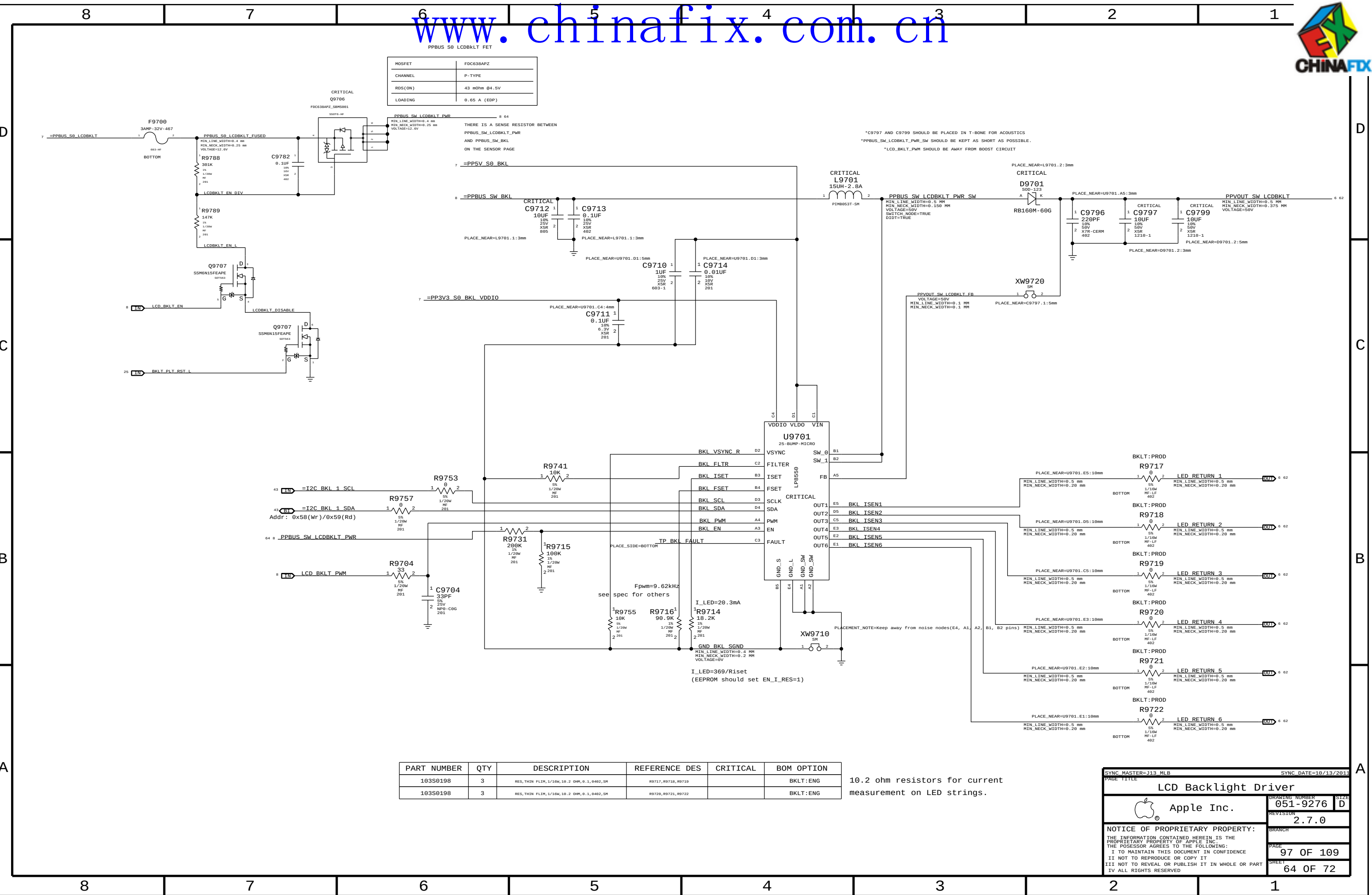
REVISION 2.7.0

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PART NUMBER	QTY	DESCRIPTION	REFERENCE DES	CRITICAL	BOM OPTION
103S0198	3	RES, THIN FLIM, 1/10W, 10.2 OHM, 0.1, 0402, SM	R9717, R9718, R9719		BKLT:ENG
103S0198	3	RES, THIN FLIM, 1/10W, 10.2 OHM, 0.1, 0402, SM	R9720, R9721, R9722		BKLT:ENG

10.2 ohm resistors for current measurement on LED strings.

SYNC MASTER=J13 MLB		SYNC DATE=10/13/2011	
PAGE TITLE		LCD Backlight Driver	
Apple Inc.		DRAWING NUMBER	051-9276
		REVISION	2.7.0
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CPU Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CPU_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CPU_27P4S	*	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	=27P4_OHM_SE	0.100 MM	0.100 MM

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_AGTL	TOP, BOTTOM	=2x_DIELECTRIC	?
CPU_AGTL	*	=STANDARD	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_8MIL	*	*	CPU_8MIL_2ANY

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_ITP	*	*	CPU_ITP_2ANY

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_COMP	CPU_COMP	*	CPU_COMP_2SELF
CPU_COMP	*	*	CPU_COMP_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_COMP_2SELF	TOP, BOTTOM	=6x_DIELECTRIC	?
CPU_COMP_2OTHER	TOP, BOTTOM	=10x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CPU_VCCSENSE	CPU_VCCSENSE	*	CPU_VCCSENSE_2SELF
CPU_VCCSENSE	*	*	CPU_VCCSENSE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CPU_VCCSENSE_2SELF	*	=4x_DIELECTRIC	?
CPU_VCCSENSE_2OTHER	*	=6x_DIELECTRIC	?

PCI-Express Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF
CLK_PCIE_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

PCIe Clock Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
CLK_PCIE	CLK_PCIE	*	CLK_PCIE_2SELF
CLK_PCIE	*	*	CLK_PCIE_2OTHER

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_PCIE_2SELF	*	=4x_DIELECTRIC	?
CLK_PCIE_2OTHER	*	=6x_DIELECTRIC	?

CPU PCIe Spacing

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_CPU_TX	PCIE_CPU_TX	*	PCIE_TX2TX
PCIE_CPU_RX	PCIE_CPU_RX	*	PCIE_RX2RX
PCIE_CPU_TX	*_CPU_TX	*	PCIE_TX20THERTX
PCIE_CPU_RX	*_CPU_RX	*	PCIE_RX20THERRX
PCIE_CPU_TX	*_CPU_RX	*	PCIE_TX2RX
PCIE_CPU_RX	*_CPU_TX	*	PCIE_RX2TX
PCIE_CPU_TX	*_TX	*	PCIE_20THERHS
PCIE_CPU_RX	*_TX	*	PCIE_20THERHS
PCIE_CPU_TX	*_RX	*	PCIE_20THERHS
PCIE_CPU_RX	*_RX	*	PCIE_20THERHS
PCIE_CPU_TX	*	*	PCIE_20THER
PCIE_CPU_RX	*	*	PCIE_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_PCH_TX	PCIE_PCH_TX	*	PCIE_TX2TX
PCIE_PCH_RX	PCIE_PCH_RX	*	PCIE_RX2RX
PCIE_PCH_TX	*_PCH_TX	*	PCIE_TX20THERTX
PCIE_PCH_RX	*_PCH_RX	*	PCIE_RX20THERRX
PCIE_PCH_TX	*_PCH_RX	*	PCIE_TX2RX
PCIE_PCH_RX	*_PCH_TX	*	PCIE_RX2TX
PCIE_PCH_TX	*_TX	*	PCIE_20THERHS
PCIE_PCH_RX	*_TX	*	PCIE_20THERHS
PCIE_PCH_TX	*_RX	*	PCIE_20THERHS
PCIE_PCH_RX	*_RX	*	PCIE_20THERHS
PCIE_PCH_TX	*	*	PCIE_20THER
PCIE_PCH_RX	*	*	PCIE_20THER

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
PCIE_PCH_TX	PCIE_PCH_TX	*	PCIE_TX2TX
PCIE_PCH_RX	PCIE_PCH_RX	*	PCIE_RX2RX
PCIE_PCH_TX	*_PCH_TX	*	PCIE_TX20THERTX
PCIE_PCH_RX	*_PCH_RX	*	PCIE_RX20THERRX
PCIE_PCH_TX	*_PCH_RX	*	PCIE_TX2RX
PCIE_PCH_RX	*_PCH_TX	*	PCIE_RX2TX
PCIE_PCH_TX	*_TX	*	PCIE_20THERHS
PCIE_PCH_RX	*_TX	*	PCIE_20THERHS
PCIE_PCH_TX	*_RX	*	PCIE_20THERHS
PCIE_PCH_RX	*_RX	*	PCIE_20THERHS
PCIE_PCH_TX	*	*	PCIE_20THER
PCIE_PCH_RX	*	*	PCIE_20THER

SOURCE: 471984_Chief_River_MS_PD6_1.0 and the spacing rule is adjusted per SI team feedback.

CPU Net Properties

NET_TYPE			
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING	
DMI_S2N	PCIE_80D	PCIE_PCH_TX	DMI S2N P<3:0> 9 17
DMI_S2N	PCIE_80D	PCIE_PCH_TX	DMI S2N N<3:0> 9 17
DMI_N2S	PCIE_80D	PCIE_PCH_RX	DMI N2S P<3:0> 9 17
DMI_N2S	PCIE_80D	PCIE_PCH_RX	DMI N2S N<3:0> 9 17
FDI_DATA	PCIE_80D	PCIE_PCH_RX	FDI DATA P<7:0> 9 17
FDI_DATA	PCIE_80D	PCIE_PCH_RX	FDI DATA N<7:0> 9 17
CPU_45S	CPU_AGTL		FDI FSYNC<1..0> 9 17
CPU_45S	CPU_AGTL		FDI LSYNC<1..0> 9 17
CPU_45S	CPU_AGTL		FDI_INT 9 17
CPU_PECT	CPU_45S	CPU_COMP	CPU PECT 10 19 41
PM_SYNC	CPU_45S	CPU_AGTL	PM SYNC 10 17
PM_MEM_PWRGD	CPU_45S	CPU_AGTL	PM MEM_PWRGD 10 17 26
CPU_45S	CPU_ITP		XDP DBRESET_L 10 23 25
CPU_45S	CPU_ITP		XDP_CPU_PRDY_L 10 23
CPU_45S	CPU_ITP		XDP_CPU_PREQ_L 10 23
CPU_27P4S	CPU_COMP		EDP_COMP 9
CPU_27P4S	CPU_COMP		CPU_PEG_COMP 9
CPU_SM_RCOMP	CPU_COMP		CPU_SM_RCOMP<0> 10
CPU_SM_RCOMP	CPU_COMP		CPU_SM_RCOMP<1> 10
CPU_SM_RCOMP	CPU_COMP		CPU_SM_RCOMP<2> 10
CPU_45S	CPU_ITP		CPU_CFG<11..0> 9 23
CPU_CATERR_L	CPU_45S	CPU_AGTL	CPU_CATERR_L 10 40
CPU_45S	CPU_AGTL		CPU_VCCIO_SEL 12
CPU_PROCHOT_I	CPU_45S	CPU_AGTL	CPU_PROCHOT_L 10 40 41 56
CPU_PWRGD	CPU_45S	CPU_AGTL	CPU_PWRGD 10 19 23
PM_THRMTRIP_I	CPU_45S	CPU_BMTI	PM_THRMTRIP_L 10 19 41
DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE	DMI_CLK100M_CPU_P 10 16
DMI_CLK100M	CLK_PCIE_80D	CLK_PCIE	DMI_CLK100M_CPU_N 10 16
DP11_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE	DP11_REF_CLKP 8 10
DP11_REF_CLK120M	CLK_PCIE_80D	CLK_PCIE	DP11_REF_CLKN 8 10
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPCPU_CLK100M_P 10 16
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPCPU_CLK100M_N 10 16
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPXDPCPU_CLK100M_P 10 23
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	ITPXDPCPU_CLK100M_N 10 23
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	XDP_CPU_CLK100M_P 23
ITPCPU_CLK100M	CLK_PCIE_80D	CLK_PCIE	XDP_CPU_CLK100M_N 23
XDP_TDI	CPU_45S	CPU_ITP	XDP_CPU_TDI 10 23
XDP_TDO	CPU_45S	CPU_ITP	XDP_CPU_TDO 10 23
XDP_TMS	CPU_45S	CPU_ITP	XDP_CPU_TMS 10 23
XDP_TCK	CPU_45S	CPU_ITP	XDP_CPU_TCK 10 23
XDP_TRST_L	CPU_45S	CPU_ITP	XDP_CPU_TRST_L 10 23
XDP_BPM_I	CPU_45S	CPU_ITP	XDP_BPM_L<3..0> 10 23
XDP_BPM_I_R_CFG	CPU_45S	CPU_ITP	XDP_BPM_L<7..4> 10 23
(XDP_BPM_I_R_CFG)	CPU_45S	CPU_ITP	XDP_OBSDATA_B<3..0> 9 23
(XDP_BPM_I_R_CFG)	CPU_45S	CPU_ITP	CPU_CFG<15..12> 9 23
(FSB_CPURST_I)	CPU_45S	CPU_ITP	XDP_CPURST_L 23
CPU_VCCSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE	CPU_VCCSENSE_P 12 56
CPU_VCCSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE	CPU_VCCSENSE_N 12 56
CPU_VCCIOSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE	CPU_VCCIOSENSE_P 12 58
CPU_VCCIOSENSE	SENSE_1T01_P2MM	CPU_VCCSENSE	CPU_VCCIOSENSE_N 12 58
CPU_AXG_SENSE	SENSE_1T01_P2MM	CPU_VCCSENSE	CPU_AXG_SENSE_P 12 56
CPU_AXG_SENSE	SENSE_1T01_P2MM	CPU_VCCSENSE	CPU_AXG_SENSE_N 12 56
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDDQ_SENSE_P 12
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VDDQ_SENSE_N 12
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_P 9
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_AXG_VALSENSE_N 9
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_P 9
CPU_VALSENSE	CPU_27P4S	CPU_VCCSENSE	CPU_VCC_VALSENSE_N 9
CPU_SVIDALERT_I	CPU_45S	CPU_COMP	CPU_VIDALERT_L 12 56
CPU_SVIDCLK	CPU_45S	CPU_COMP	CPU_VIDCLK 12 56
CPU_SVIDSOUT	CPU_45S	CPU_COMP	CPU_VIDSOUT 12 56
PCIE_CPU_MUX_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_C_P<0> 6 37
PCIE_CPU_MUX_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_C_N<0> 6 37
PCIE_CPU_MUX_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_MUX_IN_P 37
PCIE_CPU_MUX_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_MUX_IN_N 37
PCIE_CPU_MUX_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_P<0> 6 37
PCIE_CPU_MUX_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_N<0> 6 37
PCIE_CPU_MUX_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_MUX_OUT_P 37
PCIE_CPU_MUX_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_MUX_OUT_N 37
PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_R2D_C_P<1> 6 37
PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_R2D_C_N<1> 6 37
PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_P<1> 6 37
PCIE_CPU_SSD_R2D	PCIE_80D	PCIE_CPU_TX	PCIE_SSD_R2D_N<1> 6 37
PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_P<1> 6 37
PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_N<1> 6 37
PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_C_P<1> 6 37
PCIE_CPU_SSD_D2R	PCIE_80D	PCIE_CPU_RX	PCIE_SSD_D2R_C_N<1> 6 37
PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_SSD_P 6 16 37
PCIE_CLK100M_SSD	CLK_PCIE_80D	CLK_PCIE	PCIE_CLK100M_SSD_N 6 16 37
DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_P<3..0> 9 62
DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_N<3..0> 9 62
DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_F_P<3..0> 6 62
DP_INT_ML	DP_80D	DP_TX	DP_INT_ML_F_N<3..0> 6 62
DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_C_P 6 62
DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_C_N 6 62
DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_P 6 62
DP_INT_AUXCH	DP_80D	DP_AUX	DP_INT_AUX_CH_N 6 62

Note: DisplayPort tables are on Page 103

DMI/FDI

PCIe SSD

DP

SYNC MASTER=CONSTRAINTS		SYNC DATE=01/11/2012			
PAGE TITLE					
CPU Constraints					
Apple Inc.		DRAWING NUMBER	051-9276		
		SIZE	D		
		REVISION	2.7.0		
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Memory Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE			
		PHYSICAL	SPACING		
	MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK P<5..0>	8 11 27 28 32
	MEM_A_CLK	MEM_72D	MEM_CLK	MEM A CLK N<5..0>	8 11 27 28 32
	MEM_A_CTRI	MEM_45S	MEM_CTRI	MEM A CKE<3..0>	11 27 28 32
	MEM_A_CTRI	MEM_45S	MEM_CTRI	MEM A CS L<3..0>	11 27 28 32
	MEM_A_CTRI	MEM_45S	MEM_CTRI	MEM A ODT<3..0>	11 27 28 32
	MEM_A_CMD	MEM_45S	MEM_CMD	MEM A A<15..0>	11 27 28 32
	MEM_A_CMD	MEM_45S	MEM_CMD	MEM A BA<2..0>	11 27 28 32
	MEM_A_CMD	MEM_45S	MEM_CMD	MEM A RAS L	11 27 28 32
	MEM_A_CMD	MEM_45S	MEM_CMD	MEM A CAS L	11 27 28 32
	MEM_A_CMD	MEM_45S	MEM_CMD	MEM A WE L	11 27 28 32
	MEM_A_DQ_RYTF0	MEM_45S	MEM_A_DATA_0	MEM A DQ<7..0>	11 27
	MEM_A_DQ_RYTF1	MEM_45S	MEM_A_DATA_1	MEM A DQ<15..8>	11 27
	MEM_A_DQ_RYTF2	MEM_45S	MEM_A_DATA_2	MEM A DQ<23..16>	11 27
	MEM_A_DQ_RYTF3	MEM_45S	MEM_A_DATA_3	MEM A DQ<31..24>	11 27
	MEM_A_DQ_RYTF4	MEM_45S	MEM_A_DATA_4	MEM A DQ<39..32>	11 28
	MEM_A_DQ_RYTF5	MEM_45S	MEM_A_DATA_5	MEM A DQ<47..40>	11 28
	MEM_A_DQ_RYTF6	MEM_45S	MEM_A_DATA_6	MEM A DQ<55..48>	11 28
	MEM_A_DQ_RYTF7	MEM_45S	MEM_A_DATA_7	MEM A DQ<63..56>	11 28
	MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM A DQS P<0>	11 27
	MEM_A_DQS0	MEM_80D	MEM_A_DQS_0	MEM A DQS N<0>	11 27
	MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM A DQS P<1>	11 27
	MEM_A_DQS1	MEM_80D	MEM_A_DQS_1	MEM A DQS N<1>	11 27
	MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM A DQS P<2>	11 27
	MEM_A_DQS2	MEM_80D	MEM_A_DQS_2	MEM A DQS N<2>	11 27
	MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM A DQS P<3>	11 27
	MEM_A_DQS3	MEM_80D	MEM_A_DQS_3	MEM A DQS N<3>	11 27
	MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM A DQS P<4>	11 28
	MEM_A_DQS4	MEM_80D	MEM_A_DQS_4	MEM A DQS N<4>	11 28
	MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM A DQS P<5>	11 28
	MEM_A_DQS5	MEM_80D	MEM_A_DQS_5	MEM A DQS N<5>	11 28
	MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM A DQS P<6>	11 28
	MEM_A_DQS6	MEM_80D	MEM_A_DQS_6	MEM A DQS N<6>	11 28
	MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM A DQS P<7>	11 28
	MEM_A_DQS7	MEM_80D	MEM_A_DQS_7	MEM A DQS N<7>	11 28

PalPilot Spacing	"Real" Spacing
1	1
2	2
3	3
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99	99
100	100

=2x_DIELECTRIC	=2x_DIELECTRIC
=5.7x_DIELECTRIC	=3x_DIELECTRIC
=4x_DIELECTRIC	=3x_DIELECTRIC
=4x_DIELECTRIC	=3x_DIELECTRIC
=4x_DIELECTRIC	=3x_DIELECTRIC
=8.6x_DIELECTRIC	=6x_DIELECTRIC
=5.7x_DIELECTRIC	=4x_DIELECTRIC
=PWR_P2MM	=PWR_P2MM
=GND_P2MM	=GND_P2MM
=8.6x_DIELECTRIC	=6x_DIELECTRIC

ents

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
MEM_A_DQS_0	*	*	MEM_20THER
MEM_A_DQS_1	*	*	MEM_20THER
MEM_A_DQS_2	*	*	MEM_20THER
MEM_A_DQS_3	*	*	MEM_20THER
MEM_A_DQS_4	*	*	MEM_20THER
MEM_A_DQS_5	*	*	MEM_20THER
MEM_A_DQS_6	*	*	MEM_20THER
MEM_A_DQS_7	*	*	MEM_20THER
MEM_B_DQS_0	*	*	MEM_20THER
MEM_B_DQS_1	*	*	MEM_20THER
MEM_B_DQS_2	*	*	MEM_20THER
MEM_B_DQS_3	*	*	MEM_20THER
MEM_B_DQS_4	*	*	MEM_20THER
MEM_B_DQS_5	*	*	MEM_20THER
MEM_B_DQS_6	*	*	MEM_20THER
MEM_B_DQS_7	*	*	MEM_20THER

MEM_A_DATA_0			MEM_20THER
MEM_A_DATA_1	*	*	MEM_20THER
MEM_A_DATA_2	*	*	MEM_20THER

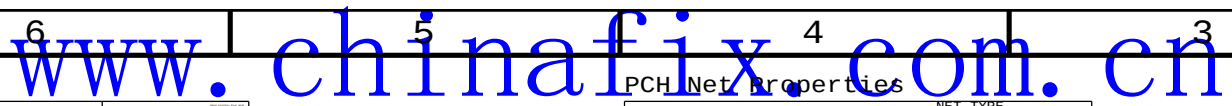
MEM_A_DATA_3	*	*	MEM_20THER
MEM_A_DATA_4	*	*	MEM_20THER
MEM_A_DATA_5	*	*	MEM_20THER
MEM_A_DATA_6	*	*	MEM_20THER

MEM_B_DATA_0	*	*	MEM_20THER
MEM_B_DATA_1	*	*	MEM_20THER

MEM_B_DATA_2			MEM_20THER
MEM_B_DATA_3	*	*	MEM_20THER
MEM_B_DATA_4	*	*	MEM_20THER

	MEM_B_C1K	MEM_72D	MEM_C1K	MEM_B CLK P<5..0>	8 11 29 38 32
	MEM_B_C1K	MEM_72D	MEM_C1K	MEM_B CLK N<5..0>	8 11 29 38 32
	MEM_B_CTRI	MEM_45S	MEM_CTRI	MEM_B CKE<3..0>	11 29 38 32
	MEM_B_CTRI	MEM_45S	MEM_CTRI	MEM_B CS L<3..0>	11 29 38 32
	MEM_B_CTRI	MEM_45S	MEM_CTRI	MEM_B ODT<3..0>	11 29 38 32
	MEM_B_CMD	MEM_45S	MEM_CMD	MEM_B A<15..0>	11 29 38 32
	MEM_B_CMD	MEM_45S	MEM_CMD	MEM_B BA<2..0>	11 29 38 32
	MEM_B_CMD	MEM_45S	MEM_CMD	MEM_B RAS L	11 29 38 32
	MEM_B_CMD	MEM_45S	MEM_CMD	MEM_B CAS L	11 29 38 32
	MEM_B_CMD	MEM_45S	MEM_CMD	MEM_B WE L	11 29 38 32
	MEM_B_DQ_RYTE0	MEM_45S	MEM_B_DATA_0	MEM_B DQ<7..0>	11 29
	MEM_B_DQ_RYTE1	MEM_45S	MEM_B_DATA_1	MEM_B DQ<15..8>	11 29
	MEM_B_DQ_RYTE2	MEM_45S	MEM_B_DATA_2	MEM_B DQ<23..16>	11 29
	MEM_B_DQ_RYTE3	MEM_45S	MEM_B_DATA_3	MEM_B DQ<31..24>	11 29
	MEM_B_DQ_RYTE4	MEM_45S	MEM_B_DATA_4	MEM_B DQ<39..32>	11 30
	MEM_B_DQ_RYTE5	MEM_45S	MEM_B_DATA_5	MEM_B DQ<47..40>	11 30
	MEM_B_DQ_RYTE6	MEM_45S	MEM_B_DATA_6	MEM_B DQ<55..48>	11 30
	MEM_B_DQ_RYTE7	MEM_45S	MEM_B_DATA_7	MEM_B DQ<63..56>	11 30
	MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM_B DQS P<0>	11 29
	MEM_B_DQS0	MEM_80D	MEM_B_DQS_0	MEM_B DQS N<0>	11 29
	MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM_B DQS P<1>	11 29
	MEM_B_DQS1	MEM_80D	MEM_B_DQS_1	MEM_B DQS N<1>	11 29
	MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM_B DQS P<2>	11 29
	MEM_B_DQS2	MEM_80D	MEM_B_DQS_2	MEM_B DQS N<2>	11 29
	MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM_B DQS P<3>	11 29
	MEM_B_DQS3	MEM_80D	MEM_B_DQS_3	MEM_B DQS N<3>	11 29
	MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM_B DQS P<4>	11 30
	MEM_B_DQS4	MEM_80D	MEM_B_DQS_4	MEM_B DQS N<4>	11 30
	MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM_B DQS P<5>	11 30
	MEM_B_DQS5	MEM_80D	MEM_B_DQS_5	MEM_B DQS N<5>	11 30
	MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM_B DQS P<6>	11 30
	MEM_B_DQS6	MEM_80D	MEM_B_DQS_6	MEM_B DQS N<6>	11 30
	MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM_B DQS P<7>	11 30
	MEM_B_DQS7	MEM_80D	MEM_B_DQS_7	MEM_B DQS N<7>	11 30
			MEM_PWR	PP1V5 S3RS0	6 7
			MEM_PWR	PP1V5 S3	6 7
			MEM_PWR	PP0V75 S3 MEM VREFCA A	27 28 31
			MEM_PWR	PP0V75 S3 MEM VREFDQ A	27 28 31

SYMC MASTER-CONSTRAINTS		SYMC DATE=01/11/2012	
PAGE TITLE			
Memory Constraints			
	Apple Inc.	DRAWING NUMBER	051-9276
		SIZE	D
	REVISION	2.7.0	
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PCH Net Properties

NET_TYPE		
ELECTRICAL_CONSTRAINT_SET	PHYSICAL	SPACING
SATA_PCH_MIX_R2D	SATA_R8D	SATA3_PCH_TY

SATA HDD R2D C P

SATA PCH MUX R2D	SATA 80D	SATA3 PCH TX	SATA SSD R2D MUX IN P	16 37
	SATA 80D	SATA3 PCH TX	SATA SSD R2D MUX IN N	37
	SATA 80D	SATA3 PCH TX	SATA SSD R2D P	37
SATA MUX SSD R2D	SATA 80D	SATA3 PCH TX		6 37

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SATA3_T2XTX	*	=2.5x_DIELECTRIC	?
SATA3_RX2RX	*	=2.5x_DIELECTRIC	?
SATA3_T2X20THERTX	*	=4x_DIELECTRIC	?
SATA3_RX20THERRX	*	=4x_DIELECTRIC	?
SATA3_T2X2RX	*	=6x_DIELECTRIC	?
SATA3_RX2TX	*	=6x_DIELECTRIC	?
SATA3_20THERHS	*	=4x_DIELECTRIC	?
SATA3_20THER	*	=3x_DIELECTRIC	?

	SATA_PCH_MUX_D2R	SATA_80D	SATA3_PCH_RX	SATA_HDD_D2R_P	16	97
	SATA_PCH_MUX_D2R	SATA_80D	SATA_HDD_D2R_N		16	97
		SATA_80D	SATA3_PCH_RX	SATA SSD D2R MUX_OUT_P	16	37
		SATA_80D	SATA3_PCH_RX	SATA SSD D2R MUX_OUT_N	16	37
	SATA_MUX_SSD_D2R	SATA_80D	SATA3_PCH_RX	SATA SSD D2R_P	6	37
	SATA_MUX_SSD_D2R	SATA_80D	SATA3_PCH_RX	SATA SSD D2R_N	6	37
	PCH_SATA_TCMP		SATA_TCMP	PCH_SATAIDCOMP	16	

D

USB_HUB1_UP	USB_80D	USB	USB HUB UP P	18	24
USB_HUB1_UP	USB_80D	USB	USB HUB UP N	18	24
USB_RT	USB_80D	USB	USB BT P	24	36
USB_RT	USB_80D	USB	USB BT N	24	36
	USB_80D	USB	USB BT_CONN P	6	36
	USB_80D	USB	USB BT_CONN N	6	36
	USB_80D	USB	USB BT_WAKE P	36	
	USB_80D	USB	USB BT_WAKE N	36	
USB_TP4D	USB_80D	USB	USB TP4D P	6	48
USB_TP4D	USB_80D	USB	USB TP4D N	6	48
	USB_80D	USB	USB TP4D_CONN P		
	USB_80D	USB	USB TP4D_CONN N		
USB_TP4D_HUB	USB_80D	USB	USB TP4D_HUB P	24	
USB_TP4D_HUB	USB_80D	USB	USB TP4D_HUB N	24	
	USB_80D	USB	USB TP4D_R P	24	48
	USB_80D	USB	USB TP4D_R N	24	48
USB_TP4D_M	USB_80D	USB	USB TP4D_M P	48	
USB_TP4D_M	USB_80D	USB	USB TP4D_M N	48	
USB_SDCARD	USB_80D	USB	USB SDCARD P	8	24
USB_SDCARD	USB_80D	USB	USB SDCARD N	8	24
USB_SMC	USB_80D	USB	USB SMC P	24	48
USB_SMC	USB_80D	USB	USB SMC N	24	48

C

USB_CAMERA	USB_80D	USB	USB_CAMERA_P	6 1B
USB_CAMERA	USB_80D	USB	USB_CAMERA_N	6 1B
			USB_EXTN_P	

USB_EXTN	USB_80D	USB	USB_EXTN_P	10	38
USB_EXTN	USB_80D	USB	USB_EXTN_N	38	38
	UART_45S	UART	SMC_DEBUGPRT_TX_L	38	40
	UART_45S	UART	SMC_DEBUGPRT_RX_L	38	40
	USB_80D	USB	USB2_EXTN_MUXED_P	38	40
	USB_80D	USB	USB2_EXTN_MUXED_N	38	38
	USB_80D	USB	USB2_EXTN_MUXED_F_P	38	38
	USB_80D	USB	USB2_EXTN_MUXED_F_N	38	38
USB3_EXTN_RX	USB_80D	USB3_PCH_RX	USB3_EXTN_RX_P	10	38
USB3_EXTN_RX	USB_80D	USB3_PCH_RX	USB3_EXTN_RX_N	10	38
USB3_EXTN_TX	USB_80D	USB3_PCH_TX	USB3_EXTN_TX_P	10	38
USB3_EXTN_TX	USB_80D	USB3_PCH_TX	USB3_EXTN_TX_N	10	38
	USB_80D	USB3_PCH_RX	USB3_EXTN_RX_F_P	38	38
	USB_80D	USB3_PCH_RX	USB3_EXTN_RX_F_N	38	38
	USB_80D	USB3_PCH_TX	USB3_EXTN_TX_F_P	38	38
	USB_80D	USB3_PCH_TX	USB3_EXTN_TX_F_N	38	38
	USB_80D	USB3_PCH_TX	USB3_EXTN_TX_C_P	38	38
	USB_80D	USB3_PCH_TX	USB3_EXTN_TX_C_N	38	38

B

USB_EXTB	USB_80D	USB	USB_EXTB_P	6 24
USB_EXTB	USB_80D	USB	USB_EXTB_N	6 24
	USB_80D	USB	USB_EXTB_EHCI_P	10 24
	USB_80D	USB	USB_EXTB_EHCI_N	18 24
	USB_80D	USB	USB_EXTB_XHCI_P	18 24
	USB_80D	USB	USB_EXTB_XHCI_N	18 24
USB3_EXTB_RX	USB_80D	USB3_PCH_RX	USB3_EXTB_RX_P	18 39
USB3_EXTB_RX	USB_80D	USB3_PCH_RX	USB3_EXTB_RX_N	18 39
	USB_80D	USB3_PCH_RX	USB3_EXTB_RX_RC_P	6 39
	USB_80D	USB3_PCH_RX	USB3_EXTB_RX_RC_N	6 39
	USB_80D	USB3_PCH_RX	USB3_EXTB_RX_CONN_P	
	USB_80D	USB3_PCH_RX	USB3_EXTB_RX_CONN_N	
USB3_EXTB_TX	USB_80D	USB3_PCH_TX	USB3_EXTB_TX_P	18 39
USB3_EXTB_TX	USB_80D	USB3_PCH_TX	USB3_EXTB_TX_N	18 39
	USB_80D	USB3_PCH_TX	USB3_EXTB_TX_C_P	6 39
	USB_80D	USB3_PCH_TX	USB3_EXTB_TX_C_N	6 39

(USB_TPAD_HUB)	USB_80D	USB	USB_EXTD_XHCI_P	18 24
(USB_TPAD_HUB)	USB_80D	USB	USB_EXTD_XHCI_N	18 24

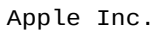
	PCH_USB_RTBIAS	PCH_USB_RTBIAS	PCH_USB_RTBIAS	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
	PCH_DTEFCLK_UNUSED	CLK_PCIE_800	CLK_PCIE	18
		CPU_455	CLK_PCIE	18

A

[illegible]

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
USB3_TX2TX	*	=2.5x_DIELECTRIC	?
USB3_RX2RX	*	=2.5x_DIELECTRIC	?
USB3_TX20THERTX	*	=4x_DIELECTRIC	?
USB3_RX20THERRX	*	=4x_DIELECTRIC	?
USB3_TX2RX	*	=6x_DIELECTRIC	?
USB3_RX2TX	*	=6x_DIELECTRIC	?
USB3_20THERHS	*	=4x_DIELECTRIC	?
USB3_20THER	*	=3x_DIELECTRIC	?

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LPC Bus Constraints								
PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP	
LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD	
CLK_LPC_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD	

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
LPC	*	=3x_DIELECTRIC	?
CLK_LPC	*	=4x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SMBus Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SMB_45S_R_50S	TOP,BOTTOM	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE	=50_OHM_SE		
SMB_45S_R_50S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SMB	*	=2x_DIELECTRIC	?

HD Audio Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
HDA_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
HDA	*	=2x_DIELECTRIC	?

SOURCE: Calpella Platform Design Guide for Ibex Peak M (DG-398905-398905_v1.5), Section 3.15

SIO Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=4x_DIELECTRIC	?

SPI Interface Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
SPI	*	=4x_DIELECTRIC	?

XDP Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
PCH_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
PCH_ITP	*	=2:1_SPACING	?

DisplayPort

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
DP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT	SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
DP_2DP	*	=3x_DIELECTRIC	?	DP_2DP	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_20THERHS	*	=4x_DIELECTRIC	?	DP_20THERHS	TOP,BOTTOM	=6x_DIELECTRIC	?
DP_20THER	*	=3x_DIELECTRIC	?	DP_20THER	TOP,BOTTOM	=4x_DIELECTRIC	?
DP_AUX	*	=3x_DIELECTRIC	?	DP_AUX	TOP,BOTTOM	=4x_DIELECTRIC	?

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
DP_TX	DP_TX	*	DP_2DP
DP_TX	*_TX	*	DP_20THERHS
DP_TX	*_RX	*	DP_20THERHS
DP_TX	*	*	DP_20THER

System Clock Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW ROUTE ON LAYER?	MINIMUM LINE WIDTH	MINIMUM NECK WIDTH	MAXIMUM NECK LENGTH	DIFFPAIR PRIMARY GAP	DIFFPAIR NECK GAP
CLK_SLOW_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD
CLK_25M_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE SPACING	WEIGHT
CLK_SLOW	*	=2x_DIELECTRIC	?
CLK_25M	*	=5x_DIELECTRIC	?

NOTE: 25MHz system clocks very sensitive to noise.

PCH Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		PHYSICAL		SPACING	
LPC_AD	LPC_45S	LPC	LPC AD<3..0>	6	16	40	42
LPC_FRAME_L	LPC_45S	LPC	LPC FRAME_L	6	16	40	42
LPC_45S	LPC_45S	LPC	LPCPLUS RESET_L	6	25	42	
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC CLK33M_SMC	25	40		
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC CLK33M_SMC_R	18	25		
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC CLK33M_LPCPLUS	6	25	42	
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	LPC CLK33M_LPCPLUS_R	18	25		
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	PCH CLK33M_PCIIN	16	25		
LPC_CLK33M	CLK_LPC_45S	CLK_LPC	PCH CLK33M_PCIOUT	18	25		
SMBUS_PCH_CLK	SMR_45S_R_50S	SMR	SMBUS_PCH_CLK	16	43		
SMBUS_PCH_DATA	SMR_45S_R_50S	SMR	SMBUS_PCH_DATA	16	43		
SMBUS_PCH_0_CLK	SMR_45S_R_50S	SMR	SML_PCH_0_CLK	16	43		
SMBUS_PCH_0_DATA	SMR_45S_R_50S	SMR	SML_PCH_0_DATA	16	43		
SMBUS_SMC_1_S0_SCI	SMR_45S_R_50S	SMR	SML_PCH_1_CLK	16	43		
SMBUS_SMC_1_S0_SDA	SMR_45S_R_50S	SMR	SML_PCH_1_DATA	16	43		
HDA_BIT_CLK	HDA_45S	HDA	HDA_BIT_CLK	6	16	39	
HDA_SYNC	HDA_45S	HDA	HDA_BIT_CLK_R	6	16	39	
HDA_SYNC	HDA_45S	HDA	HDA_SYNC	6	16	39	
HDA_RST_L	HDA_45S	HDA	HDA_SYNC_R	16			
HDA_RST_L	HDA_45S	HDA	HDA_RST_R_L	16			
HDA_SDIN0	HDA_45S	HDA	HDA_RST_L	6	16	39	
HDA_SDOUT	HDA_45S	HDA	HDA_SDIN0	6	16	39	
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT	6	16	39	
HDA_SDOUT	HDA_45S	HDA	HDA_SDOUT_R	16	25		
PM_SUS_CLK	CLK_SLOW_45S	CLK_SLOW	PM_CLK32K_SUSCLK_R	17	41		
SPT_CLK	CLK_SLOW_45S	CLK_SLOW	SMC_CLK32K	40	41		
SPT_45S	SPT_45S	SPT	SPI_CLK_R	16	42		
SPT_45S	SPT_45S	SPT	SPI_CLK	42			
SPT_45S	SPT_45S	SPT	SPI_MOSI_R	16	42		
SPT_45S	SPT_45S	SPT	SPI_MOSI	42			
SPT_45S	SPT_45S	SPT	SPI_MISO	16	42		
SPT_45S	SPT_45S	SPT	SPI_CS0_R_L	16	42		
SPT_45S	SPT_45S	SPT	SPI_CS0_L	42			
SPT_45S	SPT_45S	SPT	SPI_SMC_CLK	40	41		
SPT_45S	SPT_45S	SPT	SPI_SMC_MOSI	40	41		
SPT_45S	SPT_45S	SPT	SPI_SMC_MISO	40	41		
SPT_45S	SPT_45S	SPT	SPI_SMC_CS_L	40	41		
SPT_45S	SPT_45S	SPT	SPI_MLB_CLK	41	42	49	
SPT_45S	SPT_45S	SPT	SPI_MLB_MOSI	41	42	49	
SPT_45S	SPT_45S	SPT	SPI_MLB_MISO	41	42	49	
SPT_45S	SPT_45S	SPT	SPI_MLB_CS_L	41	42	49	
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_P	6	36		
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_N	6	36		
PCIE_AP_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_AP_R2D_C_P	16	36		
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_R2D_C_N	16	36		
PCIE_AP_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_AP_D2R_P	6	16	36	
PCIE_CLK100M_AP	CLK_PCTE_80D	CLK_PCTE	PCIE_AP_D2R_N	6	16	36	
PCIE_CLK100M_AP	CLK_PCTE_80D	CLK_PCTE	PCIE_CLK100M_AP_P	6	16	36	
PCIE_CLK100M_AP	CLK_PCTE_80D	CLK_PCTE	PCIE_CLK100M_AP_N	6	16	36	
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_P<3..0>	33			
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_N<3..0>	33			
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_P<3..0>	6	33		
PCIE_TBT_R2D	PCIE_80D	PCIE_PCH_TX	PCIE_TBT_R2D_C_N<3..0>	6	33		
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_P<3..0>	6	33		
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_N<3..0>	6	33		
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_P<3..0>	6	33		
PCIE_TBT_D2R	PCIE_80D	PCIE_PCH_RX	PCIE_TBT_D2R_C_N<3..0>	6	33		
PCIE_CLK100M_TBT	CLK_PCTE_80D	CLK_PCTE	PCIE_CLK100M_TBT_P	16	33		
PCIE_CLK100M_TBT	CLK_PCTE_80D	CLK_PCTE	PCIE_CLK100M_TBT_N	16	33		
CLK_PCTE_80D	CLK_PCTE_80D	CLK_PCTE	PEG_CLK100M_P	8	16		
CLK_PCTE_80D	CLK_PCTE_80D	CLK_PCTE	PEG_CLK100M_N	8	16		
XDP_TDI	PCH_45S	PCH_TTP	XDP_PCH_TDI	16	23		
XDP_TDO	PCH_45S	PCH_TTP	XDP_PCH_TDO	16	23		
XDP_TMS	PCH_45S	PCH_TTP	XDP_PCH_TMS	16	23		
XDP_TCK	PCH_45S	PCH_TTP	XDP_PCH_TCK	16	23		

Chipset Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		PHYSICAL		SPACING	
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK0_ML_P<3..0>	6	33		
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK0_ML_N<3..0>	6	33		
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK0_ML_C_P<3..0>	6	33		
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK0_ML_C_N<3..0>	6	33		
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK0_AUXCH_P	33			
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK0_AUXCH_N	33			
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK0_AUXCH_C_P	6	33		
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK0_AUXCH_C_N	6	33		
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK1_ML_P<3..0>	33			
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK1_ML_N<3..0>	33			
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK1_ML_C_P<3..0>	6	33		
DP_TBT_MI	DP_80D	DP_TX	DP TBTSNK1_ML_C_N<3..0>	6	33		
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK1_AUXCH_P	33			
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK1_AUXCH_N	33			
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK1_AUXCH_C_P	6	33		
DP_TBT_AUXCH	DP_80D	DP_AUX	DP TBTSNK1_AUXCH_C_N	6	33		

Clock Net Properties

ELECTRICAL_CONSTRAINT_SET		NET_TYPE		PHYSICAL		SPACING	
SYSCLK_CLK32K_RTC	CLK_SLOW_45S	CLK_SLOW	SYSCLK_CLK32K_RTC	16	25		
SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB	16	25		
SYSCLK_CLK25M_SB	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_SB_R	16			
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT	25	33		
SYSCLK_CLK25M_TBT	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_TBT_R	33			
SYSCLK_CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X1	25			
SYSCLK_CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2	25			
SYSCLK_CLK25M_XTAL	CLK_25M_45S	CLK_25M	SYSCLK_CLK25M_X2_R	25			

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PAGE TITLE			
PCH Constraints 2			
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DisplayPort Signal Constraints

NOTE: DisplayPort Physical/Spacing Constraints provided by Chipset or GPU page.

Thunderbolt SPI Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
TBT_SPI_45S	*	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=45_OHM_SE	=STANDARD	=STANDARD

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
TBT_SPI	*	=2x_DIELECTRIC	?

Thunderbolt/DP Connector Signal Constraints

PHYSICAL_RULE_SET	LAYER	ALLOW_ROUTE_ON_LAYER?	MINIMUM_LINE_WIDTH	MINIMUM_NECK_WIDTH	MAXIMUM_NECK_LENGTH	DIFFPAIR_PRIMARY_GAP	DIFFPAIR_NECK_GAP
TBDTP_80D	*	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF	=80_OHM_DIFF

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
TBDTP_TX	TBDTP_TX	*	TBDTP_TX2TX
TBDTP_RX	TBDTP_RX	*	TBDTP_RX2RX
TBDTP_TX	TBDTP_RX	*	TBDTP_TX2RX
TBDTP_RX	TBDTP_TX	*	TBDTP_TX2RX
TBDTP_TX	*_TX	*	TBDTP_20THERHS
TBDTP_RX	*_TX	*	TBDTP_20THERHS
TBDTP_TX	*_RX	*	TBDTP_20THERHS
TBDTP_RX	*_RX	*	TBDTP_20THERHS
TBDTP_TX	*	*	TBDTP_20THER
TBDTP_RX	*	*	TBDTP_20THER

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
TBDTP_TX2TX	TOP,BOTTOM	=6x_DIELECTRIC	?
TBDTP_RX2RX	TOP,BOTTOM	=6x_DIELECTRIC	?
TBDTP_TX2RX	TOP,BOTTOM	=10x_DIELECTRIC	?
TBDTP_20THERHS	TOP,BOTTOM	=10x_DIELECTRIC	?
TBDTP_20THER	TOP,BOTTOM	=6x_DIELECTRIC	?

SPACING_RULE_SET	LAYER	LINE-TO-LINE_SPACING	WEIGHT
TBDTP_TX2TX	*	=4x_DIELECTRIC	?
TBDTP_RX2RX	*	=4x_DIELECTRIC	?
TBDTP_TX2RX	*	=6x_DIELECTRIC	?
TBDTP_20THERHS	*	=6x_DIELECTRIC	?
TBDTP_20THER	*	=4x_DIELECTRIC	?

Thunderbolt/DP Net Properties

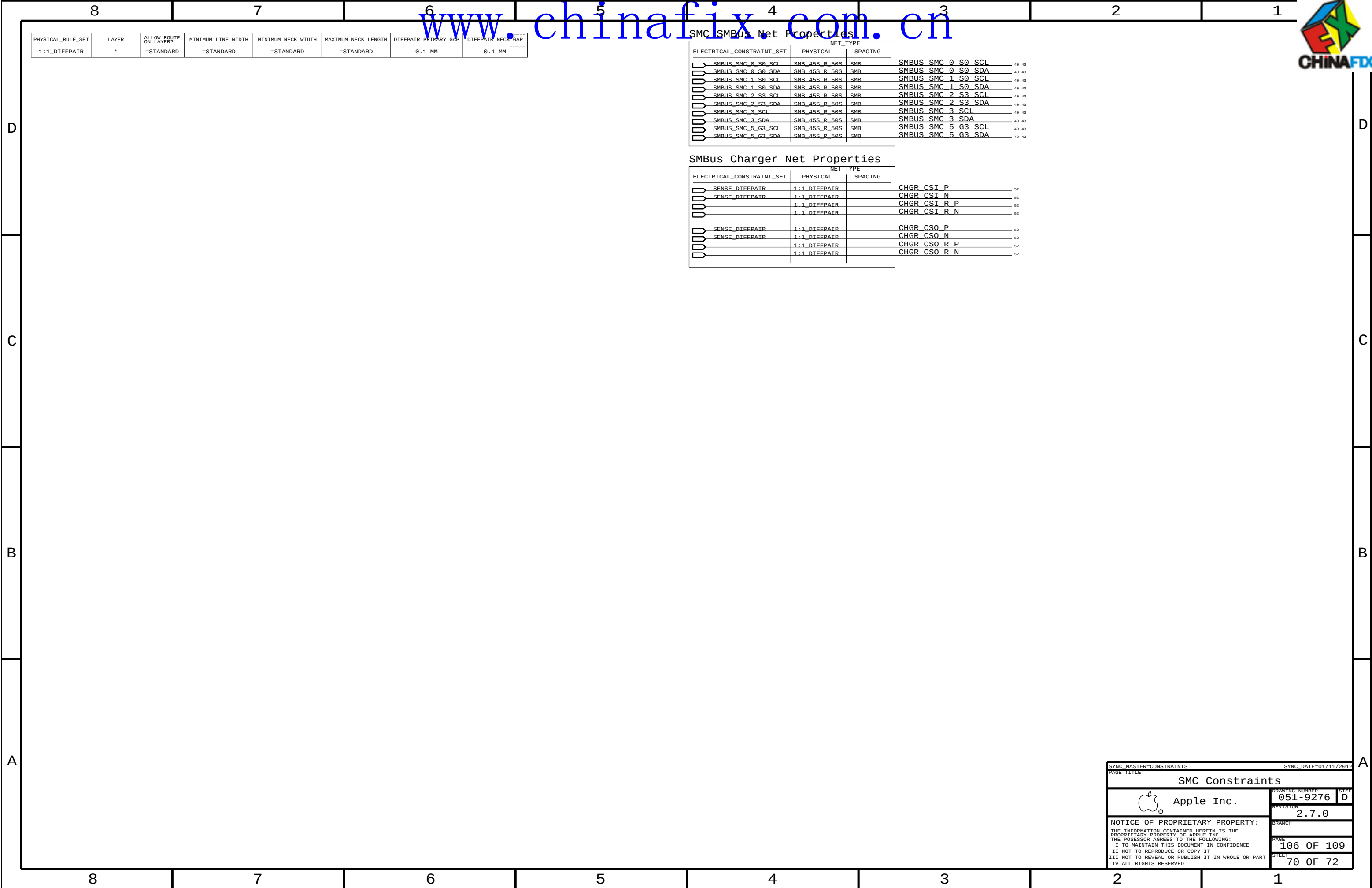
ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
TBT_A_R2D	TBDTP_80D	TBDTP_TX	TBT A R2D C P<1..0> 33
TBT_A_R2D	TBDTP_80D	TBDTP_TX	TBT A R2D C N<1..0> 33
	TBDTP_80D	TBDTP_TX	TBT A R2D P<1..0> 63
	TBDTP_80D	TBDTP_TX	TBT A R2D N<1..0> 63
DP_TBTPA_ML1	DP_80D	DP_TX	DP TBTPA ML C P<1> 33
DP_TBTPA_ML1	DP_80D	DP_TX	DP TBTPA ML C N<1> 33
DP_TBTPA_ML3	DP_80D	DP_TX	DP TBTPA ML C P<3> 33
DP_TBTPA_ML3	DP_80D	DP_TX	DP TBTPA ML C N<3> 33
	DP_80D	DP_TX	DP TBTPA ML P<3..1:2> 63
	DP_80D	DP_TX	DP TBTPA ML N<3..1:2> 63
	DP_80D	DP_TX	DP A LSX ML P<1> 63
	DP_80D	DP_TX	DP A LSX ML N<1> 63
	TBDTP_80D	TBDTP_RX	TBT A D2R C P<1..0> 63
	TBDTP_80D	TBDTP_RX	TBT A D2R C N<1..0> 63
TBT_A_D2R1	TBDTP_80D	TBDTP_RX	TBT A D2R P<1> 33
TBT_A_D2R1	TBDTP_80D	TBDTP_RX	TBT A D2R N<1> 33
TBT_A_D2R0	TBDTP_80D	TBDTP_RX	TBT A D2R P<0> 33
TBT_A_D2R0	TBDTP_80D	TBDTP_RX	TBT A D2R N<0> 33
TBT_A_AUXCH	DP_80D	DP_AUX	DP TBTPA AUXCH C P 33
TBT_A_AUXCH	DP_80D	DP_AUX	DP TBTPA AUXCH C N 33
	DP_80D	DP_AUX	DP TBTPA AUXCH P 63
	DP_80D	DP_AUX	DP TBTPA AUXCH N 63
	DP_80D	DP_AUX	DP A AUXCH DDC P 63
	DP_80D	DP_AUX	DP A AUXCH DDC N 63
	TBDTP_80D	TBDTP_RX	TBT A D2R1 AUXDDC P 63
	TBDTP_80D	TBDTP_RX	TBT A D2R1 AUXDDC N 63
TBT_B_R2D	TBDTP_80D	TBDTP_TX	TBT B R2D C P<1..0> 8 33
TBT_B_R2D	TBDTP_80D	TBDTP_TX	TBT B R2D C N<1..0> 8 33
	TBDTP_80D	TBDTP_TX	TBT B R2D P<1..0> 63
	TBDTP_80D	TBDTP_TX	TBT B R2D N<1..0> 63
DP_TBTPB_ML	DP_80D	DP_TX	DP TBTPB ML C P<3..1:2> 8 33
DP_TBTPB_ML	DP_80D	DP_TX	DP TBTPB ML C N<3..1:2> 8 33
	DP_80D	DP_TX	DP TBTPB ML P<3..1:2> 63
	DP_80D	DP_TX	DP TBTPB ML N<3..1:2> 63
	DP_80D	DP_TX	DP B LSX ML P<1> 63
	DP_80D	DP_TX	DP B LSX ML N<1> 63
	TBDTP_80D	TBDTP_RX	TBT B D2R C P<1..0> 63
	TBDTP_80D	TBDTP_RX	TBT B D2R C N<1..0> 63
TBT_B_D2R	TBDTP_80D	TBDTP_RX	TBT B D2R P<1..0> 8 33
TBT_B_D2R	TBDTP_80D	TBDTP_RX	TBT B D2R N<1..0> 8 33
TBT_B_AUXCH	DP_80D	DP_AUX	DP TBTPB AUXCH C P 8 33
TBT_B_AUXCH	DP_80D	DP_AUX	DP TBTPB AUXCH C N 8 33
	DP_80D	DP_AUX	DP TBTPB AUXCH P 63
	DP_80D	DP_AUX	DP TBTPB AUXCH N 63
	DP_80D	DP_AUX	DP B AUXCH DDC P 63
	DP_80D	DP_AUX	DP B AUXCH DDC N 63
	TBDTP_80D	TBDTP_RX	TBT B D2R1 AUXDDC P 63
	TBDTP_80D	TBDTP_RX	TBT B D2R1 AUXDDC N 63

Only used on dual-port hosts.

Thunderbolt IC Net Properties

ELECTRICAL_CONSTRAINT_SET	NET_TYPE		
	PHYSICAL	SPACING	
	DP_80D	DP_TX	DP TBTSRC ML C P<3..0> 33
	DP_80D	DP_TX	DP TBTSRC ML C N<3..0> 33
	DP_80D	DP_AUX	DP TBTSRC AUXCH C P 33
	DP_80D	DP_AUX	DP TBTSRC AUXCH C N 33
TBT_SPI_CLK	TBT_SPI_45S	TBT_SPI	TBT SPI CLK 33
TBT_SPI_MOSI	TBT_SPI_45S	TBT_SPI	TBT SPI MOSI 33
TBT_SPI_MISO	TBT_SPI_45S	TBT_SPI	TBT SPI MISO 33
TBT_SPI_CS_L	TBT_SPI_45S	TBT_SPI	TBT SPI CS L 33

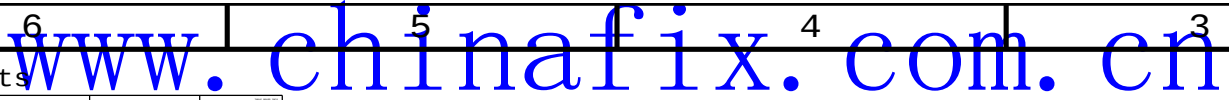
Only used on hosts supporting Thunderbolt video-in



J11/J13-Specific Net Properties

NET_SPACING_TYPE1	NET_SPACING_TYPE2	AREA_TYPE	SPACING_RULE_SET
GND	CLK_PCIE	*	GND_P2MM
GND	PCIE*	*	GND_P2MM
GND	SATA*	*	GND_P2MM
GND	USB*	*	GND_P2MM
GND	LVDS*	*	GND_P2MM
SB_POWER	CLK_PCIE	*	PwR_P2MM
SB_POWER	SATA*	*	PwR_P2MM
SB_POWER	SATA*	*	PwR_P2MM

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